Physics-Informed Neural Quantum States for Quantum Dots: Tangent-Space Geometry, Message Passing, and Wigner Molecules

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Abstract

How far can a compact, well-conditioned neural network go as a variational wavefunction for interacting electrons; what does letting the particles talk to one another actually buy; and can such a state be trained without a Markov chain? We take up these three questions for two-dimensional harmonic quantum dots—from the compact, kinetic-energy-dominated droplet at strong confinement to the dilute Wigner molecule at weak confinement—for $N \in \{2, 6, 12, 20\}$ electrons and trap frequencies ω from 1 down to 10^{-3}.

The ansatz is a Slater determinant dressed by a neural Jastrow correlator, which reshapes the amplitude, and a coordinate backflow, which reshapes the nodes. Both are made *safe by construction* for Laplacian-based training: coordinates are scaled to trap units, pair features are soft-core, and the electron–electron cusp is imposed analytically. Each learnable object comes in two forms—one that pools per-particle and pairwise features in a single pass, and one that passes messages between particles on a graph—and we read the trained states through the geometry of their log-derivatives, the quantum geometric tensor and the neural tangent kernel. Training is by residual pretraining followed by stochastic reconfiguration (SR), and, separately, by importance-sampled collocation that uses no Markov chain.

Message passing, it turns out, buys one thing: a *relational channel*. Seen through the tangent kernel it compresses the correlator onto roughly half as many effective directions as a single-pass network, and it keeps the backflow from collapsing—at the Wigner crystal a conventional backflow falls to a single collective mode and loses accuracy, while the message-passing backflow holds full rank and stays near reference quality. The same geometry governs the rest. SR helps in proportion to how ill-conditioned the tangent metric is, but only while that metric can still be estimated reliably; once the sampling that builds it collapses, inverting it does harm. And importance-sampled collocation matches SR–VMC while its sampling stays healthy, reaching a genuine Wigner molecule at $\omega = 0.0035$ with no Markov chain at any point in training—SR–VMC carries the $N = 20$ energies, while the Markov-chain-free route is demonstrated for $N \leq 12$. Energies match diffusion Monte Carlo within uncertainty for $N = 2$ and to $\sim 10^{-2}$–10^{-1}% for $N \in \{6, 12, 20\}$ wherever a DMC reference exists ($\omega \geq 0.1$); below that no benchmark is available, and we say so.

On the physics, the trained states let us map the Fermi-liquid-to-Wigner crossover from the inside. We give topology-resolved shell, bond-orientational, and stiffness diagnostics, with shell-mixture reconstructions of $g(r)$ matching the measured pair distribution to cosine similarity $0.989-1.000$: at $\omega = 10^{-3}$ the $N = 6$ dot is a single ring with a robust $(1, 5) \leftrightarrow (0, 6)$ near-degeneracy (radial mode $\approx 114\, a_0$), the $N = 12$ dot a two-shell molecule with a modal $(3, 9)$ split, and the $N = 20$ dot a three-shell molecule that collapses toward a single shell within one decade in ω. One thread runs through architecture, optimiser, and training paradigm alike: what matters is not raw capacity but staying aligned with the low-dimensional, physically collective structure of the problem—the same structure that, in real space, is the Wigner molecule [1–6].

Acknowledgements

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The $N = 20$ reference energies used for comparison in Chapter 6 are drawn from the master's thesis of Daniel Haas [7], University of Oslo (2024), and I am grateful for the availability of that work as an independent benchmark.

Contents

Part I

Introduction

Introduction

Quantum mechanics is a central pillar of modern physics. It provides the framework in which we describe atoms, molecules, solids, and nanoscale structures, and it underlies much of today's technology. At the same time, it leads directly to some of the most challenging computational problems we face. The main example is the *many-body problem*: solving the Schrödinger equation for interacting particles and extracting physically meaningful quantities from the resulting wavefunction.

In principle, the many-body wavefunction contains complete information about the system. In practice, its dimension grows exponentially with the number of particles and the size of the single-particle basis. The competition between kinetic energy, which tends to delocalize particles, and interaction and external potentials, which drive localization and correlations, takes place in a configuration space that is far too large to handle naively. Nevertheless, we need accurate results—energies, correlation functions, and phase boundaries—to understand real materials and model strongly correlated systems.

A wide range of electronic-structure methods has been developed to address this. Mean-field approaches such as Hartree–Fock replace the full many-body problem with an effective single-particle picture, where each electron moves in an average field from the others. This captures Fermi statistics and some exchange effects but leaves most correlation physics untreated. More systematic methods such as Full Configuration Interaction (FCI) are, in principle, exact within a finite basis, but their cost grows combinatorially with system size. Coupled Cluster (CC) theory provides a more efficient and size-extensive hierarchy of approximations, but it remains computationally demanding and can struggle in strongly correlated regimes or near degeneracies.

Quantum Monte Carlo (QMC) methods, including Variational Monte Carlo (VMC) and Diffusion Monte Carlo (DMC), take a different approach: they represent the wavefunction implicitly through stochastic sampling. For suitable trial states, they can produce highly accurate benchmark energies and correlations for continuum systems. However, QMC is not free of difficulties. The fermion sign problem limits direct simulations, fixed-node approximations inherit bias from the chosen trial nodes, and constructing high-quality trial wavefunctions is increasingly difficult as systems become more strongly correlated or inhomogeneous. In practice, a large part of the many-body problem becomes the problem of choosing an expressive, well-conditioned ansatz and optimising it reliably.

Machine learning and many-body wavefunctions. In parallel with these developments, machine learning—especially deep learning—has become a standard tool for representing complex, high-dimensional functions and fitting them to data. Neural networks now play a central role in domains such as computer vision, language modelling and reinforcement learning. They combine flexible parametric families with inductive biases (e.g. symmetry and locality), normalisation, and optimisation procedures that scale well with dimension.

In physics, this has led to two broad uses of neural networks. In the first, they act as *surrogates*: they learn mappings from parameters or boundary conditions to observables, or approximate solution operators of partial differential equations. Physics-informed neural networks (PINNs) and operator-learning approaches fall into this category by embedding differential operators and constraints directly into the training objective. In the second, neural networks are used as *variational ansätze* for quantum states (neural quantum states), where the network outputs amplitudes, phases or log-probabilities for many-body configurations and is trained by variational Monte Carlo.

In principle, neural networks provide highly expressive ansätze for quantum problems. In practice, several issues arise. Objectives based on local energies involve second derivatives and can be numerically delicate. Coordinate parameterizations that do not respect known structure can lead to large gradients and Laplacian spikes near electron coalescence. Generic architectures can waste capacity on degrees of freedom that are irrelevant from a physical point of view. These challenges motivate hybrid approaches in which as much known structure as possible—antisymmetry, cusp conditions, scaling, and symmetries—is built into the ansatz and its coordinates, and the network is used to learn the remaining correlation corrections.

A further ingredient concerns not *what* structure is imposed but *how* the network represents the electrons. Correlation is intrinsically relational—what one electron should do depends on where the others are—so architectures that let particles exchange information (message passing), rather than processing each coordinate independently, are a natural candidate for the correlator and for the backflow that deforms the nodes. How much this relational capacity actually buys, and why, is one of the questions this thesis takes up.

Where this work sits. Representing a quantum state by a neural network and optimising it variationally was introduced by Carleo and Troyer [8]; for continuum fermions the idea has since matured into highly accurate *ab initio* solvers such as FermiNet [9] and PauliNet [10], and into explicit neural *backflow* transformations [11]—the direct ancestor of the coordinate backflow used here. Message passing, the relational mechanism at the centre of this thesis, has recently been built into neural quantum states for extended fermion systems [12, 13]. Against that backdrop the present work is deliberately narrow and complementary. Rather than pushing a new architecture to ever larger systems, it asks *why* the relational channel helps, and uses the tangent-kernel geometry to make the answer precise; it focuses on the two-dimensional quantum-dot Wigner regime, where correlation is strongest and where topology-resolved structural diagnostics remain sparse; and it treats conditioning, the optimiser, and Markov-chain-free training as objects of study in their own right rather than fixed choices. To our knowledge these questions have not previously been addressed together for parabolic quantum dots.

Quantum dots as a controlled testbed. Two-dimensional parabolic quantum dots provide a clean and tunable setting in which to explore such ideas. They offer a simple route from Fermi-liquid–like behaviour to Wigner-molecule physics. As the trap frequency ω decreases, the single-particle level spacing collapses while Coulomb repulsion becomes dominant. Electrons then tend to localize into ring or shell patterns. In the lab frame, rotational symmetry and quantum fluctuations smear these into smooth densities; in a co-rotating frame, one recovers crystalline-like structures whose topology depends on particle number. For N=6 and N=12, classical and semiclassical studies have identified preferred shellings such as $(1, 5)$ and $(3, 9)$ and mapped parts of the crossover [1, 3, 4, 6].

From the perspective of this thesis, quantum dots are attractive for two reasons. First, they are genuinely non-trivial: Coulomb singularities, strong correlations and an actual Fermi-liquid–to–Wigner crossover appear already for moderate N. Second, they are still simple enough that accurate DMC references exist in many regimes and that detailed structural diagnostics (shell occupancies, bond order, virial ratios) are computationally feasible. For fully quantum dots, however, quantitative topology-resolved diagnostics (occupancy fractions, bond-orientational order, shell "stiffness") and a systematic picture of the Wigner regime remain relatively sparse. In addition, it is not well understood how modern neural ansätze internally represent these states.

Scope and questions. In this thesis we revisit two-dimensional harmonic quantum dots with a Slater–Jastrow(+backflow) ansatz implemented as a physics-informed neural network. The design is meant to be *safe by construction* for Laplacian-based objectives: coordinates are scaled to trap units ($\tilde{R} = \sqrt{\omega}R$), learned pair features are soft-core (avoiding spurious $1/r$ behaviour in derivatives), and the electron–electron cusp is enforced analytically rather than learned. On top of a Slater determinant of harmonic-oscillator orbitals, two learnable objects carry the correlation: a Jastrow correlator, which reshapes the amplitude, and a coordinate backflow, which reshapes the nodes. Each can be built either without message passing—encoding and pooling per-particle and pairwise features in a single pass, as in a DeepSet—or as a *message-passing* network in which the particles exchange and update information through explicit, transported edge state before a quantity is read out.

Once such an ansatz reaches reference accuracy—which, as we will see, a surprisingly compact one does—the questions worth a thesis are *why* its parts behave as they do, and what it can tell us about the electrons. We pursue four. The first three we read through a single lens: the geometry of the wavefunction's log-derivatives, captured by the quantum geometric tensor and the neural tangent kernel.

1. **Architecture.** What does message passing buy that a separable network cannot? We find it supplies a *relational channel*—a way to condition each particle on where the others are—that compresses the correlator onto fewer effective directions and keeps the backflow from collapsing to a single collective mode in the strongly correlated limit.

2. **Optimiser.** When does natural-gradient (stochastic reconfiguration) preconditioning actually change the result? Its value tracks the conditioning of the tangent metric: negligible for well-conditioned, $|\Psi|^2$-sampled VMC, decisive for the ill-conditioned collocation objective.
3. **Paradigm.** Can an accurate wavefunction be trained *without* Markov-chain Monte Carlo, using importance-sampled collocation instead—and how far into the correlated regime does that reach before the sampling loses the target?
4. **Physics.** Can the trained wavefunction serve as a scientific instrument for Wigner-molecule physics—shell topologies, angular order, and melting—and does its internal representation reflect that physics?

These are not four separate stories. The same low-dimensional, physically collective structure that message passing captures is what natural gradients exploit, what $|\Psi|^2$ sampling stays aligned with, and what ultimately crystallises into the Wigner molecule itself.

Diagnostics and relation to prior work. To address these questions, we go beyond total energies. Alongside DMC benchmarks, we use:

- automatic shell detection and shell-resolved reconstructions of the pair distribution $g(r)$, testing whether the global $g(r)$ is explained by shell geometry and combinatorics;
- bond-orientational order parameters $|\Phi_m|$ and angular Lindemann-like ratios, adapted from studies of classical Wigner and Yukawa layers [2], to quantify ring order and angular stiffness;
- the density parameter r_s, estimated from the first peak of $g(r)$ following Egger *et al.* [1], to locate our dots along the Fermi-liquid–to–Wigner scale;
- representation-focused diagnostics for the neural ansatz: entropy effective ranks of correlator and backflow features, principal component ablations, and simple linear probes that relate latent axes to physical observables (global size, radial variance, near-origin mass);
- tangent-kernel diagnostics of the ansatz itself: the quantum geometric tensor and neural tangent kernel, the effective dimension of the variational tangent space, and symmetric state overlaps used to compare architectures and training paradigms on an invariant footing.

All of these are applied as post-processing to $|\Psi|^2$ samples, so the same analysis could, in principle, be used to study classical, quantum-chemical, or other neural wavefunctions on the same footing.

Main contributions. The main contributions of this thesis are:

1. **Compact, conditioned accuracy.** A small Slater–Jastrow(+backflow) ansatz in trap units reaches DMC-level energies for N=2 and holds $\sim 10^{-2}$–10^{-1}% relative errors for $N \in \{6, 12, 20\}$ across $\omega \in \{0.1, 0.5, 1.0\}$, extrapolating smoothly to ultra-weak traps ($\omega = 10^{-3}$). Conditioning—trap units, analytic cusp, safe pair features—matters at least as much as network size for stability and accuracy in these dots.
2. **Message passing as a relational channel, made precise by the tangent kernel.** Read through the quantum geometric tensor, a message-passing correlator represents the ground state in roughly half as many effective tangent directions as a separable one, and a message-passing backflow retains full rank exactly where a conventional, single-round-message backflow collapses to a single collective mode at the Wigner crystal and loses accuracy. The two effects are one mechanism—a relational channel the physics requires—and both are invisible in the total energy until weak confinement turns the geometric deficit into an energetic one.
3. **Optimiser and paradigm, governed by the same geometry.** Natural-gradient (SR) preconditioning helps precisely when the tangent metric is ill-conditioned: negligibly for $|\Psi|^2$-VMC, decisively for collocation. And an MCMC-free, importance-sampled collocation objective matches VMC when the sampling is healthy and, with a physics-informed proposal built from the Wigner shell, reaches a genuine Wigner molecule at $\omega = 0.0035$ with no Markov-chain Monte Carlo during training.

4. **Topology-resolved Wigner diagnostics.** Across $\omega \in [1, 10^{-3}]$ we quantify the crossover using two-shell probabilities and inner-ring occupancy histograms, shell-mixture reconstructions of $g(r)$ with cosine similarity 0.989–1.000 to global data, and sector-resolved bond order and Lindemann indices. At $\omega = 10^{-3}$ the N=6 dot forms a persistent single-ring Wigner molecule with a clear $(1,5) \leftrightarrow (0,6)$ near-degeneracy, while the N=12 dot forms a two-shell molecule with a commensurate $(3,9)$ split as the dominant topology, complementing classical and quantum studies of parabolic dots [1, 3–5].

5. **Representation insight, with its gauge honestly drawn.** The correlator features occupy a low-dimensional manifold (one to three principal directions) whose leading axes act as collective coordinates for global size and radial fluctuations, with a $\phi \leftrightarrow \psi$ recomposition across regimes, while backflow spans many local modes. We take care to separate the parts of this picture that are *gauge-like*—fixed by how correlation is divided between the Jastrow and the backflow, two substitutable channels—from the invariants (energy, state overlap, and the energetic price of ablating a component) on which the architectural conclusions actually rest.

Thesis structure. The remainder of the thesis is organised as follows. The next chapter sets out the theoretical background: the many-body Schrödinger equation in harmonic traps, the electronic-structure methods we benchmark against, the physics of Wigner molecules, and the machine-learning tools used here—neural function approximators, physics-informed objectives, neural quantum states, and the tangent-kernel geometry (quantum geometric tensor and neural tangent kernel) that serves as the analytical lens for the results. A method chapter then presents the concrete Slater–Jastrow(+backflow) PINN ansatz in its separable and message-passing forms, the optimisation pipeline (residual pretraining, stochastic reconfiguration, and importance-sampled collocation), and the diagnostics used throughout. The results are given in two chapters. The first treats accuracy, the learned representation, and the Wigner-molecule physics for $N \in \{2, 6, 12, 20\}$; the second reads the architecture, optimiser, and training paradigm through the tangent kernel. A discussion and a concluding chapter follow, and the appendices develop the conditioning theory (Laplacian and Coulomb) and the collocation–backflow catch-22 that motivated the weak-form objective.

Part II

Theory

This chapter lays the foundation on which the rest of the thesis rests. We first introduce the mathematical framework of quantum mechanics. We then discuss the challenges of many-body systems and how second quantization provides a more tractable approach. Then, we will introduce the model systems that will be used throughout this thesis to benchmark and test our methods. Following this, we will present the Hartree-Fock method, a fundamental mean-field approach to approximate many-body wavefunctions. Finally, we will discuss Full Configuration Interaction (FCI) theory, which provides an exact solution to the many-body problem within a finite basis set, and Variational Monte Carlo (VMC), a stochastic method for approximating ground-state properties of quantum systems.

The focus then shifts to the essential concepts of machine learning, covering statistical learning theory, optimisation techniques, neural networks, and physics-informed neural networks (PINNs). This section will provide the theoretical underpinnings necessary for understanding how machine learning can be applied to quantum many-body problems.

Chapter 1

Quantum Theory

1.1 Mathematical Foundations

At the center of quantum mechanics lies a rich mathematical framework built upon linear algebra and functional analysis. This section introduces the essential mathematical concepts and notations that underpin quantum theory, including Dirac notation, operators, Hilbert spaces, and the representation of quantum states and observables.

1.1.1 Hilbert Spaces and Function Representations

In essence, quantum mechanics is formulated in complex vector spaces, specifically infinite-dimensional Hilbert spaces, where vectors generalise to square-integrable functions. The inner product becomes:

$$\langle\psi_i|\psi_j\rangle = \int \psi_i^*(x)\psi_j(x)dx = \delta_{ij}.$$

A function $a(x)$ expands in a basis $\{\psi_i(x)\}$:

$$a(x) = \sum_i \psi_i(x)a_i, \quad a_i = \int \psi_i^*(x)a(x)dx.$$

The Dirac delta $\delta(x-y)$ replaces the Kronecker delta in continuous spaces. A vector $|a\rangle$ in an n-dimensional Hilbert space $\mathcal{H}$ can be decomposed into a complete orthonormal basis $\{|i\rangle\}$:

$$|a\rangle = \sum_i |i\rangle\langle i|a\rangle = \sum_i a_i |i\rangle,$$

where $a_i = \langle i|a\rangle$ are complex coefficients. The completeness relation,

$$I = \sum_i |i\rangle\langle i|,$$

ensures the basis spans $\mathcal{H}$. Vectors in Dirac notation have dual "bra" counterparts $\langle a| = |a\rangle^\dagger$, enabling inner products $\langle a|b\rangle$ and outer products $|a\rangle\langle b|$. A more complete discussion on hilbert spaces are found in [14].

1.1.2 Operators and Matrix Representations

Linear operators $\hat{\mathcal{O}}$ map vectors to vectors: $\hat{\mathcal{O}}|a\rangle = |b\rangle$. Their adjoints $\hat{\mathcal{O}}^\dagger$ satisfy $\langle a|\hat{\mathcal{O}}^\dagger = \langle b|$. Hermitian operators ($\hat{\mathcal{O}}^\dagger = \hat{\mathcal{O}}$) represent observables, while unitary operators ($\hat{\mathcal{O}}^\dagger = \hat{\mathcal{O}}^{-1}$) preserve inner products [15, 16]. In a basis $\{|i\rangle\}$, operators are represented by matrices:

$$O_{ij} = \langle i|\hat{\mathcal{O}}|j\rangle.$$

Non-commutativity of operators is quantified by the commutator:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}.$$

1.1.3 Hilbert Spaces

In quantum mechanics a state is a vector $|\psi\rangle$ in a complex Hilbert space $\mathcal{H}$. A Hilbert space is a complex inner-product space that is *complete* in the norm induced by the inner product. Using the physics convention (linear in the second slot), the inner product $\langle\phi|\psi\rangle$ obeys

$$\begin{aligned} \text{conjugate symmetry:} \quad & \langle\phi|\psi\rangle = \langle\psi|\phi\rangle^* , \\ \text{linearity in kets:} \quad & \langle\phi|\alpha\psi_1 + \beta\psi_2\rangle = \alpha \langle\phi|\psi_1\rangle + \beta \langle\phi|\psi_2\rangle , \\ \text{positivity:} \quad & \langle\psi|\psi\rangle \geq 0 \text{ with equality iff } |\psi\rangle = 0. \end{aligned}$$

These allow us to define the norm $\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$. *Completeness* means every Cauchy sequence in this metric converges to a vector in $\mathcal{H}$, ensuring limits and expansions are well defined. For a more detailed explaination, refer to [15, 16].

1.1.4 States in Hilbert Space

A (pure) quantum state is a normalised vector $|\psi\rangle \in \mathcal{H}$ (global phase is physically irrelevant). Computations proceed by choosing an orthonormal (ON) basis $\{|i\rangle\}_{i=1}^{\infty}$ and expanding

$$|\psi\rangle = \sum_i c_i |i\rangle , \qquad c_i = \langle i|\psi\rangle , \qquad \sum_i |c_i|^2 = \|\psi\|^2.$$

Completeness is the resolution of the identity, $\sum_i |i\rangle\langle i| = \mathbb{I}$. For continuous ON bases (e.g. position), $\int |x\rangle\langle x| \, dx = \mathbb{I}$ and the wavefunction is the coordinate representation $\psi(x) = \langle x|\psi\rangle$.

Different ON bases are related by a unitary change of basis. Given $\{|i\rangle\}$ and $\{|\alpha\rangle\}$,

$$|\alpha\rangle = \sum_i U_{\alpha i} |i\rangle , \qquad U_{\alpha i} = \langle\alpha|i\rangle , \qquad U^\dagger U = \mathbb{I}.$$

This freedom is exploited to match the physics (symmetry, locality, boundary conditions), e.g. harmonic-oscillator/Fock–Darwin orbitals for confined charges or plane waves for periodic systems, leading to sparser representations and faster convergence.

For N particles the Hilbert space factorizes as $\mathcal{H}_N = \mathcal{H}_1^{\otimes N}$. With a single-particle ON set $\{|\varphi_p\rangle\}$, a convenient fermionic ON basis is the set of Slater determinants $\{|\Phi_I\rangle\}$ built from N distinct orbitals; any N-body state expands as

$$|\Psi\rangle = \sum_I C_I |\Phi_I\rangle , \qquad \sum_I |C_I|^2 = \|\Psi\|^2.$$

In practice one truncates to a finite active space guided by energy scales and symmetry quantum numbers, which makes many-body calculations tractable while retaining the relevant physics.

1.2 Quantum Mechanics

1.2.1 The Schrödinger Equation

In a Hilbert space $\mathcal{H}$, the time evolution of a normalised state $|\Psi(t)\rangle$ is postulated to be unitary and generated by the Hamiltonian H [15, 16]:

$$i\,\partial_t |\Psi(t)\rangle = H |\Psi(t)\rangle , \qquad (\hbar = 1).$$

If H is time–independent, solutions separate as

$$|\Psi(t)\rangle = \sum_n c_n \, e^{-iE_n t} |\psi_n\rangle , \qquad H |\psi_n\rangle = E_n |\psi_n\rangle ,$$

so $\{|\psi_n\rangle\}$ forms an orthonormal basis of stationary states.

For an N-particle system in d spatial dimensions, with coordinates $R = (\mathbf{r}_1, \ldots, \mathbf{r}_N)$, we use the many-body Hamiltonian

$$H = -\frac{1}{2}\sum_{i=1}^{N} \nabla^2_{\mathbf{r}_i} \; + \; \sum_{i=1}^{N} V_{\text{ext}}(\mathbf{r}_i) \; + \; \sum_{i<j} V_{\text{int}}(|\mathbf{r}_i - \mathbf{r}_j|),$$

and the time-independent Schrödinger equation (TISE)

$$H\,\Psi(R) = E\,\Psi(R), \qquad \int |\Psi(R)|^2\,dR = 1.$$

Identical particles restrict Ψ to the appropriate symmetry sector (symmetric for bosons, antisymmetric for fermions).

Throughout we work in atomic units ($\hbar = m_e = e = 1$) and focus on stationary problems. In the quantum-dot applications below we take $V_{\text{ext}}(\mathbf{r}) = \frac{1}{2}\omega^2 r^2$ and $V_{\text{int}}(r) = 1/r$ (in 2D this is understood with the usual Laplacian in two dimensions). Our computations therefore amount to approximating low-lying eigenpairs (E, Ψ) of the many-body Hamiltonian above using compact basis expansions and variational ansätze.

1.3 Second Quantization

1.3.1 Motivation: Why first-quantized wavefunctions don't scale

In first quantization an N-fermion state is an antisymmetric function $\Psi(x_1, \ldots, x_N)$ expanded in Slater determinants of M one-particle orbitals. The number of configurations grows as $\binom{M}{N}$ (and permanents for bosons grow even faster), making storage and operator application increasingly intractable. We therefore move to the occupation-number formalism, which encodes antisymmetry and particle indistinguishability algebraically.

1.3.2 Fock space and occupation representation

Let $\{|p\rangle\}$ be an orthonormal set of one-particle *spin-orbitals* (spatial orbital + spin). The fermionic Fock space is

$$\mathcal{F}_- \;=\; \bigoplus_{n=0}^{\infty} \wedge^n \mathcal{H}_1,$$

with vacuum $|0\rangle$ and occupation-number basis $|n_1 n_2 \ldots\rangle$, $n_p \in \{0, 1\}$. Creation/annihilation operators $a_p^\dagger, a_p$ act by adding/removing a particle in $|p\rangle$ (subject to Pauli) and obey the canonical anticommutation relations (CAR)

$$\{a_p, a_q^\dagger\} = \delta_{pq}, \qquad \{a_p, a_q\} = \{a_p^\dagger, a_q^\dagger\} = 0,$$

with number operator $\hat{n}_p = a_p^\dagger a_p$. A Slater determinant occupying the set $I = \{p_1 < \cdots < p_N\}$ is $|\Phi_I\rangle = a_{p_1}^\dagger \cdots a_{p_N}^\dagger |0\rangle$ and is represented by the bitstring $|n_1 n_2 \ldots\rangle$ with $n_p = \mathbf{1}_{p \in I}$. Unitary changes of one-particle basis $|p'\rangle = \sum_p U_{p'p} |p\rangle$ lift to $a_{p'}^\dagger = \sum_p U_{p'p} a_p^\dagger$, so physics is basis independent.

1.3.3 Operators in second quantization

Any *one-body* operator $O = \sum_i O(i)$ has matrix elements $O_{pq} = \langle p| O |q\rangle$ and becomes

$$O \;=\; \sum_{pq} O_{pq}\, a_p^\dagger a_q.$$

A *two-body* interaction $V = \frac{1}{2}\sum_{i \neq j} v(i,j)$ has

$$V \;=\; \frac{1}{2} \sum_{pqrs} \langle pq|v|rs\rangle\, a_p^\dagger a_q^\dagger a_s a_r \;=\; \frac{1}{4} \sum_{pqrs} \langle pq\|rs\rangle\, a_p^\dagger a_q^\dagger a_s a_r,$$

where $\langle pq|v|rs\rangle$ are two-electron integrals over spin-orbitals and $\langle pq\|rs\rangle = \langle pq|v|rs\rangle - \langle pq|v|sr\rangle$ denotes the antisymmetrized integral. In this form, operator algebra and particle indistinguishability are handled by the CAR, independent of N.

1.4 Model Systems

1.5 Hartree–Fock (HF)

Hartree–Fock (HF) approximates the N-fermion ground state by a single Slater determinant built from orthonormal spin–orbitals $\{|p\rangle\}_{p=1}^{M}$. In second quantization the Hamiltonian is

$$H = \sum_{pq} h_{pq}\, a_p^\dagger a_q + \frac{1}{4}\sum_{pqrs} (pq\|rs)\, a_p^\dagger a_q^\dagger a_s a_r,$$

with one-body matrix elements $h_{pq} = \langle p|\hat{h}_0|q\rangle$ and antisymmetrized two-body integrals $(pq\|rs) = (pq|rs) - (pq|sr)$ over spin–orbitals.

1.5.1 HF energy functional and the Fock operator

Let $|\Phi\rangle = \prod_{i=1}^{N} a_i^\dagger\, |0\rangle$ be a Slater determinant (occupied spin–orbitals $i, j, \dots$). Its energy is

$$E[\Phi] = \sum_i h_{ii} + \frac{1}{2}\sum_{i,j} (ij\|ij).$$

Stationarity of $E[\Phi]$ under unitary rotations in the occupied+virtual subspace gives the HF equations

$$F\, |\phi_p\rangle = \varepsilon_p\, |\phi_p\rangle\,, \qquad F = \hat{h}_0 + \sum_{j\in\mathrm{occ}} \Big(J_j - K_j\Big),$$

where J_j and K_j are the Coulomb and exchange operators generated by orbital j. In a fixed orthonormal one-particle basis $\{|\mu\rangle\}$ these become the matrix *Roothaan–Hall* equations

$$\mathbf{F}\,\mathbf{C} = \mathbf{C}\,\boldsymbol{\varepsilon}, \qquad F_{\mu\nu} = h_{\mu\nu} + \sum_{\lambda\sigma} P_{\lambda\sigma}\Big[(\mu\nu|\lambda\sigma) - \tfrac{1}{2}(\mu\lambda|\nu\sigma)\Big],$$

with density matrix $P_{\lambda\sigma} = \sum_{i\in\mathrm{occ}} C_{\lambda i} C_{\sigma i}^{*}$. (For a non-orthogonal basis S, use $\mathbf{FC} = \mathbf{SC}\boldsymbol{\varepsilon}$.)

Self-consistency (SCF). HF is solved iteratively: initialize $P^{(0)}$, build $\mathbf{F}[P^{(n)}]$, diagonalize to get $\mathbf{C}^{(n+1)}$, update $P^{(n+1)}$, and iterate until the total energy and/or P change is below tolerance. Damping or DIIS mixing stabilizes convergence. The dominant costs are building two-electron contributions ($\mathcal{O}(M^4)$) and diagonalization ($\mathcal{O}(M^3)$).

Remarks for quantum dots. For parabolic confinement $V_{\mathrm{ext}} = \frac{1}{2}\omega^2 r^2$ we typically choose Fock–Darwin (2D HO) spin–orbitals as the one-particle basis, which respect angular-momentum structure and accelerate convergence. Closed shells admit restricted HF (RHF); spin-polarized cases may require UHF. Koopmans' theorem (orbital energies as removal/addition estimates) can provide a rough diagnostic but is not used for final energetics here.

1.6 Full Configuration Interaction (FCI)

FCI gives the exact ground state *within the chosen one-particle basis* by expanding in *all* N-electron Slater determinants,

$$|\Psi\rangle = \sum_I C_I\, |\Phi_I\rangle\,, \qquad H\mathbf{C} = E\,\mathbf{C}.$$

The determinant space has dimension $\binom{M}{N}$ for fixed spin–orbitals M and particles N (or smaller with symmetry constraints). FCI thus captures the full correlation energy $\Delta E = E_{\mathrm{FCI}} - E_{\mathrm{HF}}$ in that basis and serves as the gold-standard benchmark.

1.6.1 Slater–Condon rules and sparsity

Matrix elements $\langle \Phi_I | H | \Phi_J \rangle$ are nonzero only if Φ_I and Φ_J differ by at most two spin–orbitals:

- *Diagonal:* $H_{II} = \sum_{i \in I} h_{ii} + \frac{1}{2} \sum_{i,j \in I} (ij \| ij)$.
- *Singles $i \to a$:* $H_{0,i^a} = F_{ia}$; in canonical HF orbitals, $F_{ia} = 0$ (Brillouin's theorem), so the HF determinant does not couple to singles.
- *Doubles $ij \to ab$:* $H_{0,ij^{ab}} = (ij \| ab)$.

This locality yields a very sparse Hamiltonian in the determinant basis and enables iterative eigensolvers without forming $\mathbf{H}$ explicitly.

1.6.2 Diagonalization and symmetry reduction

In practice one targets the lowest eigenpairs with Davidson or Lanczos iterations. Symmetries (N, S_z, total spin S if spin-adapted CSFs are used, and L_z for Fock–Darwin bases) block-diagonalize the problem and dramatically shrink the working space and memory. For few-electron quantum dots (e.g. $N = 2, 6$) this makes high-accuracy FCI feasible for moderate M; for larger N the determinant growth becomes the bottleneck.

1.6.3 Role in this thesis

HF provides physically meaningful orbitals and an upper bound; FCI (in the same one-particle basis) provides benchmark energies for small systems. We use these as references to assess the accuracy of our variational Slater–Jastrow/PINN results for 2D quantum dots. For larger N (where FCI is prohibitive) we rely on variational energies plus uncertainty estimates.

1.7 Quantum Dots

1.7.1 Model and units

We consider N spin-$\frac{1}{2}$ fermions in d=2 with harmonic confinement and Coulomb repulsion,

$$H \;=\; \sum_{i=1}^{N} \Big(-\tfrac{1}{2} \nabla^2_{\mathbf{r}_i} + \tfrac{1}{2} \omega^2 r_i^2 \Big) \;+\; \sum_{i<j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}, \qquad (\hbar = m_e = e = 1).$$

Spin does not appear in H (spin–independent), so the many-body eigenstates can be chosen as products of a spatial wavefunction and a spin function with well-defined S and S_z.

1.7.2 One–particle structure and shell filling

We use Fock–Darwin (2D harmonic–oscillator) spin–orbitals $\{|nm\sigma\rangle\}$, with spatial energies $E_{nm} = \omega(2n + |m| + 1)$ and spin $\sigma \in \{\uparrow, \downarrow\}$. For each *principal* index $k = 2n + |m|$ there are $(k{+}1)$ spatial states, each twofold spin-degenerate. Filling from the bottom yields the *closed-shell* electron numbers

$$N_{\text{closed}} = (K{+}1)(K{+}2), \quad K = 0, 1, 2, \ldots \quad (2, 6, 12, 20, 30, \ldots).$$

All other N correspond to a *partially filled* top shell (open shell), with a degenerate noninteracting manifold.

1.7.3 Spin structure and Slater form

Write single-particle spin-orbitals as $\varepsilon_\alpha(x) = \phi_\alpha(\mathbf{r})\, \chi_\sigma(s)$, with $x = (\mathbf{r}, s)$. Because $\chi_\uparrow$ and $\chi_\downarrow$ are orthogonal and H is spin-independent, the Slater determinant simplifies:

Closed shells (RHF). For even N with doubly occupied spatial orbitals $\{\phi_1, \ldots, \phi_{N/2}\}$, the determinant factorizes into two identical spatial determinants, one for each spin:

$$\Psi_{\text{closed}}(X) = \frac{1}{(N/2)!} \det[\phi_a(\mathbf{r}_i)]_{i\in\uparrow, a=1..N/2} \det[\phi_a(\mathbf{r}_j)]_{j\in\downarrow, a=1..N/2} \times \Xi_{S=0,S_z=0},$$

where Ξ is the singlet spin function. This is the *restricted* HF (RHF) reference and a natural starting point for correlated ansätze.

1.7.4 Kato's Cusp Conditions (2D Coulomb systems)

For Coulomb interactions the potential diverges as $1/r$ at coalescence. Finiteness of the local energy

$$E_L(R) = \frac{(H\Psi)(R)}{\Psi(R)}$$

requires that the kinetic singularity cancels the $1/r$ divergence. Kato's theorem, first showed in [17], gives the necessary short–distance behaviour of the (spherically averaged) wavefunction. Let r_{ij} be the interparticle distance, $\mu_{ij} = m_i m_j/(m_i + m_j)$ the reduced mass (in a.u.), Z_i the charge number (electron $Z = -1$, nucleus $Z = +Z_A$), and D the spatial dimension. Then

$$\frac{\partial}{\partial r_{ij}} \ln \overline{\Psi}\bigg|_{r_{ij}\to 0} = \frac{2\,\mu_{ij}\, Z_i Z_j}{D-1} \quad \text{(s-wave / opposite-spin channel).}$$

For *identical fermions* the s-wave is excluded; the leading channel is p-wave ($\ell = 1$), and the wavefunction carries a linear node at coalescence. The general like-spin cusp is then

$$\frac{\partial}{\partial r_{ij}} \ln \overline{\Psi}\bigg|_{r_{ij}\to 0} = \frac{2\,\mu_{ij}\, Z_i Z_j}{2\ell + D - 1} = \frac{2\,\mu_{ij}\, Z_i Z_j}{D+1} \quad \text{(like-spin, } \ell = 1\text{).}$$

Specializing to 2D electrons (atomic units). With $m_e = 1$ and $\mu_{ee} = 1/2$:

$$\boxed{a_{ee}^{\uparrow\downarrow} = \frac{\partial}{\partial r_{ij}} \ln \overline{\Psi}\bigg|_0 = 1\ , \qquad a_{ee}^{\uparrow\uparrow} = \tfrac{1}{3}\ .}$$

The like-spin value follows from the p-wave (linear-node) analysis carried out in full in Appendix B.2, the canonical derivation used throughout this thesis; it gives the familiar $\frac{1}{4}$ in three dimensions and $\frac{1}{3}$ here in two. The 2D like-spin cusp is therefore *not* simply half the opposite-spin value—a coincidence that holds only in 3D—and the implemented value is $1/(D+1) = \frac{1}{3}$. For electron–nucleus coalescence ($\mu_{eA} \approx 1$, $Z_i Z_j = -Z_A$):

$$\boxed{a_{eA} = \frac{\partial}{\partial r_{iA}} \ln \overline{\Psi}\bigg|_0 = -2Z_A\ .}$$

Why this matters. These conditions ensure the kinetic energy's r^{-1} singularity cancels the Coulomb r^{-1} term so that E_L remains finite as particles coalesce. Enforcing the cusp explicitly improves conditioning.

1.8 Structural diagnostics and Wigner markers

This section introduces the structural diagnostics conceptually—what each marker means and why it detects Wigner order; the precise computational procedures used in this thesis (estimators, thresholds, and stability protocol) are collected in the Analysis chapter (Chapter 5).

Unless stated otherwise, curves are accumulated in *trap units* $\tilde{\mathbf{r}} = \sqrt{\omega}\,\mathbf{r}$ so that length scales align across ω, while headline distances are reported in *Bohr* (physical units $\mathbf{r}$). Production chains are thinned; we split samples into halves for stability checks (acceptance window, JSD between halves, reproducibility under restarts).

1.8.1 Pair distribution $g(r)$, $P(r)$, and width markers

We estimate the (spin-summed) pair distribution $g(r)$ via a shell-area corrected histogram in trap units,

$$g(r_\alpha) \;=\; \frac{A_{\rm eff}}{N(N-1)}\,\frac{\big\langle \sum_{i<j}\mathbf{1}\{r_{ij}\in[r_\alpha^-,r_\alpha^+)\}\big\rangle}{2\pi r_\alpha\,\Delta r_\alpha},\qquad r_{ij}=\|\tilde{\mathbf{r}}_i-\tilde{\mathbf{r}}_j\|_2, \tag{1.1}$$

with $A_{\rm eff}=\pi R_{0.95}^2$ and $R_{0.95}$ the 95% quantile of single-particle radii (trap units). We form the normalised radial probability in d=2, $P(r)\propto r\,g(r)$, and report the mode $r_{\rm mode}$, mean $\bar r$, standard deviation σ_r, FWHM, and quantiles q_{10},q_{50},q_{90} (in Bohr). A compact width–to–scale marker is

$$\gamma \;=\; \sigma_r/r_{\rm mode}, \tag{1.2}$$

which decreases as correlations stiffen (narrow distribution around a larger typical radius).[1]

Sanity anchors at $\boldsymbol{\omega=10^{-3}}$. For N=6 and N=12, $P(r)$ exhibits sharp peaks with small γ; these are the expected Wigner-side shapes. For N=2 we additionally report the near-origin mass $\int_0^{0.25\,r_{\rm mode}}\rho_1(r)\,dr$ and $\Pr(r_{12}\le 0.25\,\mathrm{mode}(r_{12}))$, both $\ll 1$ as required by the cusp and strong repulsion.

1.8.2 Automatic shell detection and topology labeling

For each frame we sort single-particle radii $\tilde s_1\le\cdots\le\tilde s_N$ (trap units), compute gaps $g_k=\tilde s_{k+1}-\tilde s_k$, and flag a frame as *two-shell* if

$$\max_k g_k \;\ge\; \tau\,\mathrm{median}(g_k),\qquad \tau\in[2.0,3.0]\ \text{(robust)}. \tag{1.3}$$

The cut radius is the midpoint of the maximal gap; the inner multiplicity $n_{\rm in}$ defines the topology $(n_{\rm in},N-n_{\rm in})$. We report: (i) the fraction of two-shell frames, (ii) the histogram over $n_{\rm in}$ among two-shell frames, and (iii) for selected topologies (e.g. $(1,5)$ or $(3,9)$) their fraction among *all* frames.

II/IO/OO decomposition (what is being added). On two-shell frames we decompose pairs into inner–inner (II), inner–outer (IO), and outer–outer (OO), accumulate $g_{\rm II},g_{\rm IO},g_{\rm OO}$, and reconstruct

$$\tilde g(r)=w_{\rm II}g_{\rm II}+w_{\rm IO}g_{\rm IO}+w_{\rm OO}g_{\rm OO},$$

with empirical weights from pair counts. We report $\cos\angle(g,\tilde g)$ (cosine similarity); values ≈ 1 indicate that the observed $g(r)$ is explained by shell geometry rather than sampling artifacts.

Sanity anchors at $\boldsymbol{\omega=10^{-3}}$. For N=6, $w_{\rm OO}:w_{\rm IO}\approx 2:1$ and $\cos\angle(g,\tilde g)=1.000$ within numerical precision. For N=12, the two-shell fraction is 0.906–0.990 across $\tau\in\{3.0,2.5,2.0\}$ and $\cos\angle(g,\tilde g)\in[0.989,1.000]$ depending mildly on τ/seed. The *modal* inner multiplicity at $\omega=10^{-3}$ is $(3,9)$, while $(2,10)$ and $(1,11)$ retain substantial weight—consistent with a rotating two-shell Wigner molecule. For classical references to shellings and near-degeneracies in parabolic/cluster confinements, see Schweigert–Peeters and Kong–Partoens–Peeters.[3, 4]

1.8.3 Orientational order and angular "Lindemann"

After recentring each detected ring (inner/outer) to remove global rotations, with angles $\{\phi_k\}_{k=1}^n$, we use the bond–orientational order

$$\Phi_m \;=\; \frac{1}{n}\sum_{k=1}^{n} e^{\,\mathrm{i}m\phi_k},\qquad m=n, \tag{1.4}$$

so that $|\Phi_m|\in[0,1]$ measures n-fold orientational order (e.g. m=5 for $(1,5)$; m=9 for the outer ring of $(3,9)$). As a complementary, dimensionless *angular Lindemann* number we use

$$\lambda_\phi \;=\; \frac{\mathrm{std}(\Delta\phi)}{\mathbb{E}[\Delta\phi]},\qquad \Delta\phi \;=\; \text{nearest-neighbour angular spacings on the ring}. \tag{1.5}$$

Smaller λ_ϕ indicates a stiffer (more Wigner-like) ring. This follows the spirit of bond–orientational order used in 2D Coulomb/Yukawa systems,[2, 5] with the Lindemann idea adapted from melting criteria to angular spacings on a fixed-n ring.

[1] For N=2, γ is complemented by the pair–distance Lindemann $\gamma_{r_{12}}=\mathrm{std}(r_{12})/\mathrm{mode}(r_{12})$, cf. Sec. 5.1.4.

Sanity anchors at $\boldsymbol{\omega = 10^{-3}}$. For N=6, phase-aligned $(1, 5)$ and $(0, 6)$ sectors show comparable order ($|\Phi_5| \approx 0.48$, $|\Phi_6| \approx 0.52$) with $\lambda_\phi \approx 0.29$—a single-ring rotating Wigner molecule. For N=12, $(3, 9)$ frames have strong order ($|\Phi_3| \approx 0.67$, $|\Phi_9| \approx 0.50$) and small λ_ϕ (~ 0.24–0.26). $(2, 10)$ exhibits very tight inner order ($|\Phi_2| \approx 0.73$) with a softer outer ring ($|\Phi_{10}| \approx 0.42$), and $(1, 11)$ has weak outer order and a softly vibrating center—precisely the *rotating Wigner molecule* (RWM) picture of fluctuating orientations across near-degenerate shellings.[1, 6]

1.8.4 Scaling check and two-body benchmarks ($N = 2$)

To connect with the classical strong-coupling limit we fit

$$r_{\text{mode}}(\omega) \; = \; C\,\omega^{\alpha} \tag{1.6}$$

by linear regression in $\big(\log\omega, \log r_{\text{mode}}\big)$ and compare with the two-body classical scaling $\alpha_{\text{cl}} = -2/3$. At $\omega = 10^{-3}$ the r_{12} histogram is sharply peaked and the near-origin mass in ρ_1 and $p(r_{12})$ is $\ll 1$, validating the cusp implementation while confirming deep Wigner localization (see also Ref. [1] for the crossover criterion in dots).

1.8.5 Reporting and stability checks

For all scalars and histograms we report standard errors from block averages over thinned frames. We verify: (i) acceptance in the target window, (ii) Jensen–Shannon divergence and ℓ_2 distance between split halves (stationarity), (iii) robustness of topology fractions and $|\Phi_m|$ under $\tau \in [2.0, 3.0]$ and mild pre-smoothing of radii, and (iv) reproducibility of $g(r)$, $P(r)$, and bond-order metrics under independent sampler restarts. All diagnostics operate directly on saved bundles $\{\mathbf{X}_{\text{Bohr}},\, r,\, g(r)\}$.

Interpretation guide (link to Results). Across $\omega \in \{1.0, 0.5, 0.1, 0.01, 10^{-3}\}$ these diagnostics map the fluid→Wigner crossover: γ and λ_ϕ shrink, shell gaps sharpen, and shell-resolved $|\Phi_m|$ becomes nonzero after phase alignment. For N=6 this manifests as a single-ring RWM with $(1, 5)$–$(0, 6)$ near-degeneracy; for N=12 as a two-shell RWM with modal $(3, 9)$ but sizable $(2, 10)$ and $(1, 11)$ weights. Classical shelling and near-degeneracies in finite Coulomb clusters[3, 4] and quantum Wigner markers in parabolic dots[1, 2, 5, 6] provide the theoretical backdrop for reading these plots.

1.9 Variational Methods

Quantum Monte Carlo (QMC) methods offer powerful tools for solving the many-body Schrödinger equation through stochastic sampling. These methods yield accurate ground-state properties by estimating expectation values of observables with controlled approximations. Among the various QMC techniques, Variational Monte Carlo (VMC) and Diffusion Monte Carlo (DMC) are particularly notable; VMC optimizes a trial wavefunction based on the variational principle, while DMC projects out the ground state through imaginary-time evolution [18, 19].

1.9.1 Variational Monte Carlo (VMC)

The Variational Principle

The variational principle guarantees that for any normalised trial wavefunction $\Psi_T(\mathbf{R}; \boldsymbol{\alpha})$, the expectation value

$$E[\Psi_T] = \frac{\langle \Psi_T | \hat{H} | \Psi_T \rangle}{\langle \Psi_T | \Psi_T \rangle}$$

provides an upper bound to the true ground-state energy E_0. Here, $\mathbf{R} = (\mathbf{r}_1, \ldots, \mathbf{r}_N)$ and $\boldsymbol{\alpha}$ represents the set of variational parameters. Optimising $\boldsymbol{\alpha}$ minimizes $E[\Psi_T]$.

Sampling

In VMC, the probability density is defined as

$$P(\mathbf{R}) = \frac{|\Psi_T(\mathbf{R})|^2}{\int |\Psi_T(\mathbf{R})|^2 \, d\mathbf{R}},$$

and configurations are sampled using the Metropolis-Hastings algorithm, initially described in [20], with acceptance probability

$$A(\mathbf{R} \to \mathbf{R}') = \min\left(1, \frac{|\Psi_T(\mathbf{R}')|^2}{|\Psi_T(\mathbf{R})|^2} \frac{T(\mathbf{R}' \to \mathbf{R})}{T(\mathbf{R} \to \mathbf{R}')}\right),$$

where T is the proposal density (for a symmetric proposal the ratio is unity and the acceptance reduces to the Metropolis form). When a drift is added, a *quantum force*

$$\mathbf{F}(\mathbf{R}) = 2\,\nabla \ln |\Psi_T(\mathbf{R})|,$$

further guides the sampling toward regions of high probability. This leads to what is known as importance sampling. A more precise discussion of this is found in [21, 22].

Local Energy and Parameter Optimisation

The local energy is defined as

$$E_L(\mathbf{R}) = \frac{\hat{H}\Psi_T(\mathbf{R})}{\Psi_T(\mathbf{R})}.$$

If Ψ_T were exact, $E_L(\mathbf{R})$ would be constant. In practice, the variational energy is approximated by averaging E_L over many configurations:

$$E[\Psi_T] \approx \frac{1}{N_{\mathrm{MC}}} \sum_{i=1}^{N_{\mathrm{MC}}} E_L(\mathbf{R}_i).$$

Variational parameters are then optimised (using gradient descent or stochastic reconfiguration) to reduce both the energy and its variance.

1.9.2 Diffusion Monte Carlo (DMC)

Imaginary-Time Evolution and Projection

DMC refines the trial wavefunction by projecting out the ground state through imaginary-time evolution. The Schrödinger equation becomes

$$-\frac{\partial \Psi(\mathbf{R}, \tau)}{\partial \tau} = \left(\hat{H} - E_T\right)\Psi(\mathbf{R}, \tau),$$

where $\tau = it$ and E_T is a trial energy. As $\tau \to \infty$, $\Psi(\mathbf{R}, \tau)$ converges to the ground-state wavefunction provided the initial state has nonzero overlap with it.

Stochastic Implementation

DMC simulates the imaginary-time evolution using a combination of diffusion, drift, and branching:

1. **Diffusion and Drift:** Particles execute random walks, with drift guided by the quantum force $\mathbf{F}(\mathbf{R})$, steering them toward regions where Ψ_T is large.
2. **Branching/Death:** Walkers are assigned a weight,

   $$w(\mathbf{R}, \Delta\tau) = \exp\left[-\left(E_L(\mathbf{R}) - E_T\right)\Delta\tau\right],$$

   which determines whether they are replicated or removed, ensuring that the walker distribution evolves toward the ground-state probability density.

1.9.3 Applications and Numerical Considerations

Model Systems

QMC methods have been successfully applied to various quantum systems:

- **Hydrogen Atom:** A simple trial wavefunction $\Psi_T(r) = e^{-\alpha r}$ yields the exact ground state for $\alpha = 1$, illustrating zero variance.
- **Harmonic Traps:** In systems confined by harmonic potentials, Jastrow factors like $\exp[-ar_{ij}/(1 + \beta r_{ij})]$ model inter-particle repulsion while Slater determinants maintain the proper fermionic symmetry.

Computational Efficiency

Key factors in achieving efficient QMC simulations include:

- **Efficient Determinant Updates:** Algorithms for rapid updates of Slater determinants and their inverses can reduce computational cost from $\mathcal{O}(N^3)$.
- **Optimised Gradient and Laplacian Calculations:** Recursive methods and efficient numerical routines for evaluating $\nabla\Psi_T/\Psi_T$ and $\nabla^2\Psi_T/\Psi_T$ are essential.

1.9.4 Conclusion

Variational and Diffusion Monte Carlo methods together provide a robust framework for studying quantum systems. VMC offers a flexible approach for optimising trial wavefunctions, while DMC projects out the exact ground state through imaginary-time evolution. These methods, working in concert, enable accurate and efficient investigations of systems from simple atoms to strongly correlated materials. Ongoing improvements in trial wavefunction design and stochastic algorithms continue to extend the reach of QMC in modern computational physics.

Chapter 2

Machine Learning

2.1 Machine Learning as Statistical Learning

Machine learning (ML) is best understood as the study of algorithms that approximate functions from data. At its core, it is not about "intelligent machines," but about developing systematic ways to extract patterns from observations and to generalise these patterns to unseen situations. From a mathematical perspective, ML is a branch of *statistical learning theory*, which views the learning process as the minimization of an expected error (or *risk*) under uncertainty.

Formally, consider an input space $\mathcal{X} \subseteq \mathbb{R}^d$ and an output space $\mathcal{Y}$, together with an unknown joint distribution $P(X, Y)$ over pairs (x, y). The aim of learning is to construct a function $f_\theta : \mathcal{X} \to \mathcal{Y}$, parameterized by $\theta \in \Theta$, such that $f_\theta(x)$ predicts y with small error. A loss function

$$\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_{\geq 0}$$

quantifies the discrepancy between prediction $f_\theta(x)$ and truth y. The central object of learning is then the **population risk**

$$\mathcal{R}(\theta) = \mathbb{E}_{(x,y)\sim P}\big[\ell(f_\theta(x), y)\big]. \tag{2.1}$$

Since the distribution P is unknown, we only have access to a finite dataset $D = \{(x_i, y_i)\}_{i=1}^n$. The corresponding **empirical risk** is

$$\hat{\mathcal{R}}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell(f_\theta(x_i), y_i). \tag{2.2}$$

This setup is known as *empirical risk minimization (ERM)*, and it underlies virtually all modern machine learning methods [23].

2.1.1 Types of learning

Depending on the information provided in the dataset, learning problems are traditionally categorized as:

- **Supervised learning.** We are given labeled pairs (x_i, y_i), and the goal is to approximate the conditional expectation $\mathbb{E}[Y|X = x]$. Regression (continuous Y) and classification (discrete Y) are canonical examples.

- **Unsupervised learning.** Only input data $\{x_i\}$ is observed, without labels. The aim is to uncover structure, such as clustering, dimensionality reduction, or latent-variable representations.

- **Reinforcement learning.** An agent interacts with an environment, receiving observations and rewards. The task is to learn a policy that maximizes expected cumulative reward.

Although reinforcement learning plays a minor role in this thesis, supervised and unsupervised perspectives provide the theoretical foundation for the physics-informed approaches discussed later.

2.1.2 Generalisation and the bias–variance tradeoff

The essential challenge in ML is not just minimizing the empirical risk $\hat{\mathcal{R}}(\theta)$, but ensuring that this leads to a low population risk $\mathcal{R}(\theta)$ on unseen data. A model that simply memorizes the training set achieves low empirical risk but poor generalisation — a phenomenon known as *overfitting*.

Statistical learning theory formalizes this challenge via the *bias–variance tradeoff*. Given a data-generating process and a learning algorithm, the expected squared error can be decomposed into

$$\mathbb{E}\big[(f_\theta(x)-y)^2\big] = \underbrace{(\mathbb{E}[f_\theta(x)] - f^*(x))^2}_{\text{bias}^2} + \underbrace{\mathbb{E}[(f_\theta(x) - \mathbb{E}[f_\theta(x)])^2]}_{\text{variance}} + \underbrace{\sigma^2}_{\text{irreducible noise}}, \tag{2.3}$$

where $f^*(x)$ is the true regression function and σ^2 the intrinsic noise of the data. Complex models tend to have low bias but high variance, while simple models have the opposite. Successful learning algorithms balance these two contributions. This is discussed in more detail in [24].

2.2 Optimisation

The minimization of empirical risk,

$$\hat{\mathcal{R}}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell(f_\theta(x_i), y_i),$$

is an *optimisation problem*. The hypothesis class $\{f_\theta\}$ is usually nonlinear and high-dimensional, so closed-form solutions are not available. Instead, optimisation is performed iteratively by updating parameters $\theta \in \mathbb{R}^p$ according to the geometry of the loss landscape.

2.2.1 Gradient descent

The most basic algorithm is *gradient descent*. Starting from an initial guess $\theta^{(0)}$, parameters are updated by

$$\theta^{(t+1)} = \theta^{(t)} - \eta\, \nabla_\theta \hat{\mathcal{R}}(\theta^{(t)}), \tag{2.4}$$

where $\eta > 0$ is the *learning rate*. Intuitively, the gradient points in the direction of steepest increase of the loss, so subtracting it decreases the loss most rapidly in a local sense.

The choice of learning rate is crucial: if too small, convergence is slow; if too large, the algorithm may diverge.

2.2.2 Stochastic gradient descent (SGD)

For large datasets, evaluating the full gradient at every step is prohibitively expensive. Instead, one uses *stochastic gradient descent (SGD)*, which approximates the gradient on a randomly sampled mini-batch $B \subset D$:

$$\nabla_\theta \hat{\mathcal{R}}_B(\theta) = \frac{1}{|B|} \sum_{(x_i,y_i)\in B} \nabla_\theta \ell(f_\theta(x_i), y_i). \tag{2.5}$$

The update rule is then

$$\theta^{(t+1)} = \theta^{(t)} - \eta\, \nabla_\theta \hat{\mathcal{R}}_B(\theta^{(t)}). \tag{2.6}$$

SGD introduces randomness into the optimisation trajectory. Paradoxically, this noise often *helps*: it prevents the algorithm from getting trapped in poor local minima and can improve generalisation through an implicit regularization effect [23, 25].

2.2.3 Momentum and adaptive methods

Plain gradient descent treats all directions in parameter space equally. To accelerate convergence, one may use *momentum*, maintaining a velocity vector v_t:

$$v_{t+1} = \beta v_t + (1-\beta)\, \nabla_\theta \hat{\mathcal{R}}_B(\theta^{(t)}), \tag{2.7}$$

$$\theta^{(t+1)} = \theta^{(t)} - \eta v_{t+1}, \tag{2.8}$$

where $\beta \in (0,1)$ controls the contribution of past gradients. Momentum smooths oscillations and allows the optimisation to traverse valleys in the loss landscape more effectively.

Beyond momentum, *adaptive methods* such as RMSProp and Adam rescale learning rates per parameter based on historical gradient magnitudes [26]. For instance, Adam combines momentum with an adaptive normalisation:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \tag{2.9}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2, \tag{2.10}$$

$$\theta^{(t+1)} = \theta^{(t)} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}, \tag{2.11}$$

where $g_t = \nabla_\theta \hat{\mathcal{R}}_B(\theta^{(t)})$ and $\hat{m}_t, \hat{v}_t$ are bias-corrected versions of m_t, v_t. Adam is widely used in practice for its robustness across diverse problems [27].

2.2.4 Second-order methods and natural gradient

First-order methods only use gradient information. In principle, convergence can be accelerated by incorporating curvature via the Hessian $H(\theta) = \nabla_\theta^2 \hat{\mathcal{R}}(\theta)$. Newton's method updates parameters as

$$\theta^{(t+1)} = \theta^{(t)} - H(\theta^{(t)})^{-1} \, \nabla_\theta \hat{\mathcal{R}}(\theta^{(t)}). \tag{2.12}$$

However, computing and inverting the Hessian is usually infeasible for large models.

An attractive alternative is the *natural gradient* [28, 29], which replaces the Euclidean metric with the geometry of probability distributions. The key object is the *Fisher information matrix* (FIM),

$$F(\theta) = \mathbb{E}\big[\nabla_\theta \log p_\theta(x) \, \nabla_\theta \log p_\theta(x)^\top\big], \tag{2.13}$$

where p_θ denotes the model distribution. The natural gradient update is

$$\theta^{(t+1)} = \theta^{(t)} - \eta \, F(\theta^{(t)})^{-1} \, \nabla_\theta \hat{\mathcal{R}}(\theta^{(t)}). \tag{2.14}$$

This preconditioning makes updates invariant to reparameterizations of θ and often improves conditioning of the optimisation. In practice, computing F^{-1} exactly is intractable, but approximations (e.g. stochastic reconfiguration in variational Monte Carlo, Kronecker-factored approximations in deep learning) make the method viable [9].

2.2.5 The variational tangent space: geometric tensor, tangent kernel, and effective dimension

For a variational quantum state the Fisher metric takes a particularly concrete form, and that form is the analytical lens used throughout the results of this thesis. Write the per-sample log-derivatives—the *score*—of an ansatz Ψ_θ as

$$O_{ki} = \frac{\partial \log |\Psi_\theta(\mathbf{x}_k)|}{\partial \theta_i}, \qquad k = 1, \ldots, B, \;\; i = 1, \ldots, P, \tag{2.15}$$

for a batch of B configurations drawn from $|\Psi_\theta|^2$ and P parameters θ (here θ collects *all* variational parameters). Its centred Gram matrices define two dual objects. The *quantum geometric tensor* (QGT),

$$S = \tfrac{1}{B} \, O^\top O \in \mathbb{R}^{P \times P}, \tag{2.16}$$

is the Fisher–Rao metric of the state written in terms of the amplitude score. It is proportional to the Fisher matrix $F(\theta)$ of the previous subsection specialised to $p_\theta = |\Psi_\theta|^2$: because $\nabla_\theta \log |\Psi_\theta|^2 = 2 \, \nabla_\theta \log |\Psi_\theta|$, one has $F = 4S$, the factor of four being a convention absorbed into the learning rate. Stochastic reconfiguration is natural-gradient descent under this metric, $\dot{\theta} = -S^{-1} \nabla_\theta E$ [30, 31]. Its Gram dual, the *neural tangent kernel* (NTK) [32],

$$K = OO^\top \in \mathbb{R}^{B \times B}, \tag{2.17}$$

shares the nonzero spectrum $\{\lambda_a\}$ of S and is cheaper to diagonalise when $B < P$.

Two scalars read from that spectrum organise the analysis. The *condition number* $\kappa(S) = \lambda_{\max}/\lambda_{\min}$ measures how anisotropic the tangent metric is, and therefore how far a natural-gradient step departs from a Euclidean one—the quantity that decides when SR is worth its cost. The *effective dimension*

$$d_{\text{eff}}(S) = \frac{\left(\sum_a \lambda_a\right)^2}{\sum_a \lambda_a^2} \tag{2.18}$$

is the participation ratio of the spectrum: it counts how many independent tangent directions carry appreciable Fisher weight at the solution, regardless of the raw parameter count P. A small d_{eff} means the state is reached along a few collective directions; a large one means the same solution is spread across many. Because these are properties of the tangent geometry rather than of the raw parameter count, and are stable across seeds and sampling measures for a given ansatz, they let us compare architectures, optimisers, and training paradigms on a common footing— which is exactly what the second results chapter (Chapter 7) does.

One caveat must be kept in view. The participation ratio d_{eff} is computed from the QGT eigenvalues in the ordinary Euclidean parameter coordinates, and those eigenvalues are *not* invariant under an arbitrary reparameterization: rescaling a single parameter $\theta_1 \mapsto c\,\theta_1$ leaves the physical state untouched but changes d_{eff} (for a diagonal metric it moves from 2 toward 1 as $c \to \infty$). d_{eff} is therefore *not* a coordinate-free intrinsic dimension of the wavefunction; it is a property of the (architecture, parameterization, normalization) triple. We use it accordingly—as a comparison between fixed architectures under matched conventions and a common probe measure—and read a smaller d_{eff} as "this architecture concentrates its Fisher weight into fewer active directions," not as an absolute manifold dimension.

2.2.6 Optimisation challenges in deep learning

Optimisation in deep neural networks poses unique challenges:

- **Non-convexity.** Loss landscapes are highly non-convex, with many local minima and saddle points. Empirically, most local minima reached by SGD perform similarly well.
- **Vanishing and exploding gradients.** In deep networks, gradients can shrink or blow up during backpropagation. Proper initialization and activation choice are essential to mitigate this problem.
- **Flat vs. sharp minima.** Empirical evidence suggests that flat minima (regions where the loss remains low under perturbations of θ) lead to better generalisation. SGD's noise implicitly biases optimisation toward flatter minima.

Optimisation thus provides not only the means of fitting a model, but also shapes the inductive biases and generalisation behaviour of neural networks. In the following sections, we build on this foundation by examining neural network expressivity, activation functions, and initialization strategies in detail.

2.3 Feed-forward neural networks

Feed-forward neural networks are the basic building block from which the architectures of this thesis are assembled. They compose affine maps with elementwise nonlinearities to represent functions that linear models cannot. We first define the feed-forward map, and then discuss the two design choices that most affect its behaviour in physics-informed settings, where gradients and Laplacians of the network enter the loss: the activation function and the weight initialization.

2.3.1 From linear models to nonlinear function classes

Classical regression models are linear in their parameters:

$$f_\theta(x) = \theta^\top x.$$

While simple and interpretable, linear models are limited to representing affine functions. Introducing basis expansions (e.g. polynomials, Fourier series, kernels) enlarges the hypothesis space, but requires manual design.

Neural networks automate this process. A feed-forward neural network with L layers is a composition of affine maps and nonlinear activation functions:

$$h^{(0)} = x, \tag{2.19}$$

$$h^{(\ell)} = \sigma\Big(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)}\Big), \quad \ell = 1, \dots, L-1, \tag{2.20}$$

$$f_\theta(x) = W^{(L)} h^{(L-1)} + b^{(L)}. \tag{2.21}$$

Here $\sigma : \mathbb{R} \to \mathbb{R}$ is a nonlinear activation. Without nonlinearity, the network collapses to a single linear transformation; with it, the space of representable functions expands dramatically.

2.3.2 Activation functions

Classical Activation Functions

The earliest activation functions were smooth, saturating nonlinearities:

$$\sigma_{\text{sigmoid}}(x) = \frac{1}{1+e^{-x}}, \tag{2.22}$$

$$\sigma_{\tanh}(x) = \tanh(x). \tag{2.23}$$

Both map $\mathbb{R}$ to a bounded range and are differentiable. The hyperbolic tangent is zero-centered, which often improves optimisation.

Despite their smoothness, these functions suffer from the *vanishing gradient problem*. For large $|x|$, their derivatives approach zero, causing gradients to vanish during backpropagation in deep networks. As a result, learning becomes prohibitively slow.

To address vanishing gradients, the *Rectified Linear Unit* (ReLU) was introduced:

$$\sigma_{\text{ReLU}}(x) = \max(0, x). \tag{2.24}$$

ReLU is unbounded above and has derivative 1 for positive inputs, avoiding gradient saturation. This makes deep networks with ReLU trainable even with many layers.

However, ReLU introduces its own drawbacks:

- Non-smoothness at $x = 0$, which can lead to optimisation instability in contexts where derivatives or Laplacians are central (e.g. physics-informed models).
- The *dying ReLU* problem, where neurons with negative inputs become permanently inactive.

Modern activations

To combine the benefits of ReLU with smoothness, a number of *smoothed rectifiers* have been proposed:

$$\sigma_{\text{SiLU}}(x) = x \cdot \sigma_{\text{sigmoid}}(x), \qquad \text{(Sigmoid Linear Unit)} \tag{2.25}$$

$$\sigma_{\text{Swish}}(x) = x \cdot \frac{1}{1+e^{-\beta x}}, \qquad \text{(generalised SiLU with } \beta > 0) \tag{2.26}$$

$$\sigma_{\text{GELU}}(x) = x \cdot \Phi(x), \qquad \text{(Gaussian Error Linear Unit)} \tag{2.27}$$

$$\sigma_{\text{Mish}}(x) = x \cdot \tanh(\log(1+e^{x})). \tag{2.28}$$

Here $\Phi(x)$ is the standard Gaussian cumulative distribution function.

These activations are smooth (C^∞) and non-saturating, preserving gradient flow while avoiding non-differentiable kinks. Empirically, GELU, SiLU/Swish, and Mish improve optimisation stability and generalisation across a wide range of tasks.

2.3.3 Initialization

The initialization of neural network weights has a critical impact on optimisation. At the start of training, parameter values determine how signals propagate forward and backward through the network. Poor initialization can lead to exploding or vanishing activations and gradients, making optimisation unstable or ineffective. Proper schemes balance signal magnitudes across layers and preserve gradient flow, especially in deep networks.

The role of initialization

Consider a feed-forward network with layers

$$h^{(\ell)} = \sigma\left(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)}\right),$$

where $W^{(\ell)}$ has entries drawn from some distribution. If weights are too large, activations grow exponentially with depth, leading to exploding gradients. If too small, activations shrink toward zero, causing vanishing gradients. Both phenomena impede learning.

Initialization therefore aims to preserve the variance of activations and gradients across layers, such that

$$\mathrm{Var}[h^{(\ell)}] \approx \mathrm{Var}[h^{(\ell-1)}].$$

Xavier/Glorot initialization

Glorot and Bengio (2010) proposed initializing weights with variance scaled by the number of input and output units:

$$W_{ij}^{(\ell)} \sim \mathcal{U}\left(-\sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}, \sqrt{\frac{6}{n_{\text{in}} + n_{\text{out}}}}\right). \tag{2.29}$$

Here n_{in} and n_{out} denote the fan-in and fan-out of a layer. This choice balances variance across layers for activations with approximately unit derivative (e.g. tanh).

Kaiming/He initialization

For rectified activations such as ReLU, He et al. (2015) proposed *Kaiming initialization*, scaling variance by $2/n_{\text{in}}$:

$$W_{ij}^{(\ell)} \sim \mathcal{N}\left(0, \frac{2}{n_{\text{in}}}\right). \tag{2.30}$$

This accounts for the fact that ReLU activations output zero half of the time, requiring larger variance to maintain signal propagation.

Gain and activation dependence

Both Glorot and Kaiming schemes can be generalised with a *gain* factor g that depends on the activation function:

$$\mathrm{Var}[W_{ij}^{(\ell)}] = \frac{g^2}{n_{\text{in}}}.$$

For example, $g = \sqrt{2}$ is recommended for ReLU, while $g = 1$ is typical for tanh and smooth activations. Choosing the correct gain ensures that variance neither explodes nor decays through the network.

Orthogonal and random normal initialization

An alternative approach is to initialize each weight matrix as an orthogonal matrix scaled by a variance factor. Orthogonal initialization preserves the norm of activations across layers, which can stabilize very deep networks.

Simpler schemes such as sampling from a standard normal distribution without scaling are generally inferior: they do not respect the scaling needs of different architectures or activation functions, and often result in unstable training.

2.3.4 Summary

Initialization is particularly delicate in deep networks. Even small imbalances in variance can compound exponentially with depth. Smooth activations (SiLU, GELU, Mish) alleviate gradient instability but require careful scaling, as their effective slope differs from ReLU or tanh.

Key takeaways include:

- Proper scaling of variance prevents vanishing/exploding signals.
- Initialization must match the activation function (Glorot for tanh/smooth, Kaiming for ReLU).

- Orthogonal schemes can improve stability in deep networks.
- For PINNs and scientific ML, initialization is not merely a convenience but essential for stable computation of derivatives.

Initialization and activation choice thus form an inseparable pair: together they determine whether information propagates stably through the network. This will be discussed further in the context of physics-informed models in Chapter 3.

2.4 Architectures for many-body wavefunctions

The feed-forward network of the previous section maps a fixed-size input vector to an output. A many-body wavefunction is not a generic function of such a vector: identical particles impose a permutation symmetry, and correlation is intrinsically *relational*—what one particle should do depends on where the others are. A plain multilayer perceptron acting on the concatenated coordinates $(\mathbf{x}_1, \ldots, \mathbf{x}_N)$ respects neither property; it would have to learn the symmetry from data, and its parameter count would grow with N. This section introduces the two architecture classes that build the required structure in—permutation-invariant pooling and message passing—and situates the specific message-passing network used in this thesis as one instance of the latter.

2.4.1 Permutation symmetry and equivariance

For identical particles the modulus $|\Psi|$ is invariant under exchange (the sign is carried by the antisymmetric Slater reference of Chapter 3). The two learnable objects of the ansatz therefore have distinct symmetry requirements. A scalar *correlator* f must be permutation-*invariant*,

$$f(P_\sigma R) = f(R),$$

while a per-particle *backflow* displacement field $\mathbf{g} = (\mathbf{g}_1, \ldots, \mathbf{g}_N)$ must be permutation-*equivariant*,

$$\mathbf{g}(P_\sigma R) = P_\sigma\, \mathbf{g}(R),$$

for any permutation σ (here P_σ relabels the particles). Enforcing these by construction is both more data-efficient than learning them and yields a parameter count independent of N, because the same shared maps are applied to every particle and pair.

2.4.2 Permutation-invariant pooling: DeepSets

The simplest way to build an invariant function is to embed each particle independently and pool the embeddings symmetrically. This is the DeepSets construction [33]: any suitably regular permutation-invariant function admits the form

$$f(R) = \rho\Big(\textstyle\sum_i \phi(\mathbf{x}_i)\Big),$$

with a per-particle embedding ϕ, a symmetric pool $\sum_i$, and a readout ρ. A two-body extension encodes pairs, $\phi(\mathbf{x}_i, \mathbf{x}_j)$, and pools over them, capturing pairwise correlation. What characterises this class is that each particle—or pair—is embedded *independently* and only then pooled: no information is exchanged between particles before the pool. We refer to such networks as *separable*, and they serve as the non-message-passing baseline for the correlator.

2.4.3 Message passing and graph networks

To let a particle's representation depend on the others' in an iterated, higher-order way, one places the particles on a graph—nodes are particles, edges are pairs—and passes learned messages along the edges [34, 35]. In a generic round ℓ, each edge gathers information from its endpoint nodes and each node aggregates its incoming edge messages,

$$\mathbf{m}_{ij}^{(\ell)} = \phi_e\big(\mathbf{h}_i^{(\ell)}, \mathbf{h}_j^{(\ell)}, \mathbf{e}_{ij}^{(\ell)}\big), \qquad \mathbf{h}_i^{(\ell+1)} = \phi_v\Big(\mathbf{h}_i^{(\ell)},\ \mathrm{Agg}_{j\neq i}\, \mathbf{m}_{ij}^{(\ell)}\Big),$$

with shared maps ϕ_e, ϕ_v and a symmetric aggregation Agg, so that permutation equivariance holds by construction; a final readout produces an invariant scalar (correlator) or an equivariant field (backflow). Both DeepSets and message passing use pairwise information, but only message passing lets that information *propagate*: after several rounds a particle's representation reflects not just its neighbours but their neighbours' responses. This iterated propagation is what we call the *relational channel*, and asking what it buys—in the correlator and in the backflow—is one of the central questions of this thesis. Message passing has been used in exactly this spirit for neural quantum states of extended fermion systems [12, 13].

2.4.4 The CTNN: one message-passing instance

The message-passing correlator and backflow studied here are realised by a particular network we refer to as the *CTNN*: a copresheaf-style message-passing network that maintains explicit *node and edge* features and transports information between them through learned maps, rather than recomputing and discarding one-shot messages. The correlator iterates this exchange over a down/up "V-cycle"; the backflow applies a single such copresheaf round. Its concrete update equations, hidden dimensions, and parameter counts are specified in the Methods chapter (§3.5.2, Eqs. (3.13)–(3.15)). Two points of framing matter for the rest of the thesis. First, the CTNN is *one* concrete realisation of the message-passing class defined above; the conclusions we draw about the value of message passing concern the class, with the CTNN as the instance actually studied. Wherever we contrast a "message-passing" network with a "separable" one, the separable member is a DeepSets-type network and the message-passing member is the CTNN. Second, we write "copresheaf-*style*" deliberately: the name reflects the node/edge transport structure of the updates, not a claim to a formal sheaf-theoretic construction.

2.4.5 Relevance to the ansatz

Both learnable objects of the ansatz—the Jastrow correlator and the coordinate backflow—can be built in either form: as a separable DeepSets-type network, or as the message-passing CTNN. Holding everything else fixed and varying only this choice isolates the effect of the relational channel, which Chapter 7 analyses through the geometry of the tangent kernel introduced in §2.2.5.

2.5 Expressivity, approximation, and generalization

The central reason neural networks have become the dominant model class in machine learning is their *expressivity*. However, in contrast with what classical statistical theory suggests, neural networks have been found to mitigate the curse of dimensionality, generalise well despite overparameterization, and optimise effectively in highly non-convex landscapes. Understanding these phenomena has been a major focus of recent theoretical research. In this section, we review key results on the approximation capabilities of neural networks, with an emphasis on implications for physical modelling.

2.5.1 Universal approximation theorems

Foundational results established that neural networks are *universal approximators*. It has been proved that a feed-forward network with at least one hidden layer and a suitable nonlinear activation (e.g. sigmoid, ReLU, tanh) can approximate any continuous function on a compact set to arbitrary accuracy [36, 37]. Formally:

Theorem 2.1 (Universal Approximation, informal)**.** *Let f^* be continuous on a compact $K \subset \mathbb{R}^d$. Then for any $\epsilon > 0$, there exists a neural network f_θ with a single hidden layer such that*

$$\sup_{x \in K} |f_\theta(x) - f^*(x)| < \epsilon.$$

This result guarantees existence but says nothing about how many neurons are required, how they should be trained, or how generalisation occurs.

2.5.2 Depth versus width

Cybenko's theorem establishes that a *single* hidden layer with a nonpolynomial sigmoidal activation is a *universal approximator*: for every continuous f on a compact domain and every $\varepsilon > 0$ there exists a one–hidden–layer network whose output g satisfies $\|f - g\|_\infty < \varepsilon$ [36]. The result concerns *existence* of an approximant and does not provide constructive bounds on the number of units required, rates of approximation, or generalisation guarantees.

In practice, depth is introduced for efficiency, not for universality. By composing intermediate features, additional layers often reduce the number of units required to attain a target accuracy. For certain target classes, there are known separations where a one-hidden-layer network needs exponentially many neurons to match the accuracy achievable with a deep architecture of polynomial size. Hence modern models leverage depth primarily for parameter efficiency and a stronger inductive bias (via hierarchical feature reuse), while width remains the lever for raw capacity [23].

2.5.3 Curse of dimensionality

Classical approximation theory predicts that uniformly approximating a generic $f : \mathbb{R}^d \to \mathbb{R}$ requires resources that grow exponentially in d—the *curse of dimensionality* (CoD). This perspective cannot explain the strong empirical performance of deep networks on very high–dimensional tasks, prompting a search for structural conditions under which the CoD does *not* apply.

A first resolution is *low effective dimension*: if the data (and loss) live on a $d' \ll d$ manifold M and error is measured in $L^p(\mu)$ with $\operatorname{supp}\mu \subset M$, then neural networks can learn local charts and partitions of unity and achieve rates that depend on d' rather than the ambient d. This replaces ambient complexity by the intrinsic geometry of M, yielding polynomial (rather than exponential) scaling.

A second route is *spectral/Barron regularity*: for targets with finite first Fourier moment (Barron class), two–layer networks attain a *dimension–independent* $O(n^{-1/2})$ L^2–approximation error with n hidden units. Deep and compositional Barron spaces extend this beyond smooth settings, providing broad families where the CoD provably disappears.

A third line exploits *PDE structure*. For many high–dimensional linear and semilinear parabolic/elliptic problems, stochastic representations (e.g., Feynman–Kac) and NN emulations of time–stepping yield approximation bounds whose complexity scales polynomially in d, again sidestepping the CoD under standard conditions.

Depth then clarifies *why* these escapes are efficient. Certain functions admit polynomial–size deep representations but require exponentially wide shallow ones (Eldan–Shamir separations), and deep ReLU networks generate exponentially more affine regions or oscillations per parameter than single–hidden–layer models. Thus, depth supplies the compositional mechanism that turns structure (manifolds, spectral regularity, PDE priors) into dimension–tolerant approximation.

2.5.4 Generalisation in overparameterized networks

Modern networks often have far more parameters than training samples, yet they fit the data *and* perform well on new inputs. This looks paradoxical if we judge models only by parameter count. A better lens is to ask which interpolating solution the training dynamics actually pick: test error is governed less by how large the hypothesis class is and more by the *kind* of function the algorithm converges to.

Two ideas make this concrete. First, stability: if small changes in the dataset produce only small changes in the learned predictor, the train–test gap remains small even at scale,

$$\left|\mathcal{R}(\hat{\theta}) - \hat{\mathcal{R}}(\hat{\theta})\right| \text{ is controlled.}$$

Second, implicit regularization: stochastic gradient methods (with finite batches, step sizes, and momentum) do not search the space uniformly; they are biased toward predictors with lower effective complexity—e.g., larger margins, smaller norms, or flatter basins in parameter space. In very wide nets, the dynamics are well-approximated by a linearized (kernel-like) model that interpolates while implicitly minimizing a function-space norm; at realistic widths, the same bias persists in practice. This also explains *double descent*: near the interpolation threshold test error can worsen, then improves again as capacity grows and the dynamics have more low-complexity interpolants to select from.

2.5.5 Optimisation in deep networks

Training is a large-scale, nonconvex problem, yet simple first-order methods reliably drive the loss to (near) zero. Two features help. Overparameterization and modern architectures (residual connections, normalisation) shape the landscape into broad, connected valleys of low loss and maintain usable descent directions through depth. As a result, poor isolated minima are less of a practical obstacle.

Stochastic gradient descent can be viewed as gradient flow with noise whose effective "temperature" is set mainly by the learning-rate–to–batch-size ratio. A higher temperature encourages escape from sharp minima and biases the search toward wider, flatter regions that tend to generalise better; annealing the temperature over time refines the final solution. Momentum accelerates along coherent directions and damps high-frequency noise; normalisation keeps activations well-scaled so larger, stable steps are possible. In short, the optimiser does more than find zero training error—it determines *which* zero-error solution you reach, and that choice largely sets generalisation quality.

2.5.6 Implications for physical modelling

In physical sciences, target functions such as wavefunctions are continuous, smooth (except at singularities like cusps), and often structured by symmetries. Neural networks thus provide a flexible ansatz space:

1. They can approximate smooth functions and their derivatives, which is crucial when differential operators appear in the loss (as in PINNs).
2. Depth allows efficient capture of multi-scale and compositional structure, relevant for many-body interactions.
3. Constraints such as permutation invariance or antisymmetry can be incorporated to encode physics explicitly.

These theoretical guarantees justify the use of neural networks as universal, symmetry-aware function approximators, laying the groundwork for our discussions of activation functions, initialization strategies, and physics-informed extensions.

2.6 Physics-Informed Neural Networks

Traditional machine learning focuses on fitting models to labeled data. In scientific applications, however, governing equations such as differential equations encode a wealth of prior information about the system. *Physics-Informed Neural Networks (PINNs)* integrate these constraints directly into the learning process, combining data-driven approximation with analytical structure. Figure 2.1 illustrates how the behaviour of neural networks changes when working with higher order derivatives, as is common in PINNs.

2.6.1 Formulation of PINNs

Let $u : \Omega \subset \mathbb{R}^d \to \mathbb{R}$ denote the unknown solution of a partial differential equation (PDE) of the form

$$\mathcal{N}[u](x) = 0, \qquad x \in \Omega, \tag{2.31}$$

subject to boundary conditions $u(x) = g(x)$ for $x \in \partial\Omega$. Here $\mathcal{N}$ is a (possibly nonlinear) differential operator.

In a PINN, $u_\theta(x)$ is represented by a neural network, and the loss function enforces both data fitting and PDE residual minimization:

$$\mathcal{L}(\theta) = \lambda_{\text{data}} \sum_{i=1}^{n_d} |u_\theta(x_i) - y_i|^2 + \lambda_{\text{PDE}} \sum_{j=1}^{n_r} |\mathcal{N}[u_\theta](\xi_j)|^2 + \lambda_{\text{BC}} \sum_{k=1}^{n_b} |u_\theta(\zeta_k) - g(\zeta_k)|^2. \tag{2.32}$$

Here $\{x_i, y_i\}$ are observed data, $\{\xi_j\}$ are collocation points in the domain for enforcing the PDE, and $\{\zeta_k\}$ are boundary points. The weights $\lambda_{\text{data}}, \lambda_{\text{PDE}}, \lambda_{\text{BC}}$ balance the different contributions.

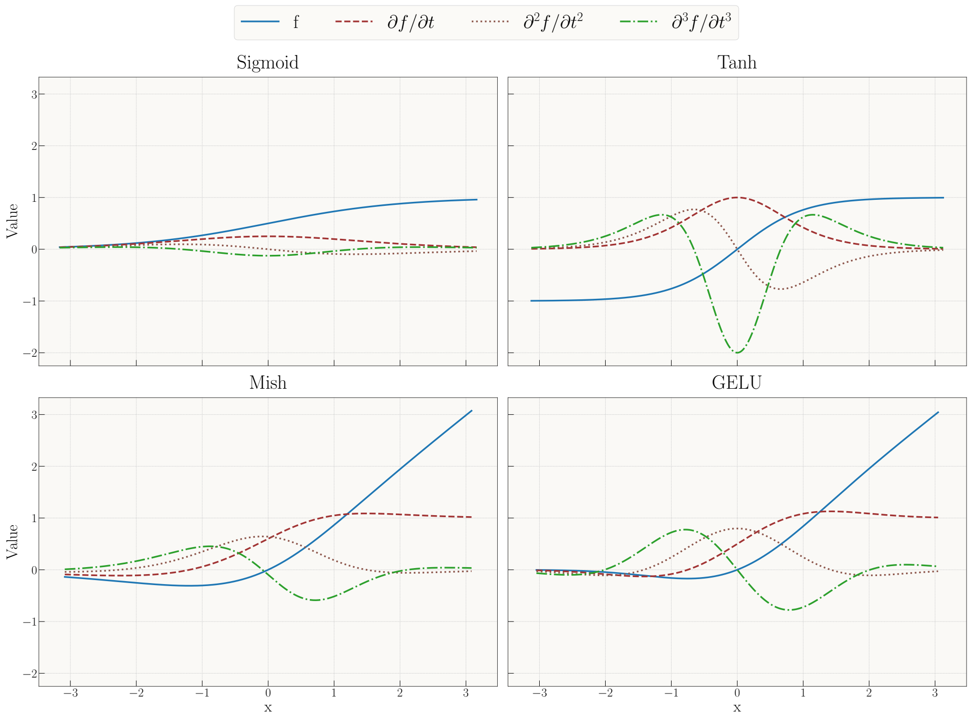


Figure 2.1: Comparison of common activation functions and their first, second, and third derivatives. Smooth activations (GELU, Mish) yield stable derivatives, while Tanh and Sigmoid have less stable higher derivatives.

2.6.2 Automatic differentiation

A key ingredient of PINNs is the use of *automatic differentiation* (AD) to evaluate ∇u_θ and Δu_θ. Unlike finite-difference approximations, AD computes derivatives exactly (up to floating-point error), making it feasible to incorporate higher-order differential operators directly into the loss. This enables end-to-end training of networks constrained by physical laws.

2.6.3 Theoretical difficulties

Although PINNs are conceptually straightforward, making them reliable and scalable raises several non-trivial challenges that go beyond classical universal approximation results.

Approximation in Sobolev spaces. Universal approximation theorems control function error in L^2/L^∞, but strong-form PINN losses depend on *derivatives* through $L[u_\theta]$. To make residuals small one must approximate in matching Sobolev norms $W^{\ell,p}$ (with ℓ the PDE order), which imposes smooth activations (e.g., tanh/SiLU for $\ell \geq 2$) and architecture choices that control higher derivatives. Low-regularity targets (discontinuities, cusps) break global $W^{\ell,p}$ control and favor weak/variational or entropy-based formulations.

Stability (residual-to-error). A small residual does not imply a small solution error unless the forward problem is (conditionally) stable. For well-posed elliptic/parabolic problems one can bound $\|u - u_\theta\|$ by $\|L[u_\theta] - f\|$, but inverse/ill-posed settings may require regularization, priors, or restricted hypothesis classes to prevent "residual overfitting" with large state error.

Sampling and generalisation. Collocation-based training minimizes a sampled surrogate of a continuum loss. Generalisation requires capacity control (e.g., bounded weights/Lipschitz loss) and sufficient sampling of interior vs. boundary/IC points; otherwise the empirical residual can be small while the population residual remains large.

Optimisation and conditioning. PINN losses couple multiple physics terms (interior, boundary, data) with disparate scales, producing ill-conditioned gradient dynamics. In the linearized/NTK view, convergence speed is governed by the spectrum of a problem-dependent kernel; large condition numbers lead to slow or stalled training.

Spectral bias. Gradient-based training tends to fit low-frequency components first, delaying convergence for solutions with sharp layers or oscillations unless the sampling, features, or architecture emphasize high-frequency content (e.g., Fourier features, windowed bases, curriculum).

Mitigating these challenges requires careful design of architectures, activations, initialization, loss weighting, and sampling strategies. These topics are explained and discussed in detail in [38]

2.6.4 Relevance to this thesis

For scientific machine learning, PINNs provide a principled framework for embedding physical structure into neural networks. In this thesis, they motivate the choice of smooth activations (for stable derivatives), careful initialization (to prevent exploding Hessians), and architectural constraints (to encode symmetry and physical laws). Thus, PINNs serve both as a methodological tool and as a theoretical lens for understanding the interplay between learning and physics.

2.6.5 Implications for scientific machine learning

For physics-informed models, generalisation takes on a slightly different meaning: the goal is not merely prediction on unseen data, but consistency with underlying physical laws. Inductive biases (symmetry, smoothness, conservation laws) are therefore not optional but essential. Implicit regularization from gradient-based optimisation complements these biases by favoring solutions that are both smooth and physically plausible. Generalisation in neural networks arises from an interplay of:

1. Algorithmic stability of gradient-based methods,
2. Implicit bias toward simple, smooth, low-complexity solutions, and
3. Architectural inductive biases encoding domain knowledge.

In scientific computing, these mechanisms explain why neural networks, despite their size, can approximate physically meaningful solutions. They also justify the careful design choices of activations, initialization, and constraints emphasized throughout this chapter.

Mitigation in practice. The challenges discussed for PINNs do not resolve themselves; they must be engineered away. In brief: use weak/variational formulations (or energy principles) when higher-order derivatives make strong forms unstable; hard–enforce essential boundary conditions and adapt weights for the rest; non-dimensionalize and balance loss terms to improve conditioning; prefer smooth activations and architectures that control derivatives; exploit symmetry, equivariance, and locality to reduce effective complexity; precondition optimisation with second-order or natural-gradient methods (e.g., L-BFGS, Stochastic Reconfiguration); and design sampling (importance/stratified, curricula in time/space) to close the generalisation gap. Together, these measures turn inductive bias into concrete numerical stability and reliable, law-consistent generalisation.

2.7 Summary and Outlook

In this chapter we have developed the theoretical foundations of modern machine learning, with particular emphasis on the aspects most relevant to scientific applications and physics-informed modelling.

We began by framing machine learning as *statistical learning*, where the goal is to minimize population risk via empirical risk minimization. This perspective highlights the central challenge of *generalisation*, formalized through the bias–variance tradeoff.

We then reviewed the role of *optimisation*, from basic gradient descent to stochastic and adaptive methods, as well as the natural gradient. Optimisation not only fits models to data but also imposes an implicit bias toward smooth and low-complexity solutions.

The expressivity of neural networks was established through universal approximation theorems and the depth–width tradeoff, showing that deep, nonlinear architectures efficiently capture compositional and structured functions. Building on this, we examined two design choices that critically affect stability: *activation functions* and *weight initialization.* Smooth modern activations (e.g. SiLU, GELU, Mish) and principled initializations (Glorot, Kaiming, orthogonal) ensure stable signal and gradient propagation, which is essential for scientific applications requiring accurate derivatives.

We introduced *physics-informed neural networks (PINNs)* as a framework for embedding differential equations and physical constraints into learning. PINNs exemplify how universal approximation, smooth activations, and stable initialization combine to produce function approximators that are not only flexible but also consistent with governing laws.

Finally, we discussed *generalisation mechanisms*, highlighting the importance of algorithmic stability, implicit regularization from gradient-based optimisation, and inductive biases encoded by architecture. For physics, such biases often correspond to symmetries, conservation laws, or invariances that dramatically reduce hypothesis complexity.

Outlook. The theoretical principles presented here provide the foundation for the architectures and diagnostics developed in this thesis. As we turn to applications in quantum systems, these ideas will guide the design of networks that balance expressivity with stability, respect physical constraints, and generalise beyond the training domain.

Part III

Methods

Chapter 3

Trial Wavefunction

In this chapter we specify the concrete modelling and algorithmic choices used in our computations. We proceed from the physical setup and trial states to objectives, with an emphasis on stability, symmetry, and differentiability—all of which are critical for Laplacian-based local-energy evaluations and stochastic reconfiguration (SR). Unless otherwise stated, we work in atomic units.

3.1 Preliminaries

Hamiltonian and units. We study N spin-$\frac{1}{2}$ fermions in two spatial dimensions confined by a harmonic trap and interacting via Coulomb repulsion,

$$H \;=\; \sum_{i=1}^{N} \Big(-\tfrac{1}{2}\nabla_{\mathbf{r}_i}^2 + \tfrac{1}{2}\,\omega^2 r_i^2 \Big) \;+\; \sum_{1\le i<j\le N} \frac{1}{|\mathbf{r}_i-\mathbf{r}_j|}, \qquad (\hbar = m_e = e = 1). \tag{3.1}$$

Spin does not appear in H; eigenstates factorize into a spatial function with good L_z and a spin function with good (S, S_z).

Coordinates and scaling. We collect positions as $R = (\mathbf{r}_1, \ldots, \mathbf{r}_N) \in \mathbb{R}^{2N}$ and use trap scaling $\tilde{\mathbf{r}}_i = \sqrt{\omega}\,\mathbf{r}_i$, $\tilde{R} = \sqrt{\omega}\,R$. This aligns typical length scales to $\mathcal{O}(1)$, improves conditioning of second derivatives, and makes the noninteracting ground state ω-independent. When useful, we also use pair distances $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ and their scaled versions $\tilde{r}_{ij} = |\tilde{\mathbf{r}}_i - \tilde{\mathbf{r}}_j|$.

Sectors and focus. We primarily report *closed-shell* cases (unique spin singlet at the noninteracting level). Open shells introduce a degenerate manifold and additional choices for the reference; we keep this to brief remarks only where needed.

Spin–orbitals and basis. One-particle spatial orbitals are Fock–Darwin functions $\{\phi_{nm}(\mathbf{r})\}$ with $E_{nm} = \omega(2n + |m| + 1)$, $n \in \mathbb{N}_0$, $m \in \mathbb{Z}$. Spin–orbitals are $\varepsilon_\alpha(x) = \phi_{nm}(\mathbf{r})\chi_\sigma(s)$, $\sigma \in \{\uparrow, \downarrow\}$.

3.2 Overview of the ansatz

Our trial state combines exact antisymmetry, an analytic short-range cusp, and a compact neural residual, while optionally evaluating the Slater determinant on backflowed coordinates:

$$\Psi_{\theta,\beta}(R) = \mathrm{SD}\big(\tilde{R}'\big)\;\exp\Big(\sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij}) + W_\theta(\tilde{R})\Big), \tag{3.2}$$

$$\tilde{R}' = \tilde{R} + \Delta_\beta(\tilde{R}). \tag{3.3}$$

Here SD enforces antisymmetry; u hard-wires the Kato cusp; W_θ (the NQS correlator) learns smooth medium/long-range structure; and Δ_β (backflow) adjusts the nodes by smoothly warping the coordinates used inside the determinants; its per-particle displacements $\Delta_{\beta,i}$ are written $\Delta\mathbf{x}_i$ in the results chapters. We deliberately feed the NQS with $\tilde{R}$ (not $\tilde{R}'$) to avoid self-reinforcing feedback and keep local-energy variance low.

Design principles. (i) *Separate guaranteed physics from learned details:* antisymmetry and the cusp are hard-wired; the network learns the rest. (ii) *Condition the geometry:* work in trap units and use soft-core pair features so that first/second derivatives are bounded. (iii) *Keep the learned object small and smooth:* once the cusp is explicit, a compact NQS suffices across (N, ω).

3.3 Slater reference

For closed shells we use a restricted product of spin-block determinants of the lowest $N/2$ harmonic oscillator orbitals (trap units):

$$\mathrm{SD}(\tilde{R}) = \det\big[\phi_a(\tilde{\mathbf{r}}_i)\big]_{i\in\uparrow,\, a\le N/2}\ \det\big[\phi_a(\tilde{\mathbf{r}}_j)\big]_{j\in\downarrow,\, a\le N/2}. \tag{3.4}$$

The correlator reweights amplitudes *within* nodal pockets; without backflow, the nodes are those of (3.4).

3.4 Neural correlator W_θ

As with the backflow, the correlator comes in two forms that we compare directly (Q1a, Chapter 7): a *separable* network that encodes particle- and pair-features independently and pools them, and a copresheaf *message-passing* (CTNN) network. We describe the separable network first—it is the baseline and also supplies the safe features and analytic cusp shared by both—and then the CTNN.

3.4.1 Separable correlator (DeepSets baseline)

The separable NQS aggregates *particlewise* and *pairwise* embeddings with permutation-invariant pooling and appends two global statistics, all in $\tilde{R}$. A core safety requirement is that pair channels satisfy $ds_k/d\tilde{r} \to 0$ as $\tilde{r} \to 0$ so that W_θ never injects spurious $1/\tilde{r}$ terms into $\nabla \log \Psi$ or $\Delta \log \Psi$.

Particle branch (DeepSets).

This part of the network is typically referred to as Deep Sets [33]. The permutation invariance of quantum many-body mechanics is naturally captured by this architecture. A neural network ϕ creates a latent embedding for each particle individually. All embeddings are pooled together to form a global representation, which is used later in the readout. Formally, we have

$$\mathbf{h}_i^\phi = \phi(\tilde{\mathbf{x}}_i) \in \mathbb{R}^{d_L}, \qquad \overline{\mathbf{h}}^\phi = N^{-1}\sum_i \mathbf{h}_i^\phi \in \mathbb{R}^{d_L}.$$

Pair branch (soft-core, even channels).

Although the particle branch could theoretically produce pair correlations, in practice it struggles to represent sharp near-field structure. We therefore add a dedicated pair branch that builds features from interparticle distances. This network processes all pairs $(i,j), i \neq j$ individually and pools the resulting latent embeddings. The problem is that raw distances $\tilde{r}_{ij}$ have unbounded derivatives at coalescence, which destabilizes Laplacian evaluations. We therefore build *safe* pair features with bounded first and second derivatives as $\tilde{r} \to 0$. With $\tilde{r}^{\mathrm{soft}} = \sqrt{\tilde{r}^2 + \varepsilon^2}$ ($\varepsilon \in [0.15, 0.30]$ in trap units), we build six "safe" features

$$\begin{gathered} s_1 = \log\big(1 + (\tilde{r}^{\mathrm{soft}}/\varepsilon)^2\big), \quad s_2 = \tfrac{\tilde{r}^2}{\tilde{r}^2+\varepsilon^2}, \quad s_3 = (\tilde{r}^{\mathrm{soft}}/\varepsilon)^2 \exp\big[-(\tilde{r}^{\mathrm{soft}}/\varepsilon)^2\big], \\ \mathrm{rbf}_g = \exp(-g\, s_1), \quad g \in \{0.25, 1, 4\}, \qquad \mathbf{u}_{ij} = [s_1, s_2, s_3, \mathrm{rbf}_{0.25}, \mathrm{rbf}_1, \mathrm{rbf}_4] \in \mathbb{R}^6. \end{gathered} \tag{3.5}$$

An MLP ψ maps $\mathbf{u}_{ij} \mapsto \mathbf{h}^{\psi}_{ij} \in \mathbb{R}^{d_L}$. A short-range gate

$$\chi(\tilde{r}) = \frac{\tilde{r}^2}{\tilde{r}^2 + r_g^2}, \quad r_g \approx 0.3, \quad \chi(0) = \chi'(0) = 0,$$

suppresses learned responses *at* coalescence, we use $\chi\, \mathbf{h}^{\psi}_{ij}$ downstream.

3.4.2 Analytic short-range cusp (shared)

The analytic cusp below is shared by both correlator variants (and is independent of the backflow). Because the safe features are necessary in the pair branch, the NQS cannot represent the Kato cusp condition exactly. We therefore add an analytic cusp term to the exponent in (3.2), written in *physical* coordinates so that its slope at contact is manifestly correct:

$$u_{\sigma\sigma'}(r) = \gamma_{\sigma\sigma'}\, r\, e^{-r/a_{\text{ho}}}, \qquad a_{\text{ho}} = 1/\sqrt{\omega}, \qquad \gamma_{\uparrow\downarrow} = \tfrac{1}{d-1}, \quad \gamma_{\uparrow\uparrow} = \tfrac{1}{d+1}. \tag{3.6}$$

Because the linear prefactor is the physical separation r, the slope at contact is $\partial_r u|_0 = \gamma_{\sigma\sigma'}$, exactly the Kato condition in general d; for $d = 2$ this gives $\gamma_{\uparrow\downarrow} = 1$ and $\gamma_{\uparrow\uparrow} = \frac{1}{3}$, the standard two-dimensional quantum-dot values, derived—including the parallel-spin case, where antisymmetry already forces a linear node—in Appendix B.2. Only the exponential taper carries the trap length a_{ho}, suppressing long-range interference so the NQS need not unlearn cusp structure at medium and long range; in trap units ($\tilde{r} = \sqrt{\omega}\, r$) the same term reads $u = (\gamma_{\sigma\sigma'}/\sqrt{\omega})\, \tilde{r}\, e^{-\tilde{r}}$.

3.4.3 Separable readout

With contributions from the particle and pair branches, we form a global representation for the final readout. In addition to the pooled embeddings $\overline{\mathbf{h}}^{\phi}$ and $\overline{\mathbf{h}}^{\psi}$, we also include two global statistics that help the network gauge overall system size and density. We average pairs by default or apply degree-normalised radial attention

$$w_{ij} = \exp[-(\tilde{r}_{ij}/r_c)^p], \quad \hat{w}_{ij} = w_{ij} \Big/ \sum_{k\neq i} w_{ik}, \quad (r_c, p) = (0.4, 6),$$

then form $g_i = \sum_{j\neq i} \hat{w}_{ij}\, \mathbf{h}^{\psi}_{ij}$ and $\overline{\mathbf{h}}^{\psi} = N^{-1} \sum_i g_i$. Two system-level scalars in trap units,

$$\mathbf{g}(\tilde{R}) = \Big[\tfrac{1}{Nd} \sum_i \|\tilde{\mathbf{x}}_i\|^2,\ \tfrac{2}{N(N-1)} \sum_{i<j} s_1(\tilde{r}_{ij})\Big] \in \mathbb{R}^2,$$

are concatenated with $\overline{\mathbf{h}}^{\phi}$ and $\overline{\mathbf{h}}^{\psi}$ and mapped to a scalar by a tiny MLP ρ:

$$W_\theta(\tilde{R}) = \rho\Big(\overline{\mathbf{h}}^{\phi} \parallel \overline{\mathbf{h}}^{\psi} \parallel \mathbf{g}(\tilde{R})\Big). \tag{3.7}$$

3.4.4 Message-passing correlator (CTNN)

The message-passing correlator (code: `CTNNJastrowVCycle`) keeps the same node/edge embeddings, safe pair features, and analytic cusp, but replaces the independent pooling of the separable network with the copresheaf transport of §2.4.4: at each round the endpoint nodes are transported into an edge update and the updated edges are transported back and aggregated onto the nodes,

$$\mathbf{h}_{ij} \mathrel{+}= F_e\big(\mathbf{h}_{ij} \parallel \rho_{v\to e}\mathbf{h}_i \parallel \rho_{v\to e}\mathbf{h}_j\big), \qquad \mathbf{h}_i \mathrel{+}= F_v\Big(\mathbf{h}_i \parallel \operatorname*{Agg}_{j\neq i} w^{\text{spin}}_{ij}\, \rho_{e\to v}\mathbf{h}_{ij}\Big), \tag{3.8}$$

with residual updates and per-round transport maps. Unlike the single-round backflow, the correlator stacks these rounds in a multiscale "V-cycle": $n_\downarrow$ down-pass rounds, a linear coarsening to a bottleneck width, then $n_\uparrow$ up-pass rounds with skip connections back to the down-pass features (defaults $n_\downarrow = n_\uparrow = 2$). A permutation-invariant readout over the final node and edge statistics returns the scalar $W_\theta(\tilde{R})$, exactly as in (3.7). Holding the Slater reference, cusp, and training fixed and swapping only the separable readout for this V-cycle isolates the relational channel studied in Q1a (Chapter 7).

3.5 Backflow

Backflow deforms the coordinates *used only inside the Slater reference* so that the nodal surface can adapt to interparticle correlations:

$$\tilde{R}' \;=\; \tilde{R} + \Delta_\beta(\tilde{R}), \qquad \Psi_{\theta,\beta}(R) \;=\; \mathrm{SD}(\tilde{R}')\,\exp\Big(\sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij}) + W_\theta(\tilde{R})\Big). \tag{3.9}$$

The correlator W_θ and cusp u are evaluated on $\tilde{R}$ to avoid feedback loops in the local energy; only the determinant "sees" $\tilde{R}'$. This choice cleanly separates (i) amplitude reweighting within nodal pockets (handled by $u + W_\theta$) from (ii) nodal *geometry* (handled by backflow).

Design goals. (i) *Stability near coalescence:* messages are built from mollified radii and short-range gates so that $\partial\Delta_\beta/\partial\tilde{r}$ stays bounded; (ii) *Permutation symmetry:* permutation equivariance holds by construction (shared maps and symmetric aggregation); (iii) *Close to identity at initialization:* a zero-initialised last layer and a positive learnable scale keep $\Delta_\beta \approx 0$ at the start. Rotational symmetry, by contrast, is only *approximate*: the displacement head reads out a d-vector from node features that include the raw coordinates, so the map is not rotation-equivariant by construction, and it is encouraged instead by the geometry-based edge features and by rotationally-augmented sampling. A global translation is not a symmetry of the trap in any case; the zero-mean projection (3.19) is a regulariser that removes a spurious centre-of-mass drift, not an imposed physical symmetry.

We study *two* backflow architectures. Both map the configuration to a per-particle displacement field and share the output bounding, spin masking, and centre-of-mass projection below; they differ only in how a particle's displacement is made to depend on the others. The conventional network is the baseline of the architecture comparison in Chapter 7; the copresheaf CTNN is the production ansatz.

3.5.1 Conventional backflow (single-round messages)

The baseline (code: `BackflowNet`) uses a single round of pairwise messages. For each ordered pair (i, j) with $i \neq j$,

$$\mathbf{m}_{ij} = \phi_\beta\big(\tilde{\mathbf{x}}_i,\ \tilde{\mathbf{x}}_j,\ \mathbf{r}_{ij},\ \tilde{r}^{\text{soft}}_{ij},\ (\tilde{r}^{\text{soft}}_{ij})^2\big)\cdot w^{\text{spin}}_{ij}, \qquad \tilde{r}^{\text{soft}}_{ij} = \sqrt{\tilde{r}^2_{ij} + \varepsilon^2}, \tag{3.10}$$

with ϕ_β a small MLP and $w^{\text{spin}}_{ij} \in \{0, 1\}$ a spin mask (same-spin only or all pairs). Messages are aggregated to a node state and a node network returns the displacement,

$$\mathbf{m}_i = \operatorname*{Agg}_{j\neq i} \mathbf{m}_{ij}, \qquad \delta\tilde{\mathbf{x}}_i = \psi_\beta\big(\tilde{\mathbf{x}}_i \,\|\, \mathbf{m}_i\big), \qquad \mathrm{Agg} \in \{\mathrm{sum}, \mathrm{mean}, \mathrm{max}\}. \tag{3.11}$$

There is no persistent edge state: each message is recomputed from the raw geometry and discarded after aggregation.

3.5.2 CTNN backflow (copresheaf)

The message-passing architecture (code: `CTNNBackflowNet`) instead maintains explicit *node and edge* features and transports information between them through learned linear maps—the copresheaf-style step of §2.4.4. Node features are embedded from the position and a spin scalar, edge features from the relative geometry,

$$\mathbf{h}^{(0)}_i = W_v\,[\tilde{\mathbf{x}}_i,\, s_i], \qquad \mathbf{h}^{(0)}_{ij} = \mathrm{MLP}_e\big(\mathbf{r}_{ij},\, |\mathbf{r}_{ij}|,\, |\mathbf{r}_{ij}|^2\big). \tag{3.12}$$

A node-to-edge transport $\rho_{v\to e}$ feeds the endpoint nodes into an edge update; an edge-to-node transport $\rho_{e\to v}$ returns the updated edges to the nodes, which are refined by a residual node update; a linear head reads out the displacement:

$$\mathbf{h}'_{ij} = \mathrm{MLP}'_e\big(\mathbf{h}^{(0)}_{ij} \,\|\, \rho_{v\to e}\mathbf{h}^{(0)}_i \,\|\, \rho_{v\to e}\mathbf{h}^{(0)}_j\big), \tag{3.13}$$

$$\mathbf{m}_i = \operatorname*{Agg}_{j\neq i} w^{\text{spin}}_{ij}\,\rho_{e\to v}\big(\mathbf{h}'_{ij}\big), \tag{3.14}$$

$$\mathbf{h}_i = \mathbf{h}^{(0)}_i + \mathrm{MLP}_v\big(\mathbf{h}^{(0)}_i \,\|\, \mathbf{m}_i\big), \qquad \delta\tilde{\mathbf{x}}_i = W_\Delta\,\mathbf{h}_i. \tag{3.15}$$

The learned transport maps $\rho_{v\to e}, \rho_{e\to v}$ are the channel whose ablation collapses the backflow at the Wigner crystal (Chapter 7). The backflow uses a single copresheaf round; the message-passing *correlator* of the architecture comparison (`CTNNJastrowVCycle`) iterates the same style of update over a down/up "V-cycle".

3.5.3 Shared output, scaling, and initialization

Both variants take inputs in trap units ($\tilde{\mathbf{x}} = \sqrt{\omega}\,\mathbf{x}$), bound the displacement, and start at the Slater nodes,

$$\Delta_\beta(\tilde{R}) \;=\; \tanh\!\big(\delta\tilde{X}\big)\cdot s_\beta, \qquad s_\beta = \mathrm{softplus}(s_\beta^{\mathrm{raw}}) > 0, \tag{3.16}$$

with the last linear layer (ψ_β or W_Δ) zero-initialised so that $\Delta_\beta \approx 0$ initially. Empirically this parameterization stabilises the SR/energy tails and protects the Laplacian.

3.5.4 Short-range behaviour and gates

As $\tilde{r}_{ij} \to 0$, raw $1/\tilde{r}$ factors can make $\nabla\cdot\Delta_\beta$ spiky. We therefore (i) mollify $\tilde{r}_{ij}$ as in (3.10), and optionally (ii) gate the message with

$$\chi_{\mathrm{bf}}(\tilde{r}_{ij}) = \frac{\tilde{r}_{ij}^2}{\tilde{r}_{ij}^2 + r_g^2}, \qquad \chi_{\mathrm{bf}}(0) = \chi_{\mathrm{bf}}'(0) = 0, \quad r_g \approx 0.3, \tag{3.17}$$

or with degree-normalised radial attention

$$w_{ij} = \exp\!\big[-(\tilde{r}_{ij}/r_c)^p\big], \qquad \hat{w}_{ij} = \frac{w_{ij}}{\sum_{k\neq i} w_{ik}}. \tag{3.18}$$

Either choice keeps the learned response smooth at coalescence and limits the BF contribution to the local energy variance.

3.5.5 Symmetries and COM neutrality

Permutation symmetry is automatic (pairwise messages and symmetric aggregation). To avoid a spurious center-of-mass (COM) drift, we project the flow to zero mean:

$$\Delta_\beta \;\leftarrow\; \Delta_\beta \;-\; \frac{1}{N}\sum_{i=1}^{N} \Delta_{\beta,i}. \tag{3.19}$$

With inputs expressed in trap units $\tilde{R}$ (already centered in practice) and with messages built from $\mathbf{r}_{ij}$, the update is effectively translation-invariant and rotation-equivariant.[1]

3.5.6 Effect on the nodes and variational safety

Because only SD is evaluated at $\tilde{R}'$, backflow changes the *nodes* but leaves the correlator exponent on $\tilde{R}$. No Jacobian term is introduced (we are not performing a change of variables of the full wavefunction). This "Feynman–Cohen style" design targets fixed-node error directly while keeping the amplitude model simple and smooth.

3.5.7 Regularization, initialization, and cost

We found the following defaults robust across (N, ω): (i) $\varepsilon \in [10^{-6}, 10^{-4}]$ (trap units) for the soft norm; (ii) bounded output tanh with a learnable scale s_β initialized in $[0.02, 0.10]$; (iii) optional gate χ_{bf} with $r_g \approx 0.3$ (trap units). Zero-initializing the last layer of ψ_β makes the ansatz start exactly at the Slater nodes and improves optimiser stability. The compute/memory cost scales as $O\big(BN^2(d+H)\big)$ for batch size B and hidden width H.

[1] Implementation note: subtracting the batchwise mean of Δ_β is a one-liner and removes the small residual COM drift seen in ablations.

3.5.8 Mapping to implementation

In the conventional network, ϕ_β and ψ_β correspond to `BackflowNet.phi` (message MLP) and `BackflowNet.psi` (node/update MLP). In the copresheaf network, the embeddings, transport maps, and update MLPs of (3.13)–(3.15) correspond to `CTNNBackflowNet`'s `node_embed`/`edge_embed`, `rho_v_to_e`/`rho_e_to_v`, `edge_update`/`node_update`, and `dx_head`. For both, aggregation is selectable (`sum|mean|max`); spin handling (`use_spin`, `same_spin_only`) produces the mask w_{ij}^{spin}; output control matches (3.16) via `out_bound="tanh"` and $s_\beta = \text{softplus}($`bf_scale_raw`$)$; and (3.19) is enforced by subtracting the batch-wise mean of the predicted Δ_β before forming $\tilde{R}'$.

Summary. Backflow provides a compact, symmetry-respecting way to move the Slater nodes toward the true correlated nodes. With mollified distances, optional short-range gates/attention, zero-mean projection, and bounded outputs, it integrates cleanly with the analytic cusp and smooth NQS correlator, reducing fixed-node error without inflating local-energy variance.

Chapter 4

Optimisation

4.1 Primary objective and two–stage training

Given a parameterized wavefunction Ψ_θ, the variational ground–state energy is

$$E(\theta) = \frac{\langle\Psi_\theta|H|\Psi_\theta\rangle}{\langle\Psi_\theta|\Psi_\theta\rangle} = \mathbb{E}_{\tilde{R}\sim\pi_\theta}\big[E_L(\tilde{R})\big], \qquad \pi_\theta(\tilde{R}) = \frac{|\Psi_\theta(\tilde{R})|^2}{\int|\Psi_\theta|^2}. \tag{4.1}$$

The local energy is evaluated in *physical* coordinates, consistently with the Hamiltonian (3.1),

$$E_L(R) = -\tfrac{1}{2}\sum_{i=1}^{N}\Big[\Delta_{\mathbf{r}_i}\ln\Psi_\theta + \|\nabla_{\mathbf{r}_i}\ln\Psi_\theta\|^2\Big] + \tfrac{1}{2}\,\omega^2\sum_{i=1}^{N} r_i^2 + \sum_{i<j}\frac{1}{r_{ij}}, \tag{4.2}$$

and the zero–variance principle holds: $\mathrm{Var}_{\pi_\theta}[E_L] = 0$ iff Ψ_θ is an eigenstate. The network *features* are computed in trap units $\tilde{\mathbf{r}} = \sqrt{\omega}\,\mathbf{r}$ for conditioning, but the local energy itself is formed in physical units, so no ω factors are absorbed into the kinetic, trap, or Coulomb terms. (In trap units the same operator reads $E_L = \omega\big[-\frac{1}{2}\sum_i(\Delta_{\tilde{\mathbf{r}}_i}\ln\Psi_\theta + \|\nabla_{\tilde{\mathbf{r}}_i}\ln\Psi_\theta\|^2) + \frac{1}{2}\sum_i \tilde{r}_i^2\big] + \sum_{i<j}\sqrt{\omega}/\tilde{r}_{ij}$; the kinetic and trap terms scale by ω and the Coulomb term by $\sqrt{\omega}$.)

Our optimisation follows two stages that mirror the implementation:

1. **Residual–based pretraining (stage I).** We fit the residual of E_L to a moving scalar target E_{eff} using the loss $\mathcal{L}_{\text{res}}(\theta) = \mathbb{E}\big[(E_L(\tilde{R}) - E_{\text{eff}})^2\big]$. We begin with a *self–target* (stabilizer) $E_{\text{eff}} = \mu$ where $\mu = \mathbb{E}[E_L]$ over the current batch. This choice is equivalent to variance minimization and prevents the network from chasing a potentially inaccurate external target before it can represent low–variance pockets. We then anneal toward a trusted reference E_{DMC} via

$$E_{\text{eff}}(\alpha) \;=\; \alpha\,E_{\text{DMC}} + (1-\alpha)\,\mu, \qquad \alpha\in[0,1], \tag{4.3}$$

   with a smooth cosine schedule $\alpha = \alpha(t)$ that ramps from α_{start} to α_{end} over a prescribed fraction of epochs (code: `alpha_start`, `alpha_end`, `alpha_decay_frac`). Setting the `objective` to `"residual"` uses $E_{\text{eff}} = \mu$; `"energy"` uses $E_{\text{eff}} = E_{\text{DMC}}$; and `"energy_var"` uses the annealed blend above.

2. **Stochastic Reconfiguration (stage II).** After residual pretraining, we refine with SR (natural–gradient VMC). Writing $O(\tilde{R}) = \partial_\theta \log\Psi_\theta(\tilde{R})$,

$$S \;=\; \mathbb{E}_\pi\big[O^\top O\big], \qquad g \;=\; 2\,\mathbb{E}_\pi\big[(E_L - \mathbb{E}_\pi[E_L])\,O\big], \tag{4.4}$$

   we solve the damped normal equations

$$(S+\lambda I)\,\Delta\theta \;=\; -\,g, \tag{4.5}$$

   using (preconditioned) conjugate gradients with restarts, a trust–region scale on the final step, and batchwise centering/whitening of O (code: `damping`, `max_param_step`, CG options).

This pipeline exploits the zero–variance principle for stable shaping of W_θ (and backflow) before applying the geometry–aware SR update.

4.2 Configuration–space sampling

Guiding principle (permutation invariance). In a many–body fermionic system, *which* particle indices are close is irrelevant; what matters is *how many* close pairs and at which length scales. Consequently, we can design samplers that (i) emphasize *configurational motifs* (clusters, shells, dimers, long tails) rather than specific labels, and (ii) deliberately permute particles and randomly rotate configurations to explore the space efficiently without biasing towards any ordering.

4.2.1 Stratified capped–simplex mixture

We draw collocation batches $X \in \mathbb{R}^{B\times N\times d}$ from a five–component mixture,

$$\text{(center, tails, mixed, shells, dimers)},$$

with nonnegative weights $w \in \Delta^4$ projected to a *capped simplex* $\{\, w : \sum_k w_k = 1,\ 0 \le w_k \le 0.3 \,\}$ to enforce coverage (code: `_project_simplex_with_caps`). Each component proposes x as follows (all in physical units; the code internally scales by $a_{\text{ho}} = 1/\sqrt{\omega}$):

- **Center:** narrow isotropic Gaussian around the trap center (samples near the density core).
- **Tails:** wide Gaussian (probes low–density outskirts and large–r behaviour).
- **Mixed:** hybrid batches mixing narrow and moderate spreads per configuration (tests robustness to inhomogeneous spreads).
- **Shells:** points placed on K hyperspherical shells with trap–scaled radii $\{r_k\}$ and occupancy $\{q_k\}$, with small radial/tangential jitter (captures ring/shell order at weak traps).
- **Dimers:** we induce a few close pairs per configuration by sampling short interparticle offsets with random directions and log–uniform distances (enforces near–coalescence coverage without singularities).

We then apply a random in–plane rotation (for $d = 2$), and a random permutation of particle indices; both preserve the target expectations by symmetry.

Adaptive mixture and shell occupancy. Per epoch we compute simple difficulty scores and update w and q by exponentiated–gradient (EG) steps with momentum, temperature, and an exploration term; w is then projected back to the capped simplex (code: `eg_eta`, `eg_temp`, `eg_momentum`, `explore_gamma`, `prob_floor`). Mixture difficulty uses the residual $(E_L - E_{\text{eff}})^2$ per component; shell difficulty uses a proxy

$$\underbrace{V_i}_{\text{harm.+Coulomb per particle}} + \underbrace{\gamma_{g^2}\,\|\nabla_{\mathbf{x}_i}\log\Psi\|^2}_{\text{stiff gradients}} + \underbrace{\gamma_{\text{cusp}}\, r_{\min}^{-1}}_{\text{coalescence pressure}},$$

aggregated by nearest shell index (code: `g2_weight`, `cusp_gamma`, safe clip `rmin_clip`). This shifts probability mass toward underfit regions *without* collapsing coverage thanks to the caps.

Hard–example injection. We maintain a small buffer of previously challenging configurations (based on residuals) and stochastically inject them, with mild multiplicative and additive jitter, into the next batch (code: `sampler_hard_*`). This provides a curriculum–like pressure while avoiding overfitting to outliers.

Robustness tweaks. We trim extreme local–energy outliers by quantiles (default 3%) *before* forming losses/updates; we also micro–batch the evaluation to keep memory bounded, and apply gradient clipping to f_θ (and backflow) parameters.

4.3 Residual–based training (stage I)

For a batch $\{\tilde{R}_b\}_{b=1}^{B}$ we compute $E_L(\tilde{R}_b)$ and the batch mean $\mu = \frac{1}{B}\sum_b E_L(\tilde{R}_b)$. The residual loss is

$$\mathcal{L}_{\text{res}}(\theta;\alpha) = \frac{1}{B}\sum_{b=1}^{B}\big(E_L(\tilde{R}_b) - E_{\text{eff}}(\alpha)\big)^2, \qquad E_{\text{eff}}(\alpha) = \alpha\, E_{\text{DMC}} + (1-\alpha)\,\mu. \tag{4.6}$$

- **Self–target warmup ($\alpha = 0$).** Minimizing $\mathbb{E}[(E_L - \mu)^2]$ is exactly variance minimization; it teaches the network to produce smooth, low–variance amplitudes inside nodal pockets and reduces sensitivity to early sampler noise.
- **Anneal to reference ($\alpha \uparrow$).** As capacity grows, we blend in the external target, gently steering the mean energy while retaining the variance pressure. The cosine ramp (code: `alpha_start`, `alpha_end`, `alpha_decay_frac`) avoids abrupt objective switches.
- **Pure energy fit ($\alpha = 1$).** If desired, we can end stage I with $E_{\text{eff}} = E_{\text{DMC}}$.

In practice we compute $\nabla_\theta \mathcal{L}_{\text{res}}$ by autodiff, use micro–batches of size `micro_batch`, apply optional gradient clipping (code: `grad_clip`), and log per–component usage and shell radii in physical units.

Local–energy derivatives in practice. We support three Laplacian backends:

1. **Exact:** chunked second partials (reference–accurate, heavier).
2. **HVP–Hutchinson:** $\Delta \log\Psi = \mathbb{E}_v[v^\top H_{\log\Psi} v]$ with Rademacher v (fast and stable; fallback to FD if a probe is non–finite).
3. **FD–Hutchinson:** centered finite differences of directional gradients (robust in corner cases; slightly noisier).

We average a small number of probes (2–16 in ablations), and trim quantiles before forming losses.

4.4 Stochastic Reconfiguration (stage II)

SR uses the covariance form of the energy gradient

$$\partial_{\theta_k} E = 2\,\mathrm{Cov}_\pi\big(E_L, O_k\big) = 2\Big(\langle E_L O_k\rangle - \langle E_L\rangle\,\langle O_k\rangle\Big), \tag{4.7}$$

and approximates a natural–gradient step by solving $(S + \lambda I)\Delta\theta = -g$. Implementation details:

- **Centering/whitening.** We center O per batch and whiten by the diagonal of S before CG to improve conditioning.
- **Trust region.** The final step is scaled to satisfy a maximum parameter norm (code: `max_param_step`).
- **Damping.** A small λ (code: `damping`) stabilizes low–variance directions.

Because our backflow only alters the Slater coordinates (and W_θ stays on safe features of $\tilde{R}$), the SR statistics remain well–behaved even when nodes shift.

4.5 Sampling paradigms and the weak-form collocation objective

The pipeline above draws its batches from an analytic proposal rather than a Markov chain, and it is worth making the choice of *sampling measure* explicit, because the results treat it as a variable in its own right. Two paradigms appear in this thesis. In *variational Monte Carlo* (VMC) the batch is drawn from $|\Psi_\theta|^2$ itself, usually by a Metropolis chain, so the estimators of the QGT and of the gradient share the very metric they are meant to precondition. In *collocation* the batch is drawn from a fixed analytic proposal $q(\mathbf{R})$ and reweighted to $|\Psi_\theta|^2$ by importance weights $w_i = |\Psi_\theta(\mathbf{R}_i)|^2/q(\mathbf{R}_i)$, so that no Markov chain is required at any point in training.

For the collocation objective we use a *weak form*: the local energy E_L enters only as a reward in a score-function (REINFORCE) gradient,

$$\nabla_\theta \langle E\rangle = 2\,\mathbb{E}_{\mathbf{R}\sim|\Psi_\theta|^2}\big[(E_L(\mathbf{R}) - b)\,\nabla_\theta \log|\Psi_\theta(\mathbf{R})|\big], \tag{4.8}$$

with a variance-reducing baseline b, and the Laplacian inside E_L is never differentiated through. This is deliberate. Differentiating the Laplacian through a backflow-shifted determinant is the source of the nodal instability analysed in Appendix C; the weak form keeps the full local-energy signal as a reward while sidestepping that pathology.

The quality of the reweighting is tracked by the effective sample size $\mathrm{ESS} = (\sum_i w_i)^2/\sum_i w_i^2$, which collapses toward 1 when q ceases to overlap $|\Psi_\theta|^2$—the failure mode that bounds the paradigm at large N and weak ω. Where a naive origin-centred proposal fails in the Wigner regime, we replace it with a *physics-informed* proposal built from the Wigner shell itself (a radial Gaussian at the classical ring radius times an angular distribution over the ring electrons), which restores a usable ESS and carries MCMC-free training deep into the correlated regime. Whether the two paradigms reach the same *state*, and where they diverge, is one of the questions of Chapter 7.

4.6 Units, scaling, and batch construction

All features are computed in trap units $\tilde{\mathbf{x}} = \sqrt{\omega}\,\mathbf{x}$ so that length scales and gates (e.g. the short–range gate in the NQS and backflow) behave consistently across ω. The sampler internally widens/narrows Gaussians and shell radii using $a_{\mathrm{ho}} = 1/\sqrt{\omega}$ in a fixed, explicit way, so that the five components cover comparable *dimensionless* regimes for $\omega \in [0.01, 1]$. Batches are i.i.d. by design; we rely on the stratified mixture (with caps, EG–adaptation, and hard–injection) rather than long MCMC chains to traverse configurations, which markedly reduces correlation and simplifies accounting.

4.7 Workflow summary

1. **Sample collocation set** X from the stratified capped–simplex mixture (center/tails/mixed/shells/dimers), then apply rotation and permutation; optionally inject hard examples.
2. **Compute** E_L using the selected Laplacian backend; trim outliers.
3. **Stage I (residual).** Minimize $\frac{1}{B}\sum_b (E_L(\tilde{R}_b) - E_{\mathrm{eff}}(\alpha))^2$ with α ramped from 0 to α_{end}; update mixture/shell weights via EG.
4. **Stage II (SR).** Form S and g, solve $(S + \lambda I)\Delta\theta = -g$, apply a trust–region step.

This procedure keeps the early dynamics variance–dominated and sampler–aware, then transitions to a geometry–aware natural–gradient refinement. Empirically it yields stable convergence across (N, ω), including weak traps where shell/dimer coverage and permutation–respecting stratification are essential.

Chapter 5

Analysis

This chapter describes *what* we measure on fresh $|\Psi_\theta|^2$ samples, *how* we compute the diagnostics, and *why* each quantity is informative. Results and numbers are presented in Chapter 6.

5.1 Structural diagnostics and Wigner markers

Unless stated otherwise, curves are accumulated in *trap units* $\mathbf{y} = \sqrt{\omega}\,\mathbf{x}$ to compare shapes across ω, while all headline distances are reported in *Bohr* (physical units $\mathbf{x}$). Production chains are thinned; we split samples into halves for stability checks (acceptance window, JSD between halves, and reproducibility under restarts). Using trap units to compare shapes across confinements is standard in parabolic dots; see the overview in [6].

5.1.1 Pair distribution and radial statistics

We estimate the pair distribution function $g(r)$ via a shell-area corrected histogram in trap units,

$$g(r_\alpha) \;=\; \frac{A_{\text{eff}}}{N(N-1)}\,\frac{\left\langle \sum_{i<j} \mathbf{1}\{r_{ij} \in [r_\alpha^-, r_\alpha^+)\}\right\rangle}{2\pi r_\alpha\,\Delta r_\alpha}, \qquad r_{ij} = \|\mathbf{y}_i - \mathbf{y}_j\|_2, \tag{5.1}$$

with $A_{\text{eff}} = \pi R_{0.95}^2$ and $R_{0.95}$ the 95% quantile of single-particle radii in trap units. This $g(r)$-based structural analysis follows established practice in mesoscopic 2D electron systems and quantum dots [1, 5, 6]. We also form the normalised *radial probability* $P(r) \propto r^{d-1} g(r)$ and report the mode r_{mode}, mean $\bar{r}$, standard deviation σ_r, and FWHM (all in Bohr). A compact width–to–scale marker is

$$\gamma \;=\; \sigma_r / r_{\text{mode}}, \tag{5.2}$$

which decreases as correlations strengthen (narrower distribution around a larger typical radius), consistent with the liquid→Wigner crossover picture [1, 5].

5.1.2 Automatic shell detection and topologies

For each frame we sort the single-particle radii $s_1 \le \cdots \le s_N$ (trap units), compute gaps $g_k = s_{k+1} - s_k$, and flag the frame as *two-shell* if the largest gap exceeds a robust threshold:

$$\max_k g_k \;\ge\; \tau\,\mathrm{median}(g_k), \qquad \tau \in [2.5, 3.5] \text{ (default 3.0)}. \tag{5.3}$$

The cut radius is the midpoint of the maximal gap; the inner multiplicity n_{in} defines the topology $(n_{\text{in}}, N - n_{\text{in}})$. Shelling and discrete ring occupancies are well known in parabolic Coulomb clusters and quantum dots [3, 4, 6]; our detector provides a reproducible, data-driven assignment in that spirit. We report: (i) the fraction of two-shell frames, (ii) the histogram over n_{in} among two-shell frames, and (iii) for selected topologies (e.g. $(1,5)$ or $(3,9)$) the fraction among *all* frames.

Shell-resolved reconstruction of $g(r)$. On two-shell frames we decompose pairs into II, IO, and OO and accumulate histograms $g_{\mathrm{II}}, g_{\mathrm{IO}}, g_{\mathrm{OO}}$. With empirical weights $w_{\mathrm{II}}, w_{\mathrm{IO}}, w_{\mathrm{OO}}$ (combinatorial counts) we reconstruct $\tilde{g}(r) = w_{\mathrm{II}} g_{\mathrm{II}} + w_{\mathrm{IO}} g_{\mathrm{IO}} + w_{\mathrm{OO}} g_{\mathrm{OO}}$ and report $\cos\angle(g, \tilde{g})$. Values ≈ 1 indicate that the observed $g(r)$ shape is explained by shell geometry rather than sampling artifacts, in line with shell-based descriptions in [3, 4, 6].

5.1.3 Orientational order and "ring stiffness"

On each detected ring (inner or outer) we center the mean angle to remove global rotations and define angles $\{\phi_k\}_{k=1}^{n}$. Bond-orientational order is

$$\Phi_m \;=\; \frac{1}{n}\sum_{k=1}^{n} e^{\,\mathrm{i}m\phi_k}, \qquad m = n, \tag{5.4}$$

so that $|\Phi_m| \in [0, 1]$ measures n-fold orientational order (e.g. m=5 for $(1, 5)$, m=9 for the outer ring of $(3, 9)$), analogous to standard bond–orientational order parameters used in 2D Coulomb/Yukawa crystals [2]. As a complementary, dimensionless *angular Lindemann* number we use

$$\lambda_\phi \;=\; \frac{\mathrm{std}(\Delta\phi)}{\mathbb{E}[\Delta\phi]}, \qquad \Delta\phi \;=\; \text{nearest-neighbour angular spacing on the ring.} \tag{5.5}$$

Smaller λ_ϕ indicates a stiffer (more Wigner-like) ring. This is a Lindemann–type orientational metric tailored to finite rings, motivated by the use of Lindemann criteria to diagnose intershell rotational (angular) melting in mesoscopic Coulomb clusters and dots [4, 5].

5.1.4 Scaling check for $N = 2$

To connect with the classical strong-coupling limit we fit

$$r_{\mathrm{mode}}(\omega) \;=\; C\,\omega^{\alpha} \tag{5.6}$$

by linear regression in $\big(\log\omega, \log r_{\mathrm{mode}}\big)$ and compare the slope with the classical reference $\alpha_{\mathrm{cl}} = -2/3$ (two charges in a harmonic trap at strong coupling; see, e.g., the reviews in [6]). Approach toward α_{cl} as $\omega \downarrow$ is one of our Wigner crossover indicators, consistent with the crossover phenomenology reported in [1, 5].

5.1.5 Reporting and stability checks

For all scalars and histograms we report standard errors calculated over thinned blocks or frames. We verify: (i) acceptance in the target window, (ii) Jensen–Shannon divergence and ℓ_2 distance between halves of production (stationarity), (iii) robustness of topology fractions and $|\Phi_m|$ under $\tau \in [2.5, 3.5]$ and mild pre-smoothing of radii, and (iv) reproducibility of $g(r)$ and $P(r)$ under independent sampler restarts. All diagnostics operate directly on saved bundles $\{\mathbf{X}_{\mathrm{Bohr}},\, r,\, g(r)\}$.

5.2 Representation analysis

We probe what the networks learn by evaluating fixed, trained models on fresh $|\Psi_\theta|^2$ samples. Unless stated otherwise, features are computed in *trap units* $\mathbf{y} = \sqrt{\omega}\,\mathbf{x}$ to match the correlator's internal scaling. All estimates use thinned MCMC frames, split into two equal blocks A and B; PCA fits on A and reports on B (and vice versa) to avoid circularity. We use `float64` throughout.

Feature taps. Let $z_\theta \in \mathbb{R}^D$ denote the concatenated correlator representation

$$z_\theta = \big[\overline{\phi} \mid \overline{\psi} \mid \mathbf{g}\big],$$

where $\overline{\phi}$ are per-particle summaries, $\overline{\psi}$ are pairwise summaries, and $\mathbf{g}$ are global/extras. The residual head is linear:

$$f_{\mathrm{base}}(z) = \rho_\theta(z) = w^\top z + b, \qquad f = f_{\mathrm{base}} + f_{\mathrm{cusp}}. \tag{5.7}$$

Backflow returns a displacement field $\Delta x \in \mathbb{R}^{N\times d}$ that enters only the Slater determinant; the correlator and cusp are evaluated on the unshifted $\tilde{R}$ (§3.5). The coupling diagnostics below additionally probe the correlator on the shifted coordinates as a counterfactual sensitivity.

5.2.1 Preprocessing and PCA

Given a sample matrix $Z \in \mathbb{R}^{K\times D}$ (rows are frames), we standardize each column using means and standard deviations estimated on block A:

$$\tilde{Z} = (Z - \mu_A) \odot \sigma_A^{-1}.$$

We perform PCA on $\tilde{Z}_A$ (SVD of the covariance) to obtain orthonormal loadings $U = [u_1, \dots, u_D]$. Scores on B are $S_B = \tilde{Z}_B U$. Unless noted, all quantities reported below are computed on the held-out block.

5.2.2 Entropy effective rank

Let $\{\lambda_i\}_{i=1}^{D}$ be the eigenvalues of $\mathrm{Cov}(\tilde{Z})$ (on the held-out block). With $p_i = \lambda_i / \sum_j \lambda_j$, the *entropy effective rank* is

$$r_{\mathrm{eff}}(Z) \;=\; \exp\Big(-\sum_{i=1}^{D} p_i \log p_i\Big),$$

which equals 1 for perfectly one-dimensional representations and increases with spectral spread.

5.2.3 PC ablations of the head

To test how many latent axes the head truly uses, we project $\tilde{Z}$ onto the top-k PCs:

$$\Pi_k = \sum_{i=1}^{k} u_i u_i^{\top}, \qquad z^{(k)} = \Pi_k \tilde{z},$$

and re-evaluate the trained linear head *without refitting*: $\hat{f}^{(k)} = \rho_\theta(z^{(k)})$. We report a *normalised* error on block B,

$$\text{rel-MAE}(k) \;=\; \frac{\mathbb{E}_B \,|\, \hat{f}^{(k)} - f_{\mathrm{base}} \,|}{\mathrm{std}_B(f_{\mathrm{base}})},$$

together with the learning curve in k.

5.2.4 PC1 block power decomposition

To attribute the leading axis to branches, we partition u_1 conformably with $z = [\overline{\phi} \,|\, \overline{\psi} \,|\, \mathbf{g}]$ and report

$$\mathrm{power}(\overline{\phi}) = \frac{\|u_{1,\overline{\phi}}\|_2^2}{\|u_1\|_2^2}, \quad \mathrm{power}(\overline{\psi}) = \frac{\|u_{1,\overline{\psi}}\|_2^2}{\|u_1\|_2^2}, \quad \mathrm{power}(\mathbf{g}) = \frac{\|u_{1,\mathbf{g}}\|_2^2}{\|u_1\|_2^2}.$$

Because PCA is performed on standardized features, these shares are comparable across blocks.

5.2.5 Linear probes from PCs to coarse physics

We regress interpretable summaries per configuration on the first k PC scores $S^{(k)}$ using ridge regression (tiny penalty $\alpha = 10^{-6}$):

$$y \approx S^{(k)}\beta, \qquad y \in \big\{\bar{r},\ \mathrm{Var}(r),\ \Pr(r < r_0),\ \text{shell contrast}\big\}.$$

We report R^2 on the held-out block; r_0 is a fixed near-contact cutoff in trap units.

5.2.6 Near-field gradient concentration (residual head)

For each frame we compute the minimum pair distance $r_{\min}$ (trap units) and the gradient of the residual head with respect to particle coordinates, $\nabla f_{\text{base}}(\mathbf{y})$. The concentration at quantile q is

$$\text{share}(q) \;=\; \frac{\mathbb{E}\big[\|\nabla f_{\text{base}}\|_2^2 \, \mathbf{1}\{r_{\min} \le Q_q\}\big]}{\mathbb{E}\big[\|\nabla f_{\text{base}}\|_2^2\big]},$$

where Q_q is the q-quantile of $r_{\min}$. Values $\gg 1$ indicate that residual stiffness is concentrated near contact (the analytic cusp is insufficient); values ≈ 1 indicate that the cusp handles most short-range structure.

5.2.7 Backflow geometry and energy-locality shares

Field spectrum. We vectorize the backflow displacement per frame, $\delta = \text{vec}(\Delta x) \in \mathbb{R}^{Nd}$, standardize across frames, and compute $r_{\text{eff}}(\Delta x)$ via Sec. 5.2.2.

Where backflow acts. On the same thinned frames we evaluate local energies with and without backflow, $E_{\text{loc}}^{\text{BF}}$ and $E_{\text{loc}}^{\text{noBF}}$, using the *same* coordinates. We form the signed correction $\Delta E_{\text{loc}} = E_{\text{loc}}^{\text{BF}} - E_{\text{loc}}^{\text{noBF}}$ and report near-field shares at quantile q of $r_{\min}$:

$$\text{NF-}\Delta E(q) \;=\; \frac{\mathbb{E}\big[|\Delta E_{\text{loc}}| \, \mathbf{1}\{r_{\min} \le Q_q\}\big]}{\mathbb{E}\big[|\Delta E_{\text{loc}}|\big]}.$$

(Using absolute values isolates magnitude; signed averages are also available.)

5.2.8 Alignment of correlator manifolds (noBF vs BF)

We collect two standardized feature matrices $\tilde{Z}^{\text{noBF}}$ and $\tilde{Z}^{\text{BF}}$ on identical frames (each standardized by its own μ, σ), compute their PC1 loadings u_1^{noBF} and u_1^{BF}, and report the cosine similarity

$$\cos\big(u_1^{\text{noBF}}, u_1^{\text{BF}}\big) \;=\; \frac{\langle u_1^{\text{noBF}},\, u_1^{\text{BF}}\rangle}{\|u_1^{\text{noBF}}\|_2 \, \|u_1^{\text{BF}}\|_2}.$$

Cosines near 1 indicate that backflow acts as a structured reweighting along the same dominant axis, rather than rotating the correlator's manifold.

5.2.9 Uncertainty and stability

For scalars we report standard errors over blocks. For PCA-based quantities and cosines we use a block bootstrap (resampling blocks with replacement) to obtain 68%/95% CIs. Stability checks include: (i) swapping train/held-out blocks for PCA, (ii) varying the near-field quantiles $q \in \{1, 5, 10\}\%$, (iii) re-running analyses on independent sampler restarts.

Reproducibility package. All diagnostics consume the saved bundles $\{\mathbf{X}_{\text{Bohr}},\, r,\, g(r)\}$ and the model snapshot, and emit JSON/CSV tables with r_{eff}, rel-MAE(k) curves, block-power shares, R^2 probes, near-field shares, $r_{\text{eff}}(\Delta x)$, and noBF/BF cosines. As in Sec. 5.1, we log JSD/ℓ_2 splits to confirm stationarity.

5.3 Tangent-kernel diagnostics

The representation diagnostics above describe the ansatz's *features*; a second family, developed with the geometry of §2.2.5, describes its *tangent space*, and it is these that carry the architecture, optimiser, and paradigm comparisons of Chapter 7. From a batch we assemble the centred score matrix $O_{ki} = \partial_{\theta_i} \log|\Psi_\theta(\mathbf{x}_k)|$ and form the quantum geometric tensor $S = O^\top O/B$ (or, when $B < P$, the tangent kernel $K = OO^\top$, which shares its nonzero spectrum). From the spectrum $\{\lambda_a\}$ we report the effective dimension $d_{\text{eff}} = (\sum_a \lambda_a)^2 / \sum_a \lambda_a^2$ and the condition number $\kappa(S) = \lambda_{\max}/\lambda_{\min}$. Cross-architecture comparisons of d_{eff} are made on a *common probe set*—all models scored on the same configurations, not on each model's own $|\Psi|^2$—so that the comparison reflects the ansatz and not its sampling measure, and we report only participation ratios that have converged in the batch size.

To ask whether two trained models are the *same* state rather than merely equal in energy, we use a symmetric, importance-sampled squared overlap,

$$\mathcal{S}^2(A, B) = \big\langle \Psi_B/\Psi_A \big\rangle_{|\Psi_A|^2} \, \big\langle \Psi_A/\Psi_B \big\rangle_{|\Psi_B|^2},$$

each factor evaluated by a log-sum-exp mean over samples of the corresponding state. It equals 1 for identical states, and on the analytic two-electron ground state the trained ansatz reproduces $\mathcal{S}^2 = 1.000000$, which anchors the tooling.

One convention must be stated, because the ratio Ψ_B/Ψ_A is signed for a fermionic state. We evaluate the ratios from $\log|\Psi|$, so the estimator is strictly a *modulus* overlap, $\langle|\Psi_B/\Psi_A|\rangle\langle|\Psi_A/\Psi_B|\rangle$. For the comparisons made here this is an accurate proxy for the signed overlap: both states share the same Slater reference, and the correlator and cusp enter as a strictly positive exponential, so the sign of Ψ is fixed by the determinant and the two states agree in sign except on the measure-suppressed set where their backflows carry a configuration across a node differently. Where the two states genuinely diverge the modulus overlap is an upper bound on the signed one, so a reported $\mathcal{S}^2 \to 0$ is if anything conservative. These quantities—together with the energetic price of ablating a component—are invariant to how correlation is divided between the Jastrow and the backflow, and it is on them, rather than on the gauge-dependent component ranks of the previous section, that the architectural conclusions of Chapter 7 rest.

Part IV

Results

Chapter 6

Quantum dots: energies, representations, and Wigner molecules

This chapter is about what the trained wavefunction *is*, and what it lets us see. It is one half of a pair. Here we ask how accurately the ansatz of Chapter 3 solves the two-dimensional harmonic quantum dot, what its internal representation looks like, and what it reveals about the electrons themselves; the companion chapter (Chapter 7) asks the complementary question of *why* the architecture, the optimiser, and the training paradigm behave as they do. The two chapters share one object and one aim: not merely to compute the ground state, but to understand it.

The quantum dot is an unforgiving benchmark, and it is worth being clear about why. The Coulomb interaction is singular where two electrons meet; the kinetic energy carries second derivatives that magnify any error in the wavefunction's curvature; and the confinement ω sweeps the physics across three decades of correlation strength—from a compact, kinetic-energy-dominated droplet at $\omega = 1$ to a dilute Wigner molecule at $\omega = 10^{-3}$, in which the electrons settle tens of oscillator lengths apart and the Coulomb repulsion, not the trap, dictates the structure. We study $N \in \{2, 6, 12, 20\}$ electrons across $\omega \in \{1.0,\ 0.5,\ 0.1,\ 0.01,\ 0.001\}$.

Three questions organise the chapter. *First, how accurate is the method?* Section 6.1 benchmarks the variational energies against diffusion Monte Carlo (DMC) wherever a reference exists—for $\omega \geq 0.1$—and is candid that below it no DMC benchmark is available for $N \geq 6$, so the strongly correlated energies rest on internal variational consistency rather than an external ground truth. *Second, what does the network learn?* Section 6.2 dissects the trained wavefunction's internal representation—a strikingly low-dimensional correlator whose *content*, but not its dimensionality, reorganises across regimes, alongside a backflow field that shares the work in a regime-dependent way. One caveat, carried over from the companion chapter and stated here at the outset, governs how these internal quantities may be read: several of them are gauge-like, fixed by the training route rather than by the state, and we take care to separate those from the findings that are invariant. *Third, can the learned wavefunction serve as a scientific instrument?* Section 6.3 turns the trained models on the physics itself, mapping the Wigner-molecule crossover in real space—shell topologies, angular order, and melting—from N=2 up to N=20.

6.1 Training protocol and energy overview

Unless otherwise stated, we first apply residual-based pretraining to obtain a well-conditioned initializer, and then run a Stochastic Reconfiguration (SR) tail to refine the last digits of the energy. In the residual phase, the network initially chooses its own target energy as a stabilizer, before we switch to the DMC reference energy (and optionally add a variance penalty). The SR phase then uses on the order of 10^3–3×10^3 iterations with batches of $\sim 3{\times}10^3$ samples per iteration.

We compare two backflow architectures throughout. Both condition each particle's displacement on the interparticle coordinates, but they do so differently: the *conventional* BackflowNet applies a single round of pairwise messages aggregated per particle, whereas the copresheaf *message-passing* network (CTNN), realised as the CTNNBackflowNet, maintains explicit node and edge features and transports information between them through learned maps (a single copresheaf round) before the displacement is read out; the contrast between them is the subject of the backflow analysis in Chapter 7.

The architecture size is identical for all N: the base PINN correlator has ≈ 54k parameters, the conventional BackflowNet adds ≈ 83k (total PINN+BF ≈ 137k), and the CTNNBackflowNet adds ≈ 182k (total PINN+CTNN ≈ 236k). All components are fully trainable; the parameter count is independent of N because particle interactions enter through permutation-equivariant features rather than per-particle weights.

6.1.1 Energies

Table 6.1 reports two-electron energies from *residual-only* training (PINN and PINN+BF). For $N{=}2$, backflow is not required to reach DMC accuracy, but it slightly reduces the variance and smooths convergence.

Table 6.1: Ground-state energies (Hartree) for two electrons in a harmonic trap using *residual-only* training. DMC references from [39]. Parentheses denote 1σ on the last digits.

ω	PINN+BF	PINN	DMC (Ref.)
1.00	2.99998(3)	2.999940(5)	3.00000(1)
0.50	1.65975(2)	1.65979(3)	1.65977(1)
0.10	0.440795(3)	0.440833(1)	0.44079(1)
0.01	0.0740724(6)	0.0741679(6)	0.073839(2)

Table 6.2 compares diffusion Monte Carlo (DMC) references with our *final* PINN+BF energies after residual pretraining and an SR tail. From the ultra–shallow two-electron case ($\omega{=}0.01$) to the many-electron systems, absolute deviations are small; for $N{=}2$ they lie within DMC uncertainties, and for $N \in \{6, 12\}$ the relative deviations are typically in the range 0.01–0.08% (see also Fig. 6.1).

Table 6.2: Residual-based pretraining + SR tail. Reference DMC and our PINN+BF and PINN+CTNN ground-state energies (Hartree) for selected (N, ω). The reference column collects the best available benchmark for each size: DMC from [39] (whose study spans $N \leq 13$) for $N \leq 12$, and the reference energies tabulated in the master's thesis of Daniel Haas [7] for $N = 20$, which lies beyond the range of [39]. Parentheses denote 1σ on the last digit(s); the % err columns are relative deviations from the reference. A dash in the reference column marks (N, ω) for which no published DMC value exists (all $\omega = 0.001$, and $\omega = 0.01$ for $N \geq 6$); the relative-error columns are correspondingly blank there, as no external ground truth is available to score against.

N	ω	Reference (DMC)	PINN+BF	% err	PINN+CTNN	% err
2	0.001	—	0.0137948(8)	—	**0.013778(1)**	—
	0.01	0.073839(2)	0.073838(1)	−0.0014	0.0738382(5)	−0.0014
	0.10	0.44079(1)	0.44079(1)	+0.0000	0.440796(4)	−0.0000
	0.50	1.65977(1)	1.65976(2)	−0.0006	1.65976(1)	−0.0006
	1.00	3.00000	2.99999(1)	−0.0003	3.00000(2)	−0.0000
6	0.001	—	0.140913(1)	—	**0.140832(1)**	—
	0.01	—	0.69125(2)	—	**0.69036(1)**	—
	0.10	3.55385(5)	3.5549(1)	+0.0295	**3.55388(5)**	+0.0008
	0.50	11.78484(6)	11.7895(4)	+0.0395	**11.7847(2)**	−0.0000
	1.00	20.15932(8)	20.1610(6)	+0.0083	**20.1585(3)**	−0.0041
12	0.001	—	0.515823(3)	—	**0.515365(4)**	—
	0.01	—	2.48620(5)	—	**2.47363(4)**	—
	0.10	12.26984(8)	12.2731(2)	+0.0266	**12.2743(2)**	+0.0364
	0.50	39.1596(1)	39.1786(8)	+0.0485	**39.1604(3)**	+0.0018
	1.00	65.7001(1)	65.717(1)	+0.0257	**65.69556(5)**	−0.0069
20	0.001	—	—	—	**1.293033(6)**	—
	0.01	—	—	—	**6.14645(5)**	—
	0.10	29.9779(1)	—	—	29.9888(2)	+0.0364
	0.50	93.8752(1)	—	—	**93.8789(5)**	+0.0040
	1.00	155.8822(1)	—	—	**155.8738(7)**	−0.0024

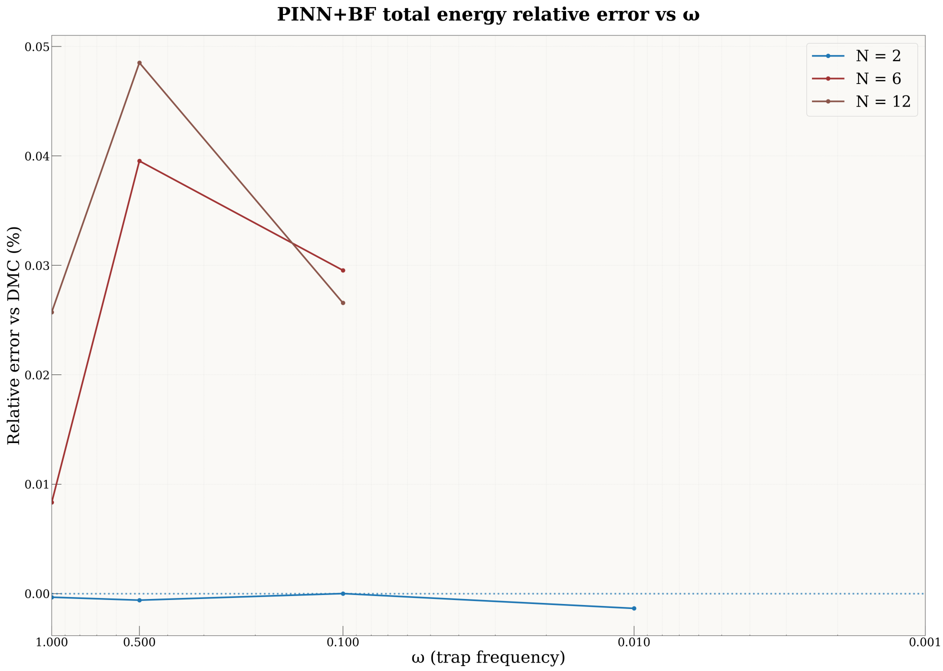


Figure 6.1: Relative deviation of PINN+BF ground-state energies from DMC references as a function of trap frequency ω for the cases where DMC data are available. For N=2 the deviations are at or below the DMC error bars ($\sim 10^{-3}$ %), while for N=6 and N=12 the errors remain in the range 2.5×10^{-2}–5×10^{-2} % across ω, with no sign of deterioration in the shallow- or tight-trap limits.

Summary. The pattern is clean wherever we can check it. For N=2 the ansatz reaches DMC accuracy at every ω for which a DMC number exists, and for $N \in \{6, 12\}$ the relative deviation stays within roughly 0.01–0.08% across the benchmarked range ($\omega \geq 0.1$; Fig. 6.1). Below $\omega = 0.1$ there is no DMC reference for $N \geq 6$, and we make no claim of absolute accuracy there; what we can say is that the energies join smoothly onto the benchmarked regime and, in the deep-Wigner limit, follow the physical energy scaling derived in §6.3.3. The efficiency is as notable as the accuracy: the SR tail converges in 10^3–3×10^3 iterations of $\sim 3{\times}10^3$ samples each, a modest optimisation budget for a fully many-body wavefunction. We attribute this stability to a small set of conditioning choices— trap-unit scaling, soft-core pair features with $ds/dr \to 0$ at coalescence, and an explicit analytic cusp—each of which keeps the gradients and the Laplacian numerically tame from the shallowest trap to the tightest.

6.2 What the networks learn

We probe the internal representations of the CTNN-augmented wavefunction across $N \in \{2, 6, 12, 20\}$ and $\omega \in \{10^{-3}, 10^{-2}, 0.1, 0.5, 1.0\}$. The analysis targets three objects: the neural correlator W_θ (the "correlator head" of Chapter 3, whose input concatenates the orbital, pair, and auxiliary branches $\overline{\phi} \| \overline{\psi} \| \mathbf{g}$), the *backflow displacement field* $\Delta\mathbf{x}$ produced by the CTNN, and the *coupling* between these two components. All diagnostics are evaluated on samples drawn from $|\Psi|^2$.

Three results organise this section. First, the correlator operates on a manifold of remarkably low dimension ($r_{\text{eff}} \lesssim 3$), whose *content*—but not dimensionality—reorganises across regimes via a $\phi \leftrightarrow \psi$ transition in the leading principal axis (§6.2.1). Second, the CTNN displacement field is a high-rank, full-dimensional correction (rank $\approx 2N$) that nonetheless leaves the correlator manifold nearly unchanged, acting as a complementary rather than redundant channel (§6.2.2). Third, PINN and CTNN switch from operating independently at strong correlation to cooperating tightly at moderate-to-weak correlation, with a sharp transition near $\omega \approx 0.1$ that coincides with the physical crossover in $\Gamma = V_{\text{int}}/T$ (§6.2.3).

How these quantities should be read. One caveat frames everything that follows, and it is better stated plainly than discovered later. The Jastrow and the backflow are partial substitutes—correlation

Table 6.3: Correlator geometry (CTNN). r_{eff}: entropy-based effective rank of Z. $|\rho|$: absolute correlation between head output and PC1 reconstruction. $\phi/\psi/\text{g}$: fraction of PC1 variance from the orbital, pair, and auxiliary branches. The ψ-dominated entries ($\psi > 0.5$) are shown in bold.

N	ω	r_{eff}	$\|\rho\|$	ϕ	ψ	g
2	0.001	1.00	1.00	0.00	**1.00**	0.00
2	0.01	1.04	0.66	0.00	**1.00**	0.00
2	0.1	1.31	0.99	0.00	**0.53**	0.46
2	0.5	1.25	1.00	0.00	**0.58**	0.42
2	1.0	1.53	0.98	0.00	0.35	**0.65**
6	0.001	1.52	0.97	0.01	0.12	**0.87**
6	0.01	1.38	0.99	0.05	0.44	**0.50**
6	0.1	2.73	0.94	**0.66**	0.21	0.14
6	0.5	2.69	0.98	0.15	**0.63**	0.22
6	1.0	2.69	0.99	0.03	**0.62**	0.35
12	0.001	1.66	0.96	**0.54**	0.13	0.33
12	0.01	1.54	0.97	0.43	0.09	**0.48**
12	0.1	3.08	0.99	0.12	**0.63**	0.26
12	0.5	2.32	0.99	0.12	**0.69**	0.19
12	1.0	1.76	0.99	0.08	**0.73**	0.19
20	0.001	2.59	0.99	0.18	**0.74**	0.08
20	0.01	1.09	0.79	0.04	**0.94**	0.01
20	0.1	2.20	0.97	0.36	0.44	0.20
20	0.5	2.30	1.00	0.17	**0.70**	0.13
20	1.0	2.32	0.99	0.18	**0.71**	0.11

that one does not capture, the other can—so how the labour is divided between them is a soft, almost flat direction of the loss, settled by the training route rather than by the state itself. Quantities that live on a single component—the backflow's rank and displacement magnitude, the per-branch loadings, the coupling between the two networks—are therefore *gauge-like* in the precise sense developed in Chapter 7: reproducible and meaningful for a given trained model, but descriptive of how *this* network chose to organise itself, not of an invariant that every optimal ansatz must share. The invariant backbone—total energy, state overlap, and the energetic price of ablating a component—is the business of the companion chapter. Read in that light, the anatomy below is not weakened by being gauge-dependent; it is a faithful dissection of one high-quality solution, and the regularities it shows across regimes are the more telling for being ones the network was free *not* to adopt.

6.2.1 Correlator geometry: a compact manifold that reorganises

We characterise the correlator feature space $Z = \overline{\phi}\|\overline{\psi}\|\mathbf{g}$ by its entropy-based effective rank $r_{\text{eff}}(Z)$, the correlation $|\rho(\text{head}, \text{PC1})|$ between the head output and its first principal component, and the power of PC1 decomposed across architectural branches. Table 6.3 reports these quantities for all CTNN configurations.

The manifold is low-dimensional everywhere. The effective rank stays below 3.1 across the entire grid, and the head output is almost perfectly reconstructed from PC1 alone ($|\rho| > 0.94$ for all $N \geq 6$ and all ω). This holds even at strong correlation where the real-space density develops pronounced shell structure—the correlator does not respond by expanding its latent dimensionality. The modest peak at N=12, ω=0.1 ($r_{\text{eff}} = 3.08$) coincides with the intermediate-correlation regime where neither the shell picture nor the crystalline picture provides a clean single-mode description.

A $\phi \leftrightarrow \psi$ transition in the leading axis. The more interesting variation is in *which branch dominates PC1.* At moderate-to-weak correlation ($\omega \geq 0.1$, $N \geq 6$), PC1 is predominantly ψ-loaded: the pair branch carries 60–73% of the variance, meaning the correlator's primary degree of freedom tracks pair correlations. At strong correlation ($\omega \leq 0.01$, $N \geq 12$), the loading shifts toward ϕ (orbital) or $\mathbf{g}$ (auxiliary) branches, with ψ dropping below 15%. The transition is sharpest at N=12, where ψ goes from 0.09 at ω=0.01 to 0.63 at ω=0.1.

Table 6.4: CTNN backflow geometry. $\|\Delta\mathbf{x}\|/\ell$: mean displacement magnitude normalised by harmonic length. Rank: effective rank of the displacement covariance. $2N-2$: maximum possible rank after centre-of-mass projection. CKA: linear centred kernel alignment between correlator features with and without backflow (1.0 = identical feature manifolds).

N	ω	$\|\Delta\mathbf{x}\|/\ell$	Rank	$2N-2$	CKA
2	0.001	0.003	1.9	2	1.000
2	1.0	0.029	2.0	2	1.000
6	0.001	0.017	10.0	10	0.999
6	0.01	0.035	9.9	10	0.998
6	0.1	0.096	10.0	10	0.995
6	0.5	0.093	10.0	10	0.991
6	1.0	0.089	10.0	10	0.979
12	0.001	0.017	21.9	22	1.000
12	0.01	0.030	21.9	22	0.998
12	0.1	0.096	21.9	22	0.993
12	0.5	0.096	21.9	22	0.993
12	1.0	0.091	21.9	22	0.995
20	0.001	0.023	37.7	38	0.999
20	0.01	0.046	37.4	38	1.000
20	0.1	0.075	37.8	38	0.997
20	0.5	0.099	37.6	38	0.990
20	1.0	0.099	37.6	38	0.991

The physical interpretation is direct. When interactions are weak relative to the trap ($\omega \gtrsim 0.1$), the non-interacting shell structure provides a good zeroth-order description, and the correlator's primary role is refining pair correlations on top of that. When interactions dominate ($\omega \lesssim 0.01$), the orbital structure itself must reorganise—which orbitals are occupied and how they mix becomes the dominant variational degree of freedom. The network discovers this distinction autonomously: regime changes appear as a rotation of a low-dimensional axis across branches, not as a growth of latent dimensionality.

6.2.2 Backflow: a high-rank field with a low-rank footprint

Table 6.4 characterises the CTNN displacement field. We report the normalised displacement magnitude $\|\Delta\mathbf{x}\|/\ell$, the effective rank of $\Delta\mathbf{x}$ (viewed as a $B \times 2N$ matrix of flattened displacements), and the CKA similarity between the correlator features with and without backflow.

Displacement magnitude scales with ω but saturates near $\|\Delta\mathbf{x}\|/\ell \sim 0.1$. At strong correlation ($\omega \leq 0.01$), displacements are small relative to the harmonic length, $\|\Delta\mathbf{x}\|/\ell \sim 0.02$–$0.05$. As the trap tightens and the system enters the moderate-to-weak correlation regime, the magnitude grows and saturates at $\sim$8–10% of ℓ for $\omega \gtrsim 0.1$. The CTNN never produces large displacements—its tanh output bound and copresheaf message-passing structure act as implicit regularisers, in contrast to the conventional BackflowNet which can exhibit displacements comparable to the system scale in poorly converged regimes (e.g. $\|\Delta\mathbf{x}\|/\ell \sim 0.95$ for N=6, ω=0.001 with BackflowNet).

Full-rank displacements, near-unity CKA. The effective rank of the displacement field is consistently $\approx 2N-2$, saturating the available spatial degrees of freedom modulo centre-of-mass conservation (confirmed by COM shifts of order 10^{-14}–10^{-17}). The backflow transformation is not a low-rank collective mode such as a breathing or rigid rotation; each particle receives a genuinely independent correction conditioned on the full configuration. This full-rank behaviour is a property of the *message-passing* backflow specifically, and it is exactly the point at which the companion chapter finds the two architectures part ways: toward the Wigner crystal a *conventional*, single-round-message backflow collapses to a single collective mode (rank $\to$ 1) and its energy degrades—its one round of pairwise messages failing to sustain the relative displacement the crystal demands—whereas the CTNN retains its full rank because its copresheaf edge-transport does sustain it (Chapter 7, §7.1.2). The high rank reported here is thus not incidental—it is the structural signature of the mechanism that makes message passing worth its cost.

Table 6.5: PINN–CTNN coupling diagnostics (CTNN ansatz). $\langle\cos\rangle$: mean cosine between ∇f and $\Delta\mathbf{x}$. Aligned: fraction of configs with $\nabla f \cdot \Delta\mathbf{x} > 0$. $|\delta f|/\sigma_f$: BF-induced change in PINN output relative to its standard deviation. →NN: fraction of particles displaced toward their nearest neighbour (0.5 = isotropic). Only $N \geq 6$ shown; N=2 displacements are near-zero for most ω.

N	ω	$\langle\cos\rangle$	Aligned	$\lvert\delta f\rvert/\sigma_f$	→NN
6	0.001	0.14	0.63	0.03	0.43
6	0.01	0.18	0.77	0.05	0.28
6	0.1	0.49	0.99	0.25	0.16
6	0.5	0.62	1.00	0.33	0.19
6	1.0	0.77	1.00	0.41	0.18
12	0.001	−0.48	0.01	0.03	0.61
12	0.01	0.22	0.82	0.05	0.29
12	0.1	0.56	1.00	0.39	0.12
12	0.5	0.65	1.00	0.49	0.20
12	1.0	0.75	1.00	0.55	0.21
20	0.001	0.32	0.97	0.06	0.44
20	0.01	0.22	0.89	0.07	0.45
20	0.1	0.39	1.00	0.25	0.17
20	0.5	0.65	1.00	0.58	0.22
20	1.0	0.78	1.00	0.69	0.25

Yet the CKA between correlator features with and without backflow exceeds 0.99 in nearly all cases (and exceeds 0.979 everywhere). The correlator manifold is almost unperturbed by the displacement field. This establishes a clean separation of concerns: the CTNN operates in the orthogonal complement of the correlator's feature space, providing configuration-dependent corrections that the globally organised correlator does not and need not represent.

6.2.3 PINN–CTNN coupling: from independence to cooperation

The previous subsection established that backflow leaves the correlator manifold intact. But are the PINN and CTNN *functionally* coupled—does the CTNN's correction depend on what the PINN computes? We measure this through three diagnostics: (i) the mean cosine similarity $\langle\cos(\nabla_{\mathbf{x}} f,\, \Delta\mathbf{x})\rangle$ between the PINN's gradient in configuration space and the CTNN displacement (gradient alignment implies the CTNN pushes along directions the PINN is sensitive to); (ii) the fraction of configurations where $\Delta\mathbf{x}$ is aligned with ∇f; and (iii) $|\delta f|/\sigma_f$, where $\delta f = f(\tilde{R} + \Delta\mathbf{x}) - f(\tilde{R})$ is the change in the correlator output when it is evaluated on the backflow-shifted coordinates, in units of the correlator's own standard deviation σ_f. Two cautions attach to this last quantity. It is a *counterfactual sensitivity*: in the production ansatz the correlator sees the unshifted $\tilde{R}$ (only the determinant sees $\tilde{R} + \Delta\mathbf{x}$; §3.5), so δf measures how much the correlator *would* respond to the displacement, not an actual term in the wavefunction. And it is a typical relative change, not a variance decomposition: $|\delta f|/\sigma_f = 0.7$ means the counterfactual change is 0.7 standard deviations, not that 70% of the variance is attributable to backflow. Table 6.5 reports these alongside the fraction of particles pushed toward their nearest neighbour.

A transition at $\omega \approx 0.1$. The data reveal a sharp regime change. At $\omega \leq 0.01$, the PINN and CTNN are weakly coupled: $\langle\cos\rangle \sim$ 0.1–0.2, the alignment fraction is well below unity, and backflow barely affects the PINN output ($|\delta f|/\sigma_f < 0.07$). In this regime the CTNN operates in a subspace largely orthogonal to the PINN's gradient—it modifies the wavefunction in directions that W_θ is insensitive to, consistent with its role being to adjust the Slater determinant's orbital positions rather than the correlator.

At $\omega \geq 0.1$, the picture changes qualitatively. The gradient cosine jumps to 0.5–0.8, the alignment fraction reaches 1.0 for $N \geq 6$, and the counterfactual output change reaches $|\delta f|/\sigma_f =$ 0.25–0.69 (i.e. up to $\sim$0.7 standard deviations of the correlator). Here the CTNN acts as a cooperative refinement: it pushes configurations along directions the PINN is maximally sensitive to, systematically correcting the correlator's residual errors. The transition is sharpest between $\omega = 0.01$ and $\omega = 0.1$, coinciding with the crossover in $\Gamma = V_{\text{int}}/T$ from $\Gamma > 10$ (interaction-dominated) to $\Gamma \sim$ 3–6 (moderate correlation).

Directional structure: correlation-hole deepening. The nearest-neighbour direction fraction provides a complementary signature. At $\omega \geq 0.1$ for $N \geq 6$, only 12–25% of particles are displaced toward their nearest neighbour. The CTNN preferentially pushes particles *apart*, actively deepening the correlation hole. This directional bias is strongest at N=12, ω=0.1 ($\rightarrow$NN = 0.12), exactly where the correlator effective rank peaks and the $\phi \leftrightarrow \psi$ transition occurs. At strong correlation ($\omega \leq 0.01$, large N), the fraction rises toward 0.5 (isotropic), because the large equilibrium inter-particle separations leave little room for further correlation-hole deepening.

6.2.4 Synthesis

The representation analysis supports a coherent picture of how the CTNN-augmented architecture partitions the variational problem across its components.

The **correlator head** W_θ forms a $\sim$1–3 dimensional manifold whose leading axis tracks the dominant global coordinate of the electron cloud. The identity of that coordinate shifts across regimes: pair correlations (ψ branch) dominate at moderate-to-weak coupling ($\omega \gtrsim 0.1$), while orbital structure (ϕ) or auxiliary features ($\mathbf{g}$) take over at strong coupling ($\omega \lesssim 0.01$). This $\phi \leftrightarrow \psi$ recomposition is the primary way the architecture adapts to changing physics; dimensionality expansion plays a negligible role.

The **CTNN backflow** provides a full-rank ($\sim 2N$), configuration-dependent displacement field that is implicitly regularised to $\sim$10% of the harmonic length. Despite its geometric complexity, it leaves the correlator manifold almost invariant (CKA > 0.98), confirming that it operates in a complementary subspace. At strong correlation, the CTNN and PINN function independently: backflow adjusts orbital positions via the Slater determinant while the correlator tracks global structure. At moderate-to-weak correlation, the two components become tightly coupled: the CTNN's displacements align with the PINN's gradient ($\langle\cos\rangle \sim$ 0.5–0.8), and the counterfactual output change reaches $|\delta f|/\sigma_f$ up to 0.70 (about 0.7 standard deviations of the correlator). The transition between these regimes is sharp and coincides with the physical crossover at $\omega \approx 0.1$.

This division of labour is architecturally desirable. It means the base correlator can robustly learn compact collective coordinates across regimes, while backflow provides an optional, fine-grained correction channel whose character—independent adjustment versus cooperative refinement—adapts automatically to the strength of electron–electron correlation. These regularities are, once again, features of one well-trained solution rather than theorems about every possible one. What the companion chapter supplies is their invariant counterpart: the same division of labour priced in energy, where deleting the backflow's inter-particle messages costs as much as deleting the displacement field entirely (Chapter 7, §7.1.2).

6.3 Wigner–molecule crossover

In a finite, rotationally symmetric harmonic trap the appropriate picture for the strongly correlated limit is a *Wigner molecule*: electrons localize in *relative* coordinates—forming concentric rings with polygon-like angular correlations—while the entire pattern may rotate freely. Consequently, lab-frame densities can appear nearly uniform even when strong internal order is present [1, 6]. The following analysis combines one-body density visualization, label-invariant shell topology, rotation-registered angular order, shell radii and separation gaps, pair-statistics validation, and Lindemann-type disorder to establish a coherent, quantitative picture of the Wigner-to-liquid crossover across $N = 2, 6, 12, 20$ and $\omega \in \{1, 0.5, 0.1, 0.01, 0.001\}$.

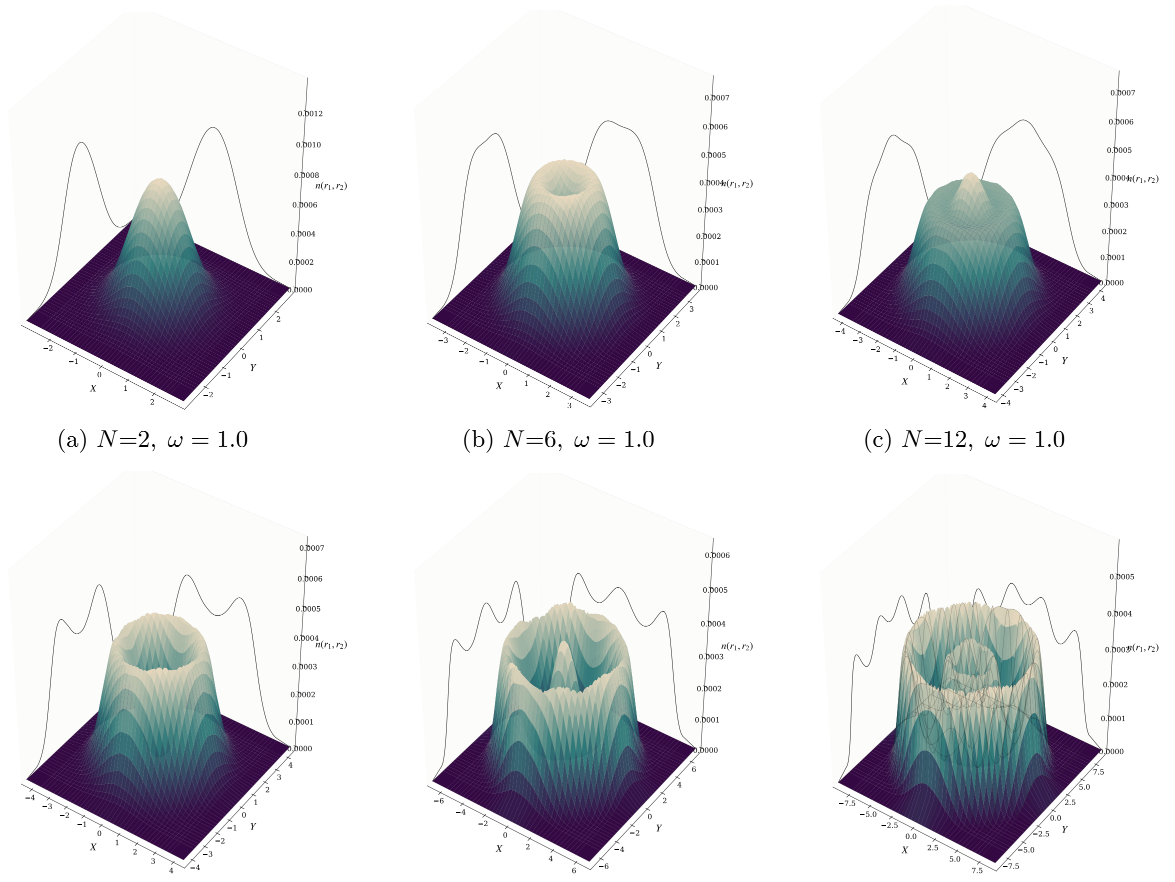
6.3.1 Observables and diagnostics

For each (N, ω) we draw $\mathbf{X} \sim |\Psi|^2$ and report (i) the pair distribution $g(r)$, (ii) the single-particle radial density $P(r) \propto r\,\rho(r)$, and summary statistics: r_{mode}, $\langle r \rangle$, σ_r, FWHM, quantiles q_{10}, q_{50}, q_{90}, and a localization ratio $\gamma = \sigma_r / r_{\mathrm{mode}}$ (smaller $\gamma \Rightarrow$ stiffer localization). For multi-electron systems we additionally monitor shell-resolved bond-orientational order $|\Phi_m|$ and a dimensionless Lindemann-type ratio for positional fluctuations on each ring [2].

Density parameter r_s. Following Egger *et al.* [1], we estimate the Wigner–Seitz parameter from the first maximum r^* of the spin-summed pair correlation, $r_s \equiv r^*/a_B^*$. The Fermi–liquid $\rightarrow$ Wigner–molecule crossover occurs near $r_s \simeq 4$ in parabolic dots; our weak-trap cases lie well beyond this threshold [1, 5].

6.3.2 One-body densities: visual crossover

Figures 6.2 and 6.3 compare one-body densities at strong confinement ($\omega = 1$) and weak confinement ($\omega = 10^{-3}$) for $N = 2, 6, 12, 20$. At $\omega = 1$ the density remains compact and the dominant change with N is progressive shell filling under the harmonic trap. At $\omega = 10^{-3}$ the density broadens dramatically and develops pronounced radial structure for all N, consistent with Coulomb-dominated localization and the onset of Wigner-molecule ordering.



(a) N=2, $\omega = 1.0$ (b) N=6, $\omega = 1.0$ (c) N=12, $\omega = 1.0$

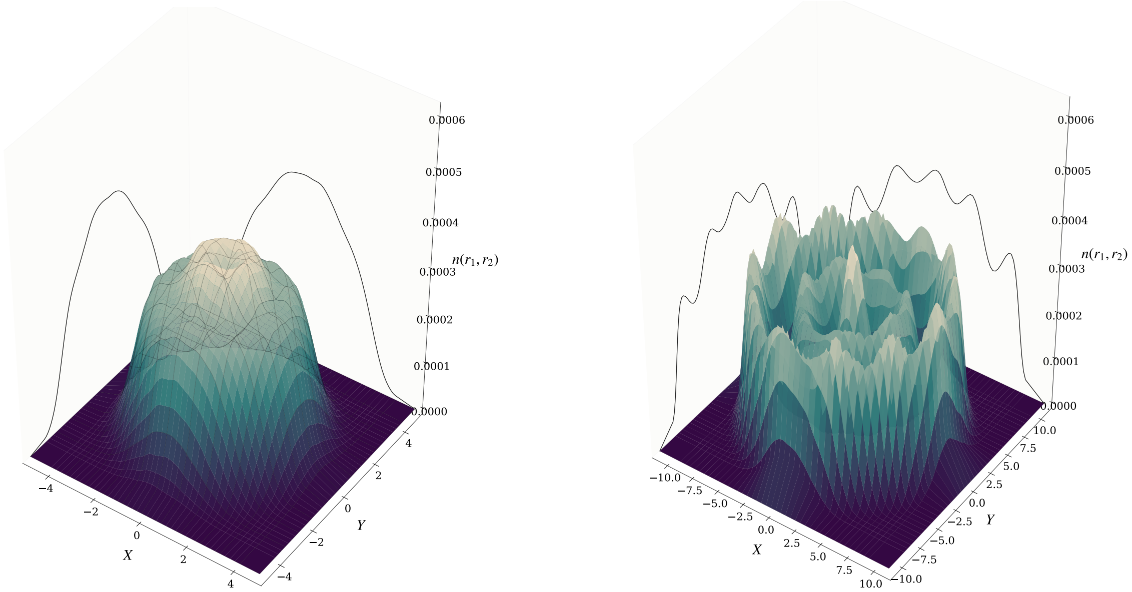
(d) N=2, $\omega = 0.001$ (e) N=6, $\omega = 0.001$ (f) N=12, $\omega = 0.001$

Figure 6.2: One-body densities for $N = 2, 6, 12$ at $\omega = 1.0$ (top) and $\omega = 0.001$ (bottom). As confinement weakens, the densities broaden and develop pronounced shell structures characteristic of Wigner molecules.

6.3.3 Energy-based crossover diagnostics

Log–log fits $E(\omega) \propto \omega^{\alpha}$ on the grid $\omega \in \{10^{-3}, 10^{-2}, 10^{-1}, 0.5, 1\}$ give $\alpha_T \simeq 1.00$, $\alpha_{V_{\mathrm{int}}} \simeq 0.66$, and $\alpha_{V_{\mathrm{trap}}} \simeq 0.81,\ 0.73,\ 0.72$ for $N = 2, 6, 12$ respectively. The kinetic energy thus scales essentially linearly with ω, while both potential terms grow sublinearly with nearly N-independent exponents. Across $\omega \in [1, 10^{-3}]$ the interaction-to-kinetic ratio $\Gamma = \langle V_{\mathrm{int}} \rangle / \langle T \rangle$ increases from $\Gamma \simeq 0.9 \to 8.7$ (N=2), $2.4 \to 25.4$ (N=6), and $3.6 \to 39.9$ (N=12), while T/V_{int} falls correspondingly (Fig. 6.4a).

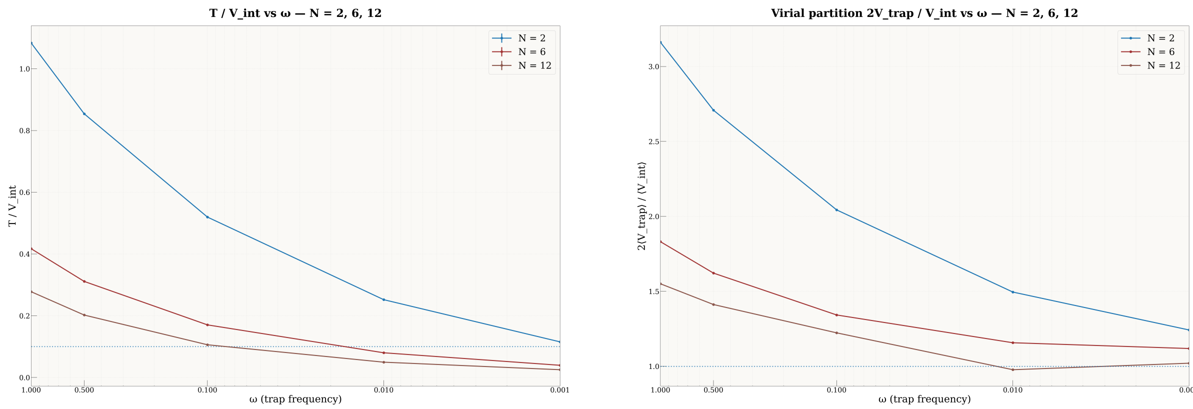
The partitioning between trap and interaction energies becomes quasi-classical at weak confinement. At $\omega = 10^{-3}$ we find $2\langle V_{\mathrm{trap}} \rangle / \langle V_{\mathrm{int}} \rangle \simeq 1.25,\ 1.12,\ 1.02$ for $N = 2, 6, 12$, so that N=12 already satisfies the classical virial condition $V_{\mathrm{int}} \approx 2V_{\mathrm{trap}}$ to within 2% (Fig. 6.4b). Using $V_{\mathrm{trap}} = \frac{1}{2}\omega^2 \sum_i \langle r_i^2 \rangle$ we extract an rms radius $r_{\mathrm{rms}} = \sqrt{2V_{\mathrm{trap}}/(N\omega^2)}$, which expands from ≈ 1.1–$2.0\,a_B^*$ at $\omega = 1$ to ≈ 70–$169\,a_B^*$ at $\omega = 10^{-3}$. Fitting $r_{\mathrm{rms}} \propto \omega^{\alpha_r}$ yields $\alpha_r \simeq -0.60$ to -0.64, i.e. $\langle r^2 \rangle \propto \omega^{-\beta}$ with $\beta \simeq 1.19$–1.28: the cloud expands *faster* than the non-interacting ω^{-1} scaling, consistent with interaction-driven swelling and shell formation.



(a) N=20, $\omega = 1.0$

(b) N=20, $\omega = 0.001$

Figure 6.3: One-body densities for N=20 at strong and weak confinement. The weak-trap case shows strong radial broadening and clear multi-shell organization, consistent with deep Wigner-molecule ordering.



(a) T/V_{int} vs. ω for N=2, 6, 12. The horizontal line marks T/V_{int}=0.1.

(b) $2V_{\text{trap}}/V_{\text{int}}$ vs. ω for N=2, 6, 12. The horizontal line marks the classical virial ratio $2V_{\text{trap}} = V_{\text{int}}$.

Figure 6.4: Energy-based Wigner diagnostics as functions of trap frequency. As ω decreases the dots become interaction dominated ($T/V_{\text{int}} \ll 1$) and the trap–interaction partition approaches the classical virial form, most clearly for N=12.

6.3.4 Radial localization and pair correlations

Table 6.6 summarizes the single-particle radial law for N=2 and Table 6.7 for N=6 across all five trap strengths. In both cases r_{mode} grows as a power law ($r_{\text{mode}} \sim \omega^{\alpha}$ with $\alpha \simeq -0.62$ for N=2, close to the classical two-body scaling $-2/3$) while the localization ratio γ shrinks, signaling stiffening of the single-particle shell.

Table 6.6: Radial diagnostics for N=2 across ω (Bohr units).

ω	r_{mode}	$\langle r \rangle$	σ_r	FWHM	q_{10}	q_{50}	q_{90}	γ
1.00	1.435	1.632	0.705	1.743	0.735	1.581	2.581	0.491
0.50	2.303	2.475	1.007	2.466	1.209	2.409	3.829	0.437
0.10	3.601	3.830	1.728	4.449	1.640	3.714	6.127	0.480
0.01	14.191	14.593	5.615	13.602	7.390	14.322	22.038	0.396
0.001	64.292	63.953	18.479	43.977	40.392	63.814	87.714	0.287

Table 6.7: Radial and angular diagnostics for N=6 across ω (Bohr; angles in radians).

ω	r_{mode}	σ_r	γ	σ(ring angles)	$\gamma_{r_{ij}}$
1.00	2.0907	0.9249	0.4424	0.7046	0.3538
0.50	3.0702	1.3648	0.4445	0.6943	0.3397
0.10	8.1686	3.5025	0.4288	0.6842	0.3149
0.01	33.1538	13.9094	0.4195	0.6497	0.2889
0.001	113.5542	47.8076	0.4210	0.6368	0.2664

At $\omega = 10^{-3}$, the N=2 pair has mode separation $r_{12} \approx 122$, Lindemann ratio $\gamma_{r_{12}} \approx 0.20$, and a relative-angle distribution sharply peaked near π ($\sigma(\pi - \Delta\phi) \approx 0.35$), establishing a dilute, strongly anti-aligned Wigner dimer. For N=6 the ring angular fluctuations and pair Lindemann both decrease monotonically, but γ plateaus near 0.42: the single shell stiffens internally even though the radial profile broadens due to the growing equilibrium radius.

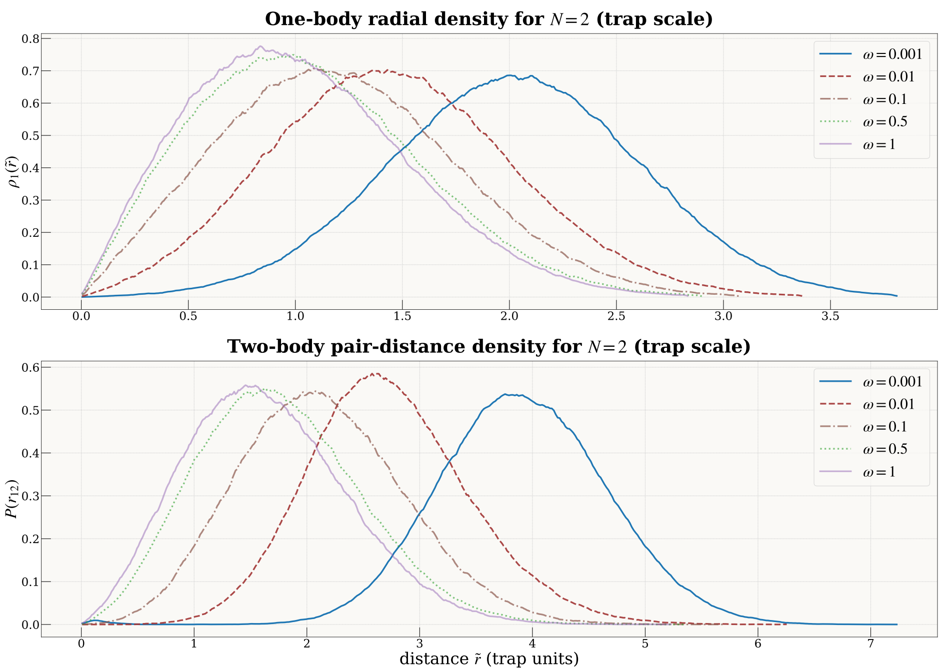


Figure 6.5: Two-electron radial densities $P(r)$ across ω.

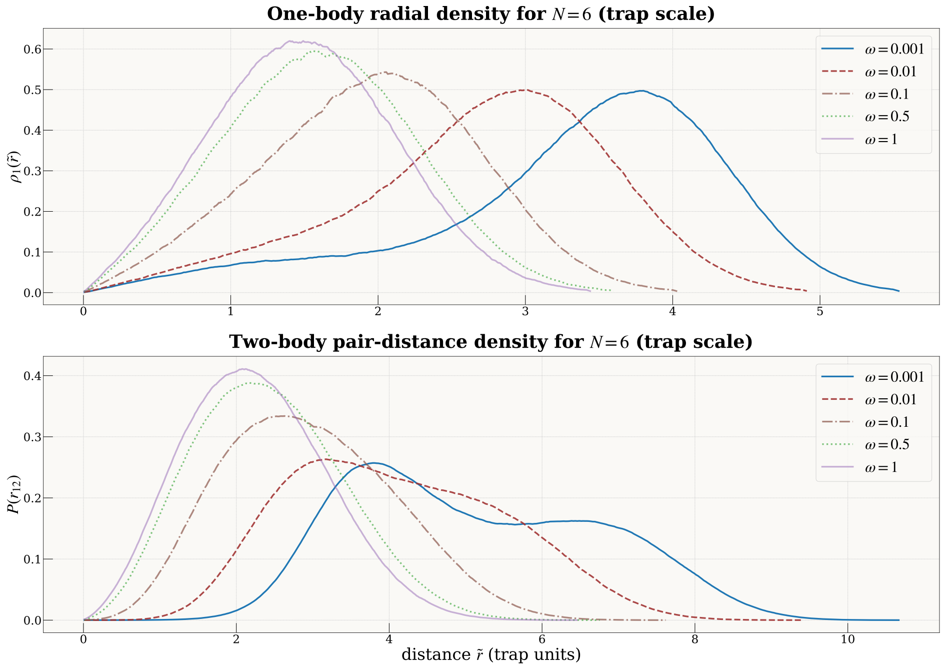


Figure 6.6: Six-electron radial densities $P(r)$ across ω.

6.3.5 Shell topology, angular order, and the rotating Wigner molecule

Because the Hamiltonian is rotationally symmetric, a Wigner molecule's internal angular order can only be observed after registering frames to a co-rotating reference. For a shell s with n_s particles we compute a rotation-registered bond-orientational parameter $|\Phi_{n_s}|$ and a corresponding Lindemann-type ratio quantifying angular stiffness. Lab-frame densities remain ring-like (the "rotating Wigner molecule" scenario), while the co-rotating frame reveals crystal-like polygon order [1, 6].

N=6: **single ring with sector mixing.** At $\omega = 0.01$ the system fluctuates between a $(1,5)$ sector (fraction ≈ 0.69) and a $(0,6)$ single-ring sector (fraction ≈ 0.31), with bond order $|\Phi_5| = 0.453 \pm 0.214$ and $|\Phi_6| = 0.466 \pm 0.206$ respectively. At $\omega = 10^{-3}$ the ring stiffens further: $(1,5)$ fraction ≈ 0.80, $(0,6) \approx 0.20$, with $|\Phi_5| = 0.597 \pm 0.224$ (Lind$\approx$0.225) and $|\Phi_6| = 0.663 \pm 0.183$ (Lind$\approx$0.205). Classically $(1,5)$ is the ground-state topology and $(0,6)$ a proximate competitor; the observed quantum mixture mirrors this hierarchy [3, 4]. In both weak-trap cases the global $g(r)$ is reproduced to numerical precision by the appropriate inner–inner / inner–outer / outer–outer (II/IO/OO) mixtures.

N=12: **two shells and commensurability.** At $\omega = 0.01$ two-shell formation is already frequent (fraction 0.761 at radial-gap threshold τ=3.0), distributed across sectors $(1,11) = 32.6\%$, $(2,10) = 25.7\%$, $(3,9) = 13.7\%$. At $\omega = 10^{-3}$ shelling becomes ubiquitous ($f_{2\text{shell}} \approx 0.99$–$0.92$ for $\tau = 2.0$–3.0) and the *commensurate* $(3,9)$ split emerges as modal (43.3% of quality-cut frames), with strong bond order on both rings: $|\Phi_3| = 0.687 \pm 0.229$ (Lind$_{\text{in}} = 0.223$), $|\Phi_9| = 0.526 \pm 0.194$ (Lind$_{\text{out}} = 0.224$). A shell-resolved $g(r)$ reconstruction (II/IO/OO mixture with combinatorial weights 0.045:0.409:0.545) reproduces the global $g(r)$ with cosine similarity 0.9982 (Fig. 6.7), confirming that the observed structure is geometric rather than a sampling artifact.

N=20: **three-shell molecule and rapid topology collapse.** At $\omega = 10^{-3}$ the dominant shell topology is $(1,7,12)$—a clear three-shell Wigner molecule with widely separated rings at radii $r_0 \approx 38$, $r_1 \approx 132$, $r_2 \approx 247$ (gaps ≈ 94 and ≈ 115). Rotation-registered angular order is strong ($\tilde{C}_{\text{global}} \approx 0.79$), while the lab-frame pinning metric remains small ($C_{\text{global}} \approx 0.06$), as expected for a freely rotating molecule. Lindemann disorder on the two occupied outer shells is low: $L_{7,\text{reg}} \approx 0.35$ and $L_{12,\text{reg}} \approx 0.27$.

Already at $\omega = 10^{-2}$, the three-shell pattern collapses to $(1,19)$: the shell radii contract by an order of magnitude ($r_0 \approx 10$, $r_1 \approx 45$), the registered angular order drops to $\tilde{C}_{\text{global}} \approx 0.20$, and the outer-shell Lindemann rises to ≈ 0.57. At $\omega = 1$ the same $(1,19)$ topology persists with compact radii ($r_0 \approx 0.55$,

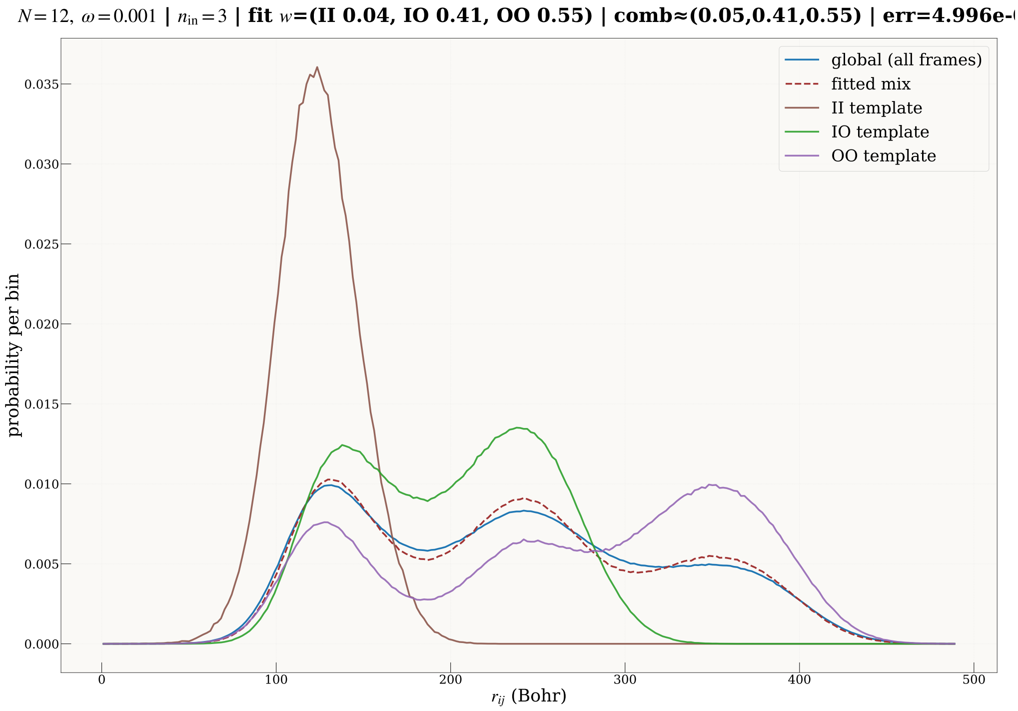


Figure 6.7: Shell-resolved reconstruction of $g(r)$ at $\omega = 10^{-3}$ for N=12. The global pair distribution is reproduced by a convex mixture of inner–inner (II), inner–outer (IO) and outer–outer (OO) histograms with weights set by the observed $(n_{\text{in}}, n_{\text{out}})$ statistics.

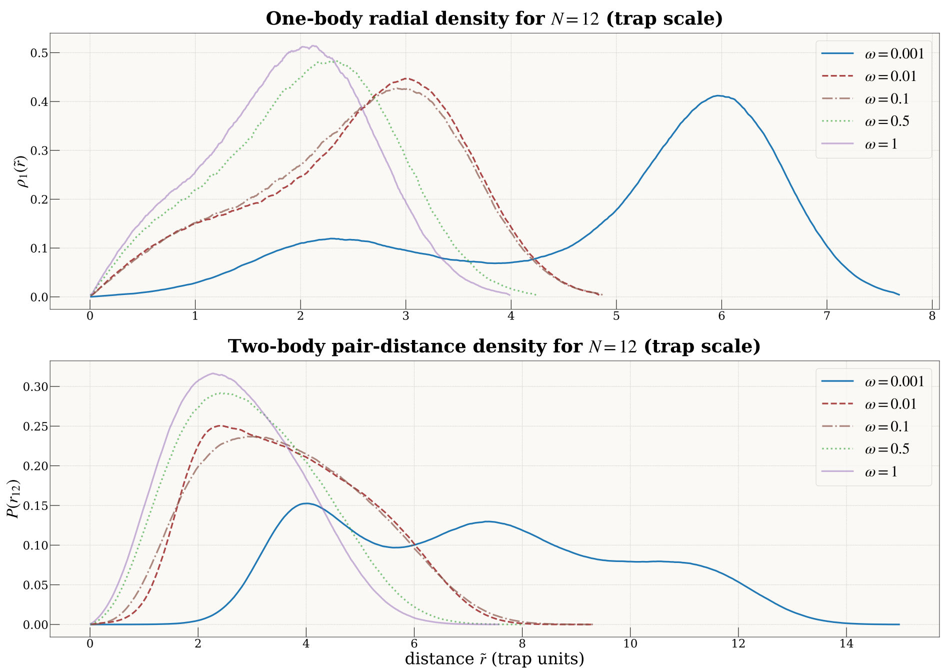


Figure 6.8: Twelve-electron radial densities $P(r)$ across ω.

$r_1 \approx 2.3$), negligible internal order ($\tilde{C}_{\text{global}} \approx 0.20$), and high disorder ($L_{\text{out}} \approx 0.69$). The sharpness of this transition—a single decade in ω separates a stiff three-shell crystal from a disordered single-shell state—is the most dramatic structural collapse observed in our system-size series.

6.3.6 Synthesis: the Wigner–molecule crossover

Across $N = 2, 6, 12, 20$ we observe a consistent, internally validated structural evolution as ω decreases from 1 to 10^{-3}:

1. **One-body densities** show dramatic spatial expansion and ring formation at $\omega = 10^{-3}$ (Figs. 6.2, 6.3).
2. **Shell topology** reveals discrete finite-N Wigner patterns at $\omega = 10^{-3}$—$(1,5)$ for N=6, $(3,9)$ for N=12, $(1,7,12)$ for N=20—consistent with classical ground-state analyses [3, 4]. Sharp topology collapses occur between 10^{-3} and 10^{-2} for the larger systems.
3. **Shell radii and separation gaps** provide a quantitative real-space counterpart: shells are widely separated and stable in the Wigner regime, while both radii and gaps shrink rapidly with increasing ω.
4. **Rotation-registered angular order** demonstrates strong polygon-like internal localization (including multi-shell ordering for N=12 and N=20). Lab-frame measures remain comparatively small, consistent with free global rotation of the molecule.
5. **Lindemann disorder** rises strongly with ω—independently confirming progressive melting—while the Wigner point supports simultaneously stiff shells (e.g. $L_{\text{out}} \approx 0.22$–$0.27$ for N=12, 20).
6. **Energy ratios** show the interaction-to-kinetic ratio Γ growing to ~ 40 for N=12 at $\omega = 10^{-3}$, and the virial condition $V_{\text{int}} \approx 2V_{\text{trap}}$ approaching its classical value.

The commensurate $(3,9)$ sector for N=12 aligns with known classical ground states, but with finite weight in competing $(1,11)$ and $(2,10)$ sectors reflecting quantum rotational mixing [5, 6]. Our sector-resolved ring-order parameters $|\Phi_m|$ are extracted *within* a fully variational quantum ansatz and measured conditionally on shell occupancy, which—to our knowledge—has not been reported before for $N \geq 12$ parabolic dots. The N=20 results extend this picture to three concentric shells and demonstrate that topology collapse, angular-order loss, and Lindemann melting all coincide within the same narrow ω window, providing a sharp quantitative trajectory from localized Wigner molecule to the denser, more weakly ordered liquid state [1, 3–5].

From physics back to method. These structural facts are also what make part of the method possible. The shell radius that grows as $\omega^{-2/3}$ and the angular order that sharpens as the trap weakens are exactly the quantities a sampler must respect to reach the Wigner regime without MCMC. Chapter 7 uses them directly: an origin-centred proposal aims where the wavefunction no longer has any mass, whereas a proposal built from the measured shell radius and angular order tracks it, and carries MCMC-free training far into the correlated regime (§7.4). What the trained model reveals about the physics is, in turn, what lets us train it more cheaply—the instrument and the method inform one another.

6.4 Collocation-assisted training

The results in Sections 6.1–6.3 rely on a two-phase protocol in which a residual-loss pretraining stage conditions the ansatz, followed by a stochastic-reconfiguration (SR) refinement phase that uses Markov-chain Monte Carlo (MCMC) sampling from $|\Psi|^2$ to optimise the variational energy. This section reports a complementary investigation: can an accurate many-body wavefunction be trained *without any MCMC sampling during training*, using instead a purely importance-sampled collocation approach?

The motivation is threefold. First, MCMC introduces autocorrelation, burn-in overhead, and mixing timescales that grow with particle number and correlation strength—particularly problematic at low ω where the wavefunction support is diffuse. Second, a collocation-only pipeline is embarrassingly parallel: each batch is drawn independently from a fixed analytic proposal, eliminating the sequential dependence inherent in MCMC chains. Third, collocation exposes the optimisation geometry more directly, making it a useful laboratory for understanding which architectural and sampling choices are load-bearing.

Two ways of posing the collocation problem must be separated at the outset, because they behave very differently. The *strong form* asks the wavefunction to satisfy the local Schrödinger equation pointwise: it minimises the variance of the local energy E_L, whose evaluation requires the Laplacian. The *weak form* instead uses a score-function (REINFORCE) gradient in which the Laplacian enters only as a reward and is never differentiated through. The distinction is not cosmetic. Appendix C shows that the strong form meets a structural obstruction as soon as it is asked to train backflow: the learning signal lives at the nodal surface, but E_L diverges there for any imperfect node, so no combination of sampling, loss, or preconditioning breaks an accuracy floor of $\sim$0.3%—and that floor is set by the Jastrow alone, with the backflow contributing almost nothing. This is a property of the strong-form objective, not of the method as a whole; it is precisely why the experiments in this section use the weak form throughout, and why the $\sim$0.3% number should not be read as the accuracy of collocation training.

Read in that light, the finding is clean. The *weak-form* objective, on the same Slater–Jastrow–backflow ansatz, recovers the variational energy to within DMC uncertainty at strong confinement and to a few tenths of a percent of our variational gold standard well into the correlated regime, for $N \in \{6, 12\}$; N=20 remains harder. And in the deep-Wigner limit a physics-informed proposal carries the same MCMC-free objective all the way to $\omega = 0.0035$ (Chapter 7, §7.4)—far past the point where a naive proposal collapses. What follows is an honest account of what makes the weak form work, and of where it, too, eventually breaks.

6.4.1 Training methodology

Objective function

The collocation objective replaces the VMC gradient estimator with a REINFORCE-style policy gradient that treats the local energy $E_L(\mathbf{R})$ as a reward signal:

$$\nabla_\theta \langle E \rangle \;=\; 2\,\mathbb{E}_{\mathbf{R}\sim|\Psi_\theta|^2}\big[\big(E_L(\mathbf{R}) - b\big)\,\nabla_\theta \log|\Psi_\theta(\mathbf{R})|\big], \tag{6.1}$$

where b is a variance-reducing baseline (running mean of E_L), and the local energy is

$$E_L(\mathbf{R}) = -\tfrac{1}{2}\big[\Delta \log \Psi(\mathbf{R}) + \|\nabla \log \Psi(\mathbf{R})\|^2\big] + V(\mathbf{R}). \tag{6.2}$$

The crucial implementation choice is that the Laplacian $\Delta \log \Psi$ enters the *reward* but is *not* differentiated through: gradients ∇_θ act only on the score function $\nabla_\theta \log|\Psi|$. This avoids the ill-conditioned second-derivative pathways identified in the catch-22 analysis (Appendix C) while preserving the full local-energy signal for optimisation.

Two robustness mechanisms stabilise the gradient:

1. **Local-energy clipping** (`clip_el`). Outlier local energies are clipped relative to a robust location–scale estimate (median and MAD), with a typical threshold of 5σ. This prevents rare pathological configurations (e.g. near nodes or coalescence points) from dominating the gradient.

2. **Reward quantile trimming** (`reward_qtrim`). The most extreme rewards are symmetrically trimmed before computing the policy gradient, reducing sensitivity to heavy-tailed noise.

An optional direct weak-form term $\beta\,\mathbb{E}[\tilde{E}]$ with $\tilde{E} = \frac{1}{2}\|\nabla \log \Psi\|^2 + V$ was also explored, but the historically successful runs set $\beta = 0$ (pure REINFORCE). A finite-difference collocation residual loss was additionally tested but proved more brittle and less competitive than the REINFORCE objective (see §6.4.4).

Proposal distribution and importance resampling

Because MCMC is avoided, the collocation points must be drawn from an analytic proposal distribution $q(\mathbf{R})$ and reweighted to approximate sampling from $|\Psi_\theta|^2$. The proposal is a mixture of isotropic Gaussians centered at the trap minimum:

$$q(\mathbf{R}) = \frac{1}{K}\sum_{k=1}^{K} \mathcal{N}\big(\mathbf{R};\, 0,\, \sigma_{f,k}^2/\omega \,\cdot\, I_{Nd}\big), \qquad \sigma_f \in \{0.8,\, 1.3,\, 2.0\}, \tag{6.3}$$

where the $1/\omega$ scaling ensures that the proposal width tracks the harmonic-oscillator length scale $\ell = 1/\sqrt{\omega}$. The three components provide coverage of the inner core (σ_f=0.8), the bulk (1.3), and the tails (2.0).

From this proposal, $M = m \times n_{\text{coll}}$ candidate points are drawn (typically m=8–16), and importance weights are computed:

$$w_i = \frac{|\Psi_\theta(\mathbf{R}_i)|^2}{q(\mathbf{R}_i)}, \qquad p_i = \frac{w_i}{\sum_j w_j}. \tag{6.4}$$

The final training batch of n_{coll} points is obtained by multinomial resampling according to $\{p_i\}$.

The quality of this procedure is monitored through the effective sample size (ESS):

$$\text{ESS} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2}, \tag{6.5}$$

which measures how many independent samples the importance-weighted batch is worth. When q has poor overlap with $|\Psi|^2$—as happens at large N or small ω—the ESS can collapse toward 1, concentrating all weight on a single sample and destroying the gradient signal.

Two additional weight-space controls were developed to manage this:

- **Weight tempering.** The importance weights are raised to a fractional power $\alpha < 1$ before normalisation, flattening the weight distribution and increasing the effective ESS at the cost of introducing a controlled bias.
- **Log-weight clipping.** The upper tail of $\log w$ is clipped at a specified quantile (e.g. the 99th percentile), preventing a single extreme sample from dominating the batch.

ESS-adaptive oversampling

For difficult regimes where the ESS is persistently low, an adaptive mechanism was introduced: if the achieved ESS falls below a specified floor (e.g. 0.1% of the batch), the oversampling multiplier m is increased in steps of 2 up to a maximum (typically 16), and the resampling is retried. This allows the system to automatically allocate more proposal samples when the overlap with $|\Psi|^2$ is poor, without manual tuning per regime.

Instability rollback

An instability-rollback mechanism monitors two quantities between successive checkpoints: the absolute relative error from the DMC reference, and the magnitude of energy jumps in units of the running standard deviation. If either exceeds a threshold, the model is reverted to the previous checkpoint and the learning rate is decayed by a factor (typically 0.85–0.92). This prevents catastrophic divergence in regimes where gradient noise is high, at the cost of slower progress.

Optimiser comparison: Adam vs. CG-SR

Two optimiser families were compared extensively. **Adam** (with separate learning rates for the Jastrow and backflow parameters, typically $\text{lr} = 5\times10^{-4}$ and $\text{lr}_{\text{jas}} = 5\times10^{-5}$) provided stable convergence across all regimes and was the most robust default.

Conjugate-gradient stochastic reconfiguration (CG-SR)—a natural-gradient method that preconditions the gradient with the inverse Fisher information matrix, solved approximately via conjugate gradients with Tikhonov damping—produced faster convergence and better final energies at moderate-to-strong confinement ($\omega \geq 0.1$). However, CG-SR was found to be catastrophically unstable for ultra-low ω (≤ 0.01): the Fisher matrix becomes extremely ill-conditioned in the weakly confined regime, causing parameter updates to diverge regardless of damping schedule (§6.4.4).

Cascade warm-start

A crucial strategy for difficult regimes was *cascade warm-starting*: networks trained at one confinement strength are used to initialise training at a neighboring, more difficult value. A typical cascade follows the sequence

$$\omega : 1.0 \;\to\; 0.5 \;\to\; 0.1 \;\to\; 0.01 \;\to\; 0.001, \tag{6.6}$$

where each stage inherits both the Jastrow and (when applicable) backflow parameters from the converged checkpoint of the previous stage. This exploits the observation that the geometric correlation patterns learned at higher ω transfer effectively to lower confinement, providing a far better starting point than random initialisation.

6.4.2 Architecture variants

Three ansatz families were explored within the collocation framework, all built on the same Slater–Jastrow core: $\Psi(\mathbf{R}) = \det[\phi_i(\mathbf{r}_j + \Delta\mathbf{r}_j)] \cdot e^{J(\mathbf{R})}$.

Jastrow-only (no backflow). The Jastrow factor is parameterised by a graph neural network (`CTNNJastrowVCycle`) with ∼25k parameters. It operates on node features (single-particle orbital values) and edge features (pair-wise distances with soft-core regularisation). The V-cycle variant uses a coarse-to-fine message-passing scheme. This family was the fastest to train and the most robust, serving both as a standalone ansatz and as an initialiser for backflow runs.

Coordinate backflow + Jastrow (BF). The `CTNNBackflowNet` adds a copresheaf-style message-passing network that predicts electron displacement vectors $\Delta\mathbf{r}_j$ fed into the Slater determinant. Edge features in the backflow network scale as $\mathcal{O}(N^2 \times h_{\mathrm{bf}})$ where h_{bf} is the hidden dimension, creating a memory bottleneck at large N. For N=6 and N=12, the default h_{bf}=128 was used (total ∼49k parameters); for N=20, reduced variants with $h_{\mathrm{bf}} \in \{48, 64\}$ were necessary. The BF+Jastrow family produced the best results at N=6 and N=12.

Neural Pfaffian. The Pfaffian ansatz replaces the Slater determinant with a more general antisymmetric form $\mathrm{Pf}[A_{ij}]$, which in principle offers richer nodal topology. Representative results: E=20.261 ± 0.004 and E=20.245 ± 0.005 at N=6, ω=1.0 (both ∼0.4% above DMC), compared to +0.009% for BF+Jastrow. The Pfaffian's greater expressivity was not exploitable within the current training pipeline; the simpler BF+Jastrow family consistently outperformed it.

6.4.3 Results

Table 6.8 presents the best energies achieved by collocation-assisted training across all (N, ω) combinations studied. All energies are evaluated by independent VMC with ∼30k MCMC samples after training (a "heavy evaluation"), ensuring that the reported numbers are not biased by the training-time importance sampling.

Table 6.8: Best ground-state energies from collocation-assisted training. "Multi-stage" denotes the best result from earlier campaigns using hand-tuned continuation chains; "Campaign (best)" denotes the best single result from the 30-run reliability campaign for N=6 (Table 6.9), and the best result across all independent training runs for N=12 (no dedicated reliability campaign was conducted for N=12). All values are VMC-evaluated (≥20k samples) at the best checkpoint. The reference column is the DMC energy where one exists and, for (N, ω) with no published DMC value (N=6 at $\omega \le 0.01$; see Table 6.2), our own converged variational (PINN+CTNN) energy; the corresponding % err is then a measure of internal consistency, not of absolute accuracy. The tabulated N=6 low-ω errors are the *reliability-campaign* bests and are deliberately conservative: the best single runs we observed reached +0.06–0.13% at $\omega \le 0.1$ (§6.4.5). All tabulated rows use the backflow+Jastrow (BF) architecture; the memory-limited N=20 case is discussed separately in Appendix D.4. Parentheses denote 1σ statistical uncertainty on the last digit(s).

N	ω	Reference	Multi-stage	Campaign (best)	% err	Arch
6	1.0	20.15932(8)	20.1613(23)	20.1620(56)	+0.013	BF
	0.5	11.78484(6)	11.7847(20)	11.7850(48)	+0.002	BF
	0.1	3.55385(5)	3.5571(4)	3.5632(7)	+0.263	BF
	0.01	0.69036	0.69141(8)	0.69200(1)	+0.237	BF
	0.001	0.140832	0.14118(1)	0.14118(1)	+0.250	BF
12	1.0	65.7001(1)	65.712(4)	65.706(4)	+0.009	BF
	0.5	39.1596(1)	39.171(4)	39.169(4)	+0.024	BF
	0.1	12.2698(1)	12.285(1)	12.2824(6)	+0.102	BF

N=6: near-reference accuracy across all ω. For six electrons, collocation training reaches DMC-level accuracy where DMC exists—at $\omega \geq 0.5$ the errors are $\lesssim 0.01\%$, inside the DMC statistical uncertainty. At the most extreme confinements, $\omega \in \{0.01, 0.001\}$, where the electrons are separated by $\sim$30–120 a_0^* (a few oscillator lengths) and no DMC reference is available, the energies stay within 0.1–0.25% of our converged variational gold standard; this is agreement between two independent training paradigms on the same ansatz, not a certificate of absolute accuracy. The best result at ω=1.0 is E=20.1613 ± 0.0023, an error of +0.009%, indistinguishable from DMC.

These results are achieved entirely without MCMC during training. The only MCMC appears in the final heavy evaluation, where it serves to certify rather than to train the wavefunction. The training itself uses 4096 importance-resampled collocation points per epoch with an oversampling factor of 8, requiring $\sim$32k proposal evaluations per gradient step—modest compared to the $\sim 3\times10^3$ MCMC samples per step in the SR-based approach.

The best N=6 result at ω=1.0 was obtained through a three-stage continuation chain (see §6.4.1): an initial BF+Jastrow joint REINFORCE run (900 epochs, lr=5×10^{-4}), followed by a low-LR continuation (750 epochs, lr=3×10^{-4}), and a final hard-focus stage (500 epochs, lr=2×10^{-4}). Each stage inherited the full checkpoint from the previous one. The monotonic improvement across stages—+0.08%, +0.05%, +0.009%—demonstrates that patient multi-stage polishing is effective within the collocation framework.

N=12: sub-0.03% at strong confinement. For twelve electrons, the BF+Jastrow collocation results track DMC closely at $\omega \geq 0.5$, with errors of +0.018% and +0.028% at ω=1.0 and 0.5 respectively. At ω=0.1 the error increases to +0.122%, reflecting the growing difficulty of importance sampling in 24 dimensions at weak confinement. The ESS at ω=0.1 is typically 5–15 per epoch (out of 4096 kept points), indicating that the Gaussian mixture proposal still maintains usable overlap with $|\Psi|^2$.

Attempts to extend to $\omega \leq 0.01$ at N=12 were unsuccessful: both direct starts and cascade warm-starts from ω=0.1 failed to converge, with the ESS collapsing to 1 and the rollback mechanism rejecting all gradient updates. This marks the boundary of the current collocation approach for N=12.

N=20: bounded by memory, not by principle. Twenty electrons in two dimensions push the collocation pipeline against a hardware wall rather than a conceptual one. The backflow's edge-feature tensor scales as $\mathcal{O}(N^2 \times h_{\mathrm{bf}})$, which at N=20 forces the hidden dimension, the collocation budget, or the oversampling factor down far enough that the gradient quality suffers—so much so that a Jastrow-only ansatz, freed of that cost, trains better than the backflow ansatz it should in principle beat. Because the limitation is one of memory and gradient noise rather than of the method, we report the N=20 collocation results, and the reversal that exposes the bottleneck, in Appendix D.4 rather than here.

6.4.4 What worked and what did not

The collocation programme explored a large space of methods. This section documents the key positive and negative findings.

Positive: pure REINFORCE over mixed objectives. Setting β=0 in the hybrid loss (pure REINFORCE, no direct weak-form term) was consistently more stable and produced better final energies than any nonzero β. The direct term $\tilde{E}=\frac{1}{2}\|\nabla \log \Psi\|^2+V$ provided a cheaper but noisier gradient signal; at β>0 it interfered with the REINFORCE gradient by pulling toward a local-kinetic minimum that did not correspond to the variational minimum.

Positive: cascade warm-start for low ω. Cold-start training at ω=0.01 never converged for $N\geq6$. Cascade initialisation from ω=0.1 made the ω=0.01 regime accessible, reducing the initial error from $\sim$30% (random init) to $\sim$5% (warm-started) and enabling subsequent optimisation to reach +0.15% within $\sim$1000 epochs.

Positive: ESS-adaptive oversampling + rollback. The combination of ESS-adaptive oversampling (automatically increasing the proposal budget when overlap is poor) and instability rollback (reverting to a good checkpoint when the energy spikes) was essential for the ultra-low-ω results. Without these mechanisms, training at ω=0.001 diverged within a few hundred epochs.

Negative: CG-SR at ultra-low ω. Natural-gradient preconditioning via CG-SR produced excellent results at $\omega \geq 0.1$ (fastest convergence, best final energies for N=6, 12). However, at $\omega \leq 0.01$ the Fisher information matrix becomes extremely ill-conditioned: the eigenvalues span many orders of magnitude because the wavefunction's sensitivity to parameters varies enormously across the diffuse, weakly confined state. CG-SR updates in this regime produced errors of +4–36% for N=6 at ω=0.01 and ω=0.001, compared to +0.15% and +0.24% from Adam with per-parameter adaptive learning rates. This demonstrates a sharp *optimiser regime boundary*: natural gradients are beneficial when the Fisher matrix is well-conditioned, but catastrophically wrong when it is not.

Negative: Langevin proposal refinement. To address the ESS catastrophe at high N, we implemented overdamped Langevin dynamics on the proposal samples: $\mathbf{R}' = \mathbf{R} + \varepsilon \nabla_{\mathbf{R}} \log |\Psi(\mathbf{R})|^2 + \sqrt{2\varepsilon}\, \boldsymbol{\eta}$, applied for K=10–20 steps before importance resampling. The intent was to push proposal samples toward high-$|\Psi|^2$ regions without running a full MCMC chain.

This approach was systematically worse than standard importance resampling in every configuration tested:

- N=20, ω=0.1: Langevin VMC error +153% vs. standard +5.7% (catastrophic divergence).
- N=20, ω=1.0: Langevin +5.2% vs. standard +1.5%.
- N=6, ω=0.001: Langevin +0.30% vs. Adam +0.24%.

The failure mechanism is that K=10–20 Langevin steps from a Gaussian starting point in Nd=40 dimensions do not equilibrate to $|\Psi|^2$. The resulting non-equilibrium sample distribution biases the REINFORCE gradient, which in turn pushes the wavefunction to concentrate further around the biased distribution—a positive-feedback loop that rapidly destroys the ansatz. Langevin refinement would require either much longer chains ($K \gtrsim 100$, negating the computational advantage over MCMC) or a Metropolis–Hastings acceptance step (reintroducing MCMC).

Negative: finite-difference residual loss. The FD collocation loss, which computes E_L via graph-safe finite differences and penalises its spread, was tested extensively. Representative BF results at N=6, ω=1.0: E=20.204 ± 0.002 (+0.22%), compared to +0.009% from REINFORCE. FD training was more sensitive to step-size h and noise in the Laplacian estimate, and never matched the REINFORCE objective for final accuracy.

Negative: Neural Pfaffian. Despite its richer antisymmetric structure, the Pfaffian ansatz underperformed the simpler BF+Jastrow family in all collocation experiments. The best Pfaffian result at N=6, ω=1.0 was E=20.245 ± 0.005 (+0.43%), more than 40× worse than BF+Jastrow. The Pfaffian's larger parameter space amplified the gradient noise from importance sampling, making it harder to optimise reliably.

Negative: backflow at N=20. Backflow training at N=20 was severely limited by memory. The edge-feature tensor scales as $\mathcal{O}(N^2 \times h_{\mathrm{bf}})$, requiring h_{bf} to be reduced from 128 to 48–64 for N=20. This reduced-capacity backflow achieved only +18% error at ω=1.0, compared to +1.45% for the Jastrow-only ansatz. The gradient signal at N=20 is too noisy for the additional backflow parameters to be trained effectively; the simpler Jastrow network with its smaller parameter count permits higher oversampling and better gradient quality.

6.4.5 Reliability analysis: the gradient-quality chain

The results in Table 6.8 report the *best* energies obtained by collocation training, selected from multiple runs on different seeds. A natural question is whether these results are *typical* or whether they reflect rare fortunate outcomes from a high-variance process. This subsection reports a systematic 30-run reliability campaign designed to answer that question, and reveals a simple causal chain that explains both the successes and failures of the MCMC-free collocation approach.

Experimental design

All runs use the BF+Jastrow ansatz (∼49k parameters) initialised from the same pretrained checkpoint. The grid is N=6 across $\omega \in \{1.0, 0.5, 0.1, 0.01, 0.001\}$ with three random seeds per cell (42, 137, 314), giving $5 \times 2 \times 3 = 30$ runs. Two training recipes are compared head-to-head:

- **Config A (baseline):** Adam optimiser with lr=5×10^{-4}, $\mathrm{lr}_{\mathrm{jas}}$=$5\times10^{-5}$, 4096 collocation points, 8× oversampling, local-energy clipping at 5σ, no reward normalisation, no rollback, fixed Gaussian mixture proposal (Eq. 6.3).
- **Config B (robust):** Identical to Config A plus four stabilisation mechanisms: 16× oversampling (doubling the proposal budget), reward normalisation (standardising the REINFORCE reward to zero mean and unit variance before the gradient step), instability rollback (reverting to the previous checkpoint when the energy jumps exceed 3σ, with learning-rate decay of 0.95), and an adaptive Gaussian mixture proposal (8 GMM components refitted every 30 epochs to track the evolving $|\Psi|^2$).

All runs train for 1200 epochs. Evaluation uses a 20 000-sample VMC probe every 50 epochs, with the best-VMC checkpoint restored for a final 20 000-sample IS evaluation.

Main results

Table 6.9 presents the complete results of the 30-run campaign. For each (ω, recipe) combination, three seeds provide a mean, standard deviation, and worst-case error.

Table 6.9: Reliability campaign: relative energy errors (% above DMC) for N=6 across five trap frequencies. Each cell shows the final IS-evaluated error for one (ω, recipe, seed) combination. Bold entries mark the best result at each ω. "Mean" and "CV" are the mean error and coefficient of variation across three seeds.

ω	Recipe	Seed 42	Seed 137	Seed 314	Mean	CV
1.0	Baseline	+0.073	+0.045	+0.060	0.059	0.24
	Robust	+0.032	**+0.013**	+0.105	0.050	0.97
0.5	Baseline	+0.106	+0.060	+0.108	0.091	0.30
	Robust	**+0.002**	+0.112	+0.047	0.054	1.04
0.1	Baseline	+0.419	+0.547	+0.427	0.464	0.15
	Robust	+0.325	**+0.263**	+0.270	0.286	0.12
0.01	Baseline	+0.500	+0.379	+0.746	0.542	0.34
	Robust	+0.289	+0.282	**+0.237**	0.269	0.10
0.001	Baseline	+3.876	+7.187	+4.733	5.265	0.33
	Robust	**+0.250**	+0.331	+0.263	0.281	0.15

Several patterns are immediately apparent. First, the robust recipe dramatically improves the low-ω regimes: at ω=0.001 the mean error drops from +5.3% (baseline) to +0.28% (robust), a factor of 19 improvement. Second, the best single result at every ω is a robust run, achieving +0.013% at ω=1.0, +0.002% at ω=0.5, +0.263% at ω=0.1, +0.237% at ω=0.01, and +0.250% at ω=0.001. Third, the robust recipe exhibits higher seed-to-seed variance (CV ∼ 1.0) at high ω but dramatically lower variance at low ω, suggesting that its stabilisation mechanisms have the most impact precisely where sampling is hardest.

The gradient-quality chain

The reliability campaign exposes a simple causal chain that governs the quality of collocation-trained wavefunctions:

$$\boxed{\text{proposal mismatch} \xrightarrow{\text{causes}} \hat{k}\text{ inflation} \xrightarrow{\text{causes}} \text{gradient noise} \xrightarrow{\text{causes}} \text{optimisation stochasticity} \xrightarrow{\text{causes}} \text{run-to-run vari}} \tag{6.7}$$

Each link in this chain can be verified quantitatively from the training logs. We now explain each component and the evidence supporting each causal link.

Step 1: proposal–target mismatch. As described in §6.4.1, the REINFORCE estimator in Eq. (6.1) requires samples from $|\Psi_\theta|^2$, but collocation draws from the analytic Gaussian mixture proposal $q(\mathbf{R})$ in Eq. (6.3) and reweights via importance sampling. The quality of this reweighting depends on how well q overlaps with $|\Psi_\theta|^2$.

For the harmonic quantum dot, $|\Psi|^2$ concentrates on a manifold whose geometry depends on ω: at ω=1.0, electrons occupy a compact cloud of radius of order the oscillator length $\ell = 1/\sqrt{\omega}$, and the three-component Gaussian mixture in Eq. (6.3) provides adequate coverage. At ω=0.001, the electron cloud extends to $\sim 100\, a_0^*$ (several oscillator lengths, $\ell \approx 31.6\, a_0^*$), the pair correlation function develops sharp shell structure (Section 6.3), and $|\Psi|^2$ concentrates on a thin correlated submanifold that the isotropic Gaussian proposal covers inefficiently.

The mismatch manifests directly in the importance weights. At ω=1.0, weights are relatively uniform: the effective sample size is typically ESS $\sim$ 5000–6000 out of $\sim$65k proposal samples (16× oversampling of 4096 points), and the Pareto shape parameter $\hat{k}$ of the weight tail is 0.4–0.5, indicating a well-behaved thin-tailed distribution. At ω=0.001, the ESS collapses to 1–6 (out of the same 65k proposal), and $\hat{k}$ rises above 1.0, entering the regime of infinite-variance importance weights.

Step 2: $\hat{k}$ inflation and gradient quality. The Pareto shape parameter $\hat{k}$ of the importance weight distribution, estimated via the PSIS diagnostic of Vehtari et al. (2024), provides a precise measure of gradient reliability. When $\hat{k} < 0.7$, the importance-weighted gradient estimator has finite variance and the central limit theorem applies normally. When $\hat{k} > 1.0$, the variance of the estimator is infinite and the gradient signal is dominated by a few extreme samples.

Table 6.10 summarises the average $\hat{k}$ across the campaign. The correspondence between $\hat{k}$ and final accuracy is striking.

Table 6.10: Average Pareto $\hat{k}$ of the importance weight distribution during training, by regime and recipe. Lower is better; $\hat{k} < 0.7$ is considered reliable, $\hat{k} > 1.0$ indicates infinite-variance weights.

	$\hat{k}$ (avg)		Mean error (%)	
ω	Baseline	Robust	Baseline	Robust
1.0	0.75	0.50	0.059	0.050
0.5	0.83	0.54	0.091	0.054
0.1	1.48	0.70	0.464	0.286
0.01	2.63	1.39	0.542	0.269
0.001	3.13	1.86	5.265	0.281

The robust recipe reduces $\hat{k}$ by approximately a factor of 2 at every ω. At ω=1.0 this takes $\hat{k}$ from 0.75 (borderline) to 0.50 (reliable), resulting in a modest improvement. At ω=0.1 it takes $\hat{k}$ from 1.48 (broken) to 0.70 (borderline), corresponding to the transition from mediocre (+0.46%) to marginal (+0.29%) accuracy. At ω=0.001 even the robust recipe has $\hat{k}$=1.86, but the adaptive proposal and rollback mechanisms prevent the catastrophic divergence seen in the baseline (+5.3%).

Step 3: from gradient noise to optimisation stochasticity. When $\hat{k}$ is large, most of the gradient signal comes from a few samples with extreme importance weights. Which samples are extreme depends on the random proposal draw at each epoch—a function of the random seed. Two seeds that happen to draw different extreme samples will follow different optimisation trajectories, even though they share the same architecture and hyperparameters.

This is not the usual seed dependence of stochastic gradient descent, where the law of large numbers ensures that gradient estimates are unbiased and the trajectory variance decreases with batch size. When $\hat{k} > 1$, the variance of the gradient estimator is *infinite*: increasing the batch size reduces the variance sublinearly (or not at all), and extreme samples can dominate the update indefinitely. The result is optimisation trajectories that diverge qualitatively across seeds, explaining the large run-to-run variance observed at low ω.

Step 4: the evaluation bottleneck. An additional subtlety is that the final energy evaluation itself uses importance sampling from the same proposal distribution. Even if the model has found an excellent wavefunction, a noisy evaluation can report a misleading energy.

The data expose this directly. Table 6.11 compares the VMC-best energy seen during training (a 20 000-sample probe evaluated by straightforward Monte Carlo from the model's own $|\Psi|^2$) with the final IS evaluation that uses the Gaussian proposal. At high ω where IS is reliable, the two estimates agree closely. At low ω, the gap between them grows, and its sign is unpredictable—the IS evaluation can overestimate or underestimate the true energy.

Table 6.11: Gap between VMC-best checkpoint energy (direct sampling from $|\Psi|^2$) and final IS evaluation, for selected runs. Positive gap means IS reports higher (worse) energy than the VMC probe saw.

ω	Recipe	Seed	VMC-best (%)	Final IS (%)	Gap (%)
1.0	Baseline	314	+0.001	+0.060	+0.059
1.0	Robust	137	+0.002	+0.013	+0.011
0.5	Robust	42	+0.037	+0.002	−0.035
0.1	Robust	137	+0.136	+0.263	+0.127
0.01	Baseline	42	+0.320	+0.500	+0.180
0.001	Baseline	314	+4.114	+4.733	+0.619

The first row is particularly instructive: this baseline run with seed 314 achieved a VMC-best energy of +0.001%—essentially exact—during training, but the IS evaluation reports +0.060%. The wavefunction is excellent; the evaluation is noisy. This demonstrates that some nominally "bad" runs may contain excellent wavefunctions measured poorly, and some nominally "good" results may benefit from evaluation noise in the favorable direction.

Summary of the chain. The causal chain in Eq. (6.7) provides a unified explanation for the full landscape of collocation results. At high ω, the proposal matches the target well ($\hat{k} \sim 0.5$), gradients are reliable, optimisation is stable, and all seeds converge to near-DMC accuracy. At low ω, the proposal-to-target mismatch grows, $\hat{k}$ inflates, gradients become noisy, optimisation becomes seed-dependent, and the evaluation itself becomes unreliable. The robust recipe acts on multiple links of this chain simultaneously: higher oversampling improves the raw ESS, reward normalisation stabilises the gradient scale, rollback prevents catastrophic divergence, and the adaptive proposal reduces the fundamental mismatch. No single mechanism is sufficient; each addresses a different link in the chain.

Chapter 7

Architecture, optimiser, and paradigm through the tangent kernel

Our ansatz is a closed-shell Slater determinant dressed by two learnable objects: a *Jastrow*-type neural correlator W_θ, which reshapes the amplitude, and a coordinate *backflow* $\Delta\mathbf{x}$, which reshapes the nodes. Each can be built in one of two ways—without message passing, encoding and pooling per-particle and pairwise features in a single feed-forward pass (as in a DeepSet), or as a copresheaf *message-passing* (CTNN) network that maintains explicit node and edge features and transports information between them through learned maps before a quantity is read out (the correlator iterates this exchange over a V-cycle; the backflow applies a single round). The previous chapter showed that the message-passing ansatz reaches diffusion Monte Carlo (DMC) quality wherever a DMC reference exists ($\omega \geq 0.1$), and matches our own variational gold standard in the strongly correlated regime below it, where no DMC benchmark is available. That settles *whether* the ansatz works and leaves the question worth a chapter: *why* message passing helps at all. We answer it for both learnable objects through a single lens, and then turn the same lens on the optimiser and on the training paradigm.

One idea, one lens. The thesis of this chapter is that message passing buys one thing—a **relational channel**, a way to condition each particle's contribution on where the others are—and that this single mechanism explains three superficially separate design choices. We read all three through the geometry of the per-sample log-derivative (score) matrix

$$O_{ki} \;=\; \frac{\partial \log |\Psi_{\boldsymbol{\theta}}(\boldsymbol{x}_k)|}{\partial \theta_i}, \qquad k = 1, \dots, B, \;\; i = 1, \dots, P, \tag{7.1}$$

whose Gram matrices are the quantum geometric tensor (QGT / Fisher metric) $S = O^\top O/B$ and the neural tangent kernel $K = OO^\top$. The central diagnostic is the *effective dimension* of the variational tangent space, the participation ratio of the QGT spectrum $\{\lambda_a\}$,

$$d_{\text{eff}}(S) \;=\; \frac{\left(\sum_a \lambda_a\right)^2}{\sum_a \lambda_a^2}, \tag{7.2}$$

which counts how many independent tangent directions carry Fisher weight at the solution, regardless of the parameter count P. We read it alongside the local-energy variance $\mathrm{Var}(E_L)$ (zero only for an exact eigenstate, hence a measure of wavefunction quality), the sampling efficiency (effective sample size ESS), and the condition number $\kappa(S)$.

The three questions. Q1 (architecture). What does a message-passing correlator/backflow learn that a separable one cannot? **Q2 (optimiser).** What does stochastic reconfiguration (SR, natural gradient) do to the update, and where does it beat Adam? **Q3 (paradigm).** What changes between $|\Psi|^2$-sampled VMC and importance-sampled collocation?

Discipline. Cross-architecture d_{eff} comparisons use a *common probe set* (all models scored on the same configurations, not on each model's own $|\Psi|^2$); we report only participation ratios converged in the sample count B; and we anchor the tooling on the analytic two-electron ground state, where the ansatz reproduces $E = 3.000127$ Ha ($+0.004\%$) with squared overlap 1.000000. Backflow comparisons are seeded

($\geq$ 2 seeds) and held at reference quality. One point of honesty about that reference is worth stating plainly, because every relative error in this chapter depends on it: an external DMC benchmark exists only for $\omega \geq 0.1$. In the strongly correlated regime below, we quote errors against our own converged variational energies (the $|\Psi|^2$-sampled SR result), and, in the deep-Wigner cascade of §7.4, cross-check against the physical energy-scaling law rather than any ground truth. Relative errors below $\omega = 0.1$ are therefore measures of internal consistency, not of absolute accuracy, and we read them as such.

Headline findings. **(Q1a, Jastrow)** A message-passing correlator compresses the ground state into $\sim 2\times$ fewer effective tangent directions than a separable DeepSet at weak coupling (1.40 ± 0.17 vs 3.25 ± 0.03 at $\omega = 1$, $\sim 11\sigma$, seeded), and those directions are the physical collective modes of H (breathing, correlation-hole, relative excitation) to $R^2 = 1.00$ at the $N = 2$ anchor. **(Q1b, backflow)** At the Wigner crystal the *conventional* backflow collapses to a single collective displacement mode (rank $10 \to 1$ at $N = 6$, $22 \to 1$ at $N = 12$) and its energy degrades to +0.5%, while the message-passing backflow keeps full rank $\sim N$ and stays at $\sim 0.02\%$; the gain there is Coulomb (not kinetic) correlation, and the inter-particle messages account for the entire backflow contribution. **(Q2)** SR whitens the tangent kernel; its value tracks $\kappa(S)$—negligible for the well-conditioned VMC metric, decisive for the ill-conditioned collocation objective. **(Q3)** VMC and collocation reach the *same state* when the importance sampling is healthy ($N = 2$: overlap > 0.997) and *diverge* when it is not ($N = 6$ Wigner: ESS $\to 0.11\%$ of the batch, overlap $\to 0$ under Adam). A cross-cutting caveat, itself a result: component-level quantities (rank, $|\Delta \mathbf{x}|$) are gauge-like; energy, overlap, and ablation are the invariants.

7.1 Q1 — Architecture: message passing supplies a relational channel

Message passing helps in both learnable objects, in two complementary ways that are the same statement about the tangent kernel: in the Jastrow it *compresses* the correlation into fewer effective directions; in the backflow it keeps the displacement from *collapsing* to a single direction. We take them in turn.

7.1.1 Q1a — In the Jastrow: message passing compresses the correlator

At $N = 6$, $\omega = 1$ the two Jastrow families are *energetically indistinguishable*: trained to DMC quality across a 20k–164k parameter ladder, every model lands within ±0.1% of the reference, and the best raw energy belongs to a 164k DeepSet—twice the CTNN parameter count. Total energy is a blunt discriminator; both have saturated the variational floor.

The difference is geometric. On a common probe set the CTNN reaches the ground state in $d_{\text{eff}} \approx 1.2$–$1.7$ (three seeds), while every DeepSet needs $d_{\text{eff}} \approx 2.9$–$3.6$—a factor of ~ 2. The gap is robust to the probe measure ($\lesssim 0.2$ whether scored on CTNN, DeepSet, or pooled points) and does not shrink as the DeepSet is given more capacity (the 164k DeepSet sits at 2.9, no lower than the 20k one). The compression is architectural, not a matter of size or sampling mass. It is also a well-posed comparison despite d_{eff} not being a reparameterization-invariant absolute dimension (§2.2.5): both families are compared under matched normalization conventions and a common probe measure, so the factor-of-two gap reflects how each architecture spends its tangent directions, not a change of coordinates.

One subtlety should be dispatched here, because the synthesis (§7.5) will caution that quantities split between the Jastrow and the backflow are gauge-like. That caution does not reach this comparison: it is between Jastrow families with *no backflow present*, so there is no correlation to divide between two substitutable channels. What remains is a comparison of how two architectures spend their tangent directions on the same ground state, and it is stable across seeds and across the probe measure—a property of the correlator, not an accounting artefact.

Seeded, at weak coupling the separation is $d_{\text{eff}} = 1.40 \pm 0.17$ (CTNN) vs 3.25 ± 0.03 (DeepSet), $\sim 11\sigma$. The gap is a *weak-coupling* effect: it closes toward the Wigner crystal, where both families reach ~ 3.7 but occupy *different* tangent subspaces, with the CTNN's carrying the lower $\text{Var}(E_L)$. A single-seed dimension "inversion" at the crystal, reported in an earlier draft, did *not* survive seeding and is retracted here as a worked cautionary result (see §7.5).

Naming the manifold. At the exact $N = 2$ anchor the leading tangent directions are not abstract: an operator decomposition identifies them, to $R^2 = 1.00$, as the physical collective modes of H—a breathing mode (∂_ω), a correlation-hole mode, and the first relative excitation. The message-passing correlator finds the ground state by moving along the handful of coordinates the Hamiltonian actually couples; the DeepSet spreads the same solution over more directions because it pools independently-encoded pairs rather than routing them through shared, iterated messages.

7.1.2 Q1b — In the backflow: message passing prevents rank collapse

The same relational channel is decisive, in a sharper form, in the backflow. Here we fix everything but the backflow architecture (conventional single-round messages vs. the copresheaf CTNN) and train both to a Woodbury-SR endpoint. At $\omega = 1$ both reach reference quality, but they diverge as the trap weakens (Table 7.1): the CTNN backflow holds $\sim 0.02\%$ error across all N, ω, while the conventional backflow degrades to $+0.5\%$ at $N = 6, \omega = 0.01$ and $\sim 6\times$ the CTNN error at $N = 20, \omega = 0.01$ ($+0.69\%$ vs. $+0.12\%$).

Table 7.1: Q1b. Relative energy error (%), seed-averaged.

	$N=6$		$N=12$		$N=20$	
ω	CTNN	conv	CTNN	conv	CTNN	conv
1.0	−0.03	+0.06	−0.01	+0.04	+0.06	+0.07
0.1	−0.05	+0.16	+0.01	+0.16	+0.02	+0.18
0.01	−0.04	+0.58	−0.01	+0.54	+0.12	+0.69

The energy gap has a sharp structural signature. Let the *backflow rank* be the participation ratio of the displacement field $\Delta\mathbf{x}$ (how many independent directions the backflow moves electrons along). At large ω both architectures use $\sim N$ modes. Toward Wigner the *conventional* backflow collapses to a single collective mode (rank $\to 1$), while the CTNN backflow retains full rank $\sim N$ (Table 7.2); the collapse sharpens with size ($10 \to 1$ at $N = 6$, $22 \to 1$ at $N = 12$, $35 \to 1$ at $N = 20$). The tangent dimension tells the same story from the other side: toward Wigner d_{eff} grows for CTNN ($\sim 4 \to 10$) and collapses for the conventional backflow ($\sim 6 \to 1$). The collapse to *exactly* rank 1 is seed-robust at every size, and the CTNN backflow holds full rank $\sim N$ throughout ($\sim 10/21/36$ at $N = 6/12/20$).

Table 7.2: Q1b. Backflow displacement rank (participation ratio), seed-averaged.

	$N=6$		$N=12$		$N=20$	
ω	CTNN	conv	CTNN	conv	CTNN	conv
1.0	8.8	10.4	20.8	22.2	36.9	35.4
0.1	7.7	10.4	18.2	21.7	33.8	37.6
0.01	9.8	**1.0**	20.8	**1.0**	35.2	**1.0**

Why, mechanistically. Ablating the backflow ($\Delta\mathbf{x} = 0$, sampler re-optimised) and splitting the recovered energy into kinetic and Coulomb parts gives two seed-stable facts. (i) A **kinetic→Coulomb crossover**: at $\omega = 1$ the backflow buys kinetic energy; at $\omega = 0.01$ the gain is entirely Coulomb ($\Delta V_{\text{Coul}} \approx +0.017$, $\Delta T \approx 0$). "Message passing buys kinetic energy" is a high-ω statement only. (ii) **At Wigner the transport maps *are* the backflow**: within the CTNN, zeroing the inter-particle transport maps ($\rho_{v\to e}, \rho_{e\to v}$) costs as much as deleting the whole displacement. Both architectures condition each particle on the relative coordinates $\mathbf{r}_{ij}$, so neither is blind to inter-particle structure; the difference is that the conventional backflow uses a single round of pairwise messages aggregated per particle, whereas the CTNN maintains and transports explicit edge state across the graph. Empirically the single-round backflow does not *sustain* a full-rank relative displacement at the crystal—it collapses to a single collective displacement mode (rank 1)—while the CTNN's copresheaf transport holds the full-rank response, and the ablation shows that this transport carries essentially the entire effect.

7.1.3 The unified reading of Q1

Q1a and Q1b are one statement about how relational information is *routed.* The baselines are not blind to inter-particle structure—the DeepSet pools independently-encoded pairs, and the conventional backflow aggregates a single round of pairwise messages—but neither maintains a persistent, transported edge representation: each pairwise term is computed and discarded. The copresheaf CTNN does maintain one, and moves it back and forth between nodes and edges through learned transport maps $(\rho_{v\to e}, \rho_{e\to v})$; the message-passing correlator iterates this exchange over a V-cycle, while the backflow applies a single such copresheaf round. Where a compact representation exists this *compresses* the correlation onto fewer tangent directions (Jastrow: smaller d_{eff}); where the physics demands a genuinely collective, full-rank response the single-round baseline instead collapses to a single collective displacement mode (backflow: rank 1), which the copresheaf transport avoids. Both effects are visible only in the tangent-kernel geometry, not in the total energy, until weak confinement makes the geometric deficit an energetic one.

7.2 Q2 — Optimiser: SR earns its keep where the metric is ill-conditioned

Stochastic reconfiguration preconditions the gradient by the inverse QGT, $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\, S^{-1}\nabla_{\boldsymbol{\theta}} E$; geometrically it *whitens* the tangent kernel. At $N = 2$ this is exact: the SR step aligns with the imaginary-time direction to numerical precision. The question is where whitening changes the outcome.

Fixing the ansatz and varying only the optimiser from a common stage-3 base: **for $|\Psi|^2$-VMC, SR and Adam are indistinguishable**—exact at $N = 2$ (var $\sim 10^{-9}$), and at $N = 6$ the energy gap $E_{\text{Adam}} - E_{\text{SR}}$ is $\lesssim 0.002$ across ω and both seeds. The VMC metric is well enough conditioned ($\kappa(S)$ moderate, and $|\Psi|^2$ sampling is self-preconditioning) that Adam already moves along the right directions. **For weak-form collocation, SR is essential**: there $E_{\text{Adam}} - E_{\text{SR}}$ is large and positive (e.g. $+2.9$ at $N = 6, \omega = 1$), because the importance weights make the objective badly conditioned and plain Adam fails to converge it.

The honest verdict is therefore not “SR wins” or “SR ties” but **SR matters precisely when the tangent metric is ill-conditioned**. Its value is a function of $\kappa(S)$, set by the ansatz and the sampling measure, not an intrinsic property of the optimiser—which is exactly why an earlier low-ω VMC sweep saw a modest SR advantage that never grew into a decisive win.

The claim needs one sharpening, because “more ill-conditioned $\Rightarrow$ more useful SR” cannot hold without bound: inverting S requires S itself to be *reliably estimated*. SR helps when the tangent metric is anisotropic but its empirical estimate is trustworthy; once the sampling that builds S collapses—as the importance weights do deep in the Wigner regime—the empirical S becomes noisy and rank-deficient, and preconditioning by its inverse amplifies that noise rather than the signal. This is exactly the behaviour seen at ultra-weak confinement, where CG-SR on the collocation objective becomes unstable (Chapter 6, §6.4.4): not a counterexample to the conditioning story but its completion—SR pays off in the window where the metric is both ill-conditioned and estimable, and fails once estimation itself breaks down.

One further boundary deserves to be drawn explicitly. The VMC equivalence of Adam and SR is established here at $N = 2$ and $N = 6$; we have not verified that it survives to $N = 12$ and $N = 20$, where the conditioning of the VMC metric could in principle worsen with dimension. The mechanism we propose—that $|\Psi|^2$ sampling keeps $\kappa(S)$ moderate and so lets Adam’s diagonal preconditioning suffice—is a prediction there, not yet a measurement, and testing it is the obvious extension.

7.3 Q3 — Paradigm: the sampling measure sets conditioning, variance, and reach

Between $|\Psi|^2$-VMC and importance-sampled collocation the wavefunction family is identical; only the measure that defines the loss changes. Three consequences, one statement.

$|\Psi|^2$ is self-preconditioning; the covering proposal is not. Sampling from $|\Psi|^2$ makes the estimator of S and of the gradient share the same measure, which conditions the update. A fixed covering proposal instead drives a monotone *ESS collapse* as N and $1/\omega$ grow: at $N = 6, \omega = 0.01$ the effective sample size falls to 0.11% of the batch (about 5 of 4096 points). When ESS collapses the gradient estimator is degenerate—a hard limit, not undertraining.

Same state, until the sampling loses the target. We test whether the two paradigms reach the same *state* (not merely the same energy) via the symmetric overlap $\mathcal{S}^2 = \langle \Psi_B/\Psi_A \rangle_{|\Psi_A|^2} \langle \Psi_A/\Psi_B \rangle_{|\Psi_B|^2}$. At $N = 2$ they are identical ($\mathcal{S}^2 > 0.997$ for every ω, both optimisers, both seeds): the paradigm is efficiency-only, collocation paying $\sim 5\times$ the variance for the same wavefunction. At $N = 6$ they *diverge*: under Adam $\mathcal{S}^2$ falls $0.85 \to 0.77 \to\approx 0$ from $\omega = 1$ to 0.01 (both seeds), tracking the ESS collapse; SR partially rescues it ($\mathcal{S}^2 \approx 0.95$) but does not close the gap. Weak-form collocation thus has a **domain of validity**—equivalent to VMC when the importance sampling is healthy, degrading as correlation strengthens. This couples Q3 to Q2: the paradigm that needs SR is the one that breaks down at scale. An adaptive proposal that tracks $|\Psi|^2$ is the natural remedy, pursued in §7.4.

Why the collocation energy readout is biased. For an exact eigenstate the local energy $E_L = \hat{H}\Psi/\Psi$ is constant over configuration space, so $\mathrm{Var}(E_L) \to 0$ as the ansatz approaches the ground state—the zero-variance principle, and the reason $\mathrm{Var}(E_L)$ is a sharp quality measure. That cancellation is a property of the *full* local energy, in which the kinetic (Laplacian) and potential terms combine to a constant; it is not a property of the Laplacian term alone. The collocation objective does *not* discard the Laplacian: it is a score-function (REINFORCE) estimator that *forms* the full E_L as a detached reward and differentiates only $\log|\Psi|$ (§6.4.1). What it gives up is the *measure*: the reward is averaged over the analytic proposal with importance weights (and, in the hardest regimes, tempered weights), so the importance-sampled energy *readout* is a biased estimate of $\langle E_L \rangle_{|\Psi|^2}$ and must not be used as the convergence criterion. The reliable criteria are $\mathrm{Var}(E_L)$ and the state overlap, both evaluated on $|\Psi|^2$ samples.

7.4 The collocation frontier: a physics-informed proposal for the Wigner regime

The ESS collapse of §7.3 is a limitation of the *proposal*, not of collocation itself, and it can be substantially pushed back. This section reports a physics-informed proposal that extends MCMC-free collocation deep into the Wigner regime, the mechanism that limits it, and the wall that remains.

7.4.1 The origin-centred proposal misses the Wigner shell

Every proposal in the standard pipeline is a mixture of *origin-centred* Gaussians $q(\mathbf{R}) = \frac{1}{K}\sum_k \mathcal{N}(\mathbf{R}; 0, \sigma_{f,k}^2/\omega)$. But as $\omega \to 0$ the electrons localise into a Wigner molecule on a *shell* far from the origin: the classical $N = 6$ minimum is the $(1,5)$ configuration—one central electron, five on a ring at radius $1.334\,\omega^{-2/3}$ [3, 4]—which in oscillator units sits at $1.96/2.87/4.22\,\ell$ for $\omega = 0.1/0.01/0.001$, moving outward as the trap weakens, past the reach of a width-2ℓ origin Gaussian. The proposal aims where $|\Psi|^2$ has no mass; hence the collapse.

7.4.2 The Wigner-ring proposal

We replace the origin mixture with a *Wigner-ring* proposal: a radial Gaussian at the shell radius times an angular distribution for the ring electrons, plus a central Gaussian. On trusted states this lifts the effective sample size by 15–47× over the origin mixture (Table 7.3), turning a collapsed sampler into a usable one—without any training, a cheap decisive test. Fitting the ring to $|\Psi|^2$ samples (the simplest adaptive form) doubles the ESS again; a $(1,5)+(0,6)$ mixture *hurts* at $N = 6$, since the state is $(1,5)$ and the extra channel dilutes the proposal mass.

Table 7.3: ESS (of 4096) against a trusted $N = 6, \omega = 0.01$ state. The origin mixture is already collapsed; the ring proposal covers the shell.

proposal	ESS
origin-Gaussian mixture (standard)	1.5
Wigner ring $(1,5)$, hand-tuned	22.5
Wigner ring $(1,5)$, fitted to $|\Psi|^2$	47.5

7.4.3 Angular crystallisation is the deep-Wigner wall

Why does the ring work at $\omega = 0.01$ but fail at $\omega = 0.001$? The molecule *crystallises angularly* as the trap weakens. The nearest-neighbour angular-gap standard deviation of the five ring electrons is $40.8°/33.3°/25.5°/6.7°$ at $\omega = 1/0.1/0.01/0.001$: a "ring liquid" at moderate coupling becomes an angular crystal (electrons pinned near 72° apart) at $\omega = 0.001$. A proposal with independent, uniform angles almost never lands on the crystalline configuration—so the angular distribution must crystallise too. This is captured MCMC-free by modelling the five gaps (which sum to 2π) as Dirichlet(α): an exact tractable density whose single concentration α interpolates liquid (α small) to crystal (α large), fitted to the measured gap variance. This diagnosis *explains* the collocation wall that is otherwise only observed.

7.4.4 An MCMC-free cascade reaches $\omega = 0.0035$

Combining the ring proposal (radial Gaussian × Dirichlet gaps) with importance oversampling, a *continuous* adaptive re-fit of the proposal to the current importance-weighted batch (every few steps, no MCMC), and a gentle ω-cascade seeded from a converged $\omega = 0.01$ state, MCMC-free collocation produces states that lie on the Wigner energy-scaling curve down to $\omega = 0.0035$—four cascade rungs, with heavy-VMC energies within 1–2% of the scaling estimate (Table 7.4), well past where the origin proposal collapses (ESS ~ 1 even at $\omega = 0.01$). The essential ingredient is the continuous re-fit: it keeps ESS healthy ($\sim 10^2$) *while* the energy descends, where a static or slowly-updated proposal collapses to ESS ~ 1 as the wavefunction moves away from it.

Table 7.4: MCMC-free Wigner-ring cascade, $N = 6$: heavy-VMC energy vs. the Wigner-scaling estimate $E(\omega) \approx 0.690\,(\omega/0.01)^{0.689}$. Good to $\omega = 0.0035$.

ω	E	scaling est.	ratio
0.0077	0.577	0.577	1.00
0.0059	0.481	0.480	1.00
0.0045	0.400	0.398	1.01
0.0035	0.340	0.335	1.02

7.4.5 The remaining wall: a fine-tuned Slater–Jastrow balance, not the sampler

Below $\omega = 0.0035$ the cascade becomes unstable, and—informatively—the instability is *non-monotonic* across proposal variants: wide-exploratory, tempered-tracking, and outward-anchored proposals each converge some rungs and diverge others. That pattern locates the obstacle not in the proposal but in the warm-start, and four transfer diagnostics (taking the converged $\omega = 0.0035$ rung down to $\omega = 0.0027$, a 23% step) pin down its mechanism precisely.

First, the frontier state is genuine: at its own $\omega = 0.0035$ the rung is a bona-fide Wigner molecule, with radial density peaked at 3.34ℓ (classical ring 3.42ℓ) and a *single* basin—initialising the sampler on the shell converges to the same 3.34ℓ. Transferred to $\omega = 0.0027$, however, the wavefunction collapses to $\sim 1\ell$, and two tests show *where* the collapse lives. It is not the sampler: initialising the Metropolis walkers directly on the classical shell (3.57ℓ) still relaxes inward to 1.1ℓ, so the collapsed density is a property of the ansatz, not of sampler initialisation. And it is not the backflow: its displacement is small ($|\Delta\mathbf{x}| \approx 0.13\ell$, radially inward) and essentially *preserved* across the transfer ($0.15\ell \to 0.13\ell$)—it scales correctly with ℓ but is far too small to bridge the 2.4ℓ gap between the Slater core ($\sim 1\ell$) and the Wigner shell.

The mechanism is a delicate cancellation. The bare harmonic-oscillator determinant suppresses the shell by $\exp(-\omega r^2/2) \sim \exp(-5.6)$ in amplitude ($\sim 10^{-5}$ in density); the Wigner peak exists only because the Jastrow amplifies it back by a comparable factor, with the backflow contributing a small inward nudge. A 23% change in ω perturbs both the Slater confinement and the Jastrow correlation length, the fine-tuned balance tips, and the density falls back to the non-interacting shell—from which the collocation re-fit cannot reliably climb out (hence the non-monotonic proposal behaviour). This rules out the natural fix of an ω-robust warm-start by coordinate rescaling: because the ansatz relaxes inward even when handed shell configurations, no coordinate transform can preserve the state. Reaching $\omega < 0.0035$ would require per-ω re-optimisation against a shell-*amplitude*-anchored objective, or a genuinely ℓ-covariant parametrisation—a distinct problem left to future work. The contribution here is concrete: a physics-informed proposal that carries MCMC-free collocation from the origin-proposal wall down to a verified Wigner molecule at $\omega = 0.0035$, and a precise account of the angular-crystallisation and Slater–Jastrow-balance effects that bound it.

7.5 Synthesis and discussion

The three axes are one geometry. **Architecture** sets the shape of the tangent space: message passing compresses it (Jastrow) and keeps it from collapsing (backflow), because it supplies the relational channel the physics needs. **Optimiser** is about inverting the metric on that space: SR pays off exactly when the metric is ill-conditioned. **Paradigm** is about which measure defines the metric and the gradient: $|\Psi|^2$ self-preconditions and stays on target, the covering proposal loses conditioning and eventually the state. The single cross-cutting discriminator, robust where dimension and rank are not, is $\mathrm{Var}(E_L)$—lower for the message-passing ansatz at every ω.

A gauge caveat, itself a result. Component-level quantities—backflow rank, displacement magnitude $|\Delta\mathbf{x}|$, per-component d_{eff}—are **gauge-like**. The Jastrow and backflow are substitutes, so how correlation is split between them is a nearly-flat, path-dependent direction of the loss; those quantities depend on the training route, not on the state. Every claim that reversed during this project was of that kind (the retracted d_{eff} inversion at the crystal; a mislabelled backflow; an artefactual rank collapse from joint-from-scratch training). The reliable measurements—energy, state overlap, and ablation (delete a component, price it in energy)—are invariant to the gauge, and the conclusions above rest only on those.

7.6 Conclusions

Message passing is not a uniformly "better network"; it buys one specific thing, a relational channel, whose value the tangent kernel makes precise. In the Jastrow that channel compresses the ground state onto the few collective modes the Hamiltonian couples; in the backflow it sustains a full-rank, relative displacement where a single-round-message network collapses to a single collective displacement mode and its energy fails—at the Wigner crystal, and by capturing Coulomb rather than kinetic correlation. The optimiser and paradigm choices around this ansatz are governed by the same geometry: SR is needed only for the ill-conditioned collocation objective, and collocation equals VMC only until its importance sampling loses the target at scale. What message passing, natural-gradient preconditioning, and $|\Psi|^2$ sampling have in common is that each keeps the variational optimisation aligned with the low-dimensional, physically-collective structure of the problem.

Part V

Discussion

Chapter 8

Discussion

8.1 Discussion of energy results

The energy benchmarks in Tables 6.1 and 6.2 show that the Slater–Jastrow PINN, with and without backflow, achieves DMC-level accuracy for two-electron dots and maintains sub-0.1% errors for $N \in \{6, 12\}$ across a wide range of trap frequencies. At the same time, it extrapolates in a controlled way into ultra-weak traps ($\omega = 10^{-3}$) where no DMC data are available.

Two electrons: architecture test and SR refinement. For N=2, residual-only training provides a clean test of the bare architecture. Across $\omega \in \{1.0, 0.5, 0.1, 0.01\}$, both the plain PINN and PINN+BF track DMC references at the $\sim 10^{-4}$ Ha level or better (Table 6.1), with deviations fully compatible with quoted uncertainties except at the shallowest pretraining point $\omega = 0.01$. There, residual-only PINN+BF remains variational but sits $\sim 3 \times 10^{-4}$ Ha above DMC, indicating a small residual bias from the PDE-based training alone.

The SR tail in Table 6.2 removes this bias. For N=2 and all $\omega \geq 0.01$, the SR-refined PINN+BF energies are statistically indistinguishable from DMC, with relative deviations $\lesssim 10^{-3}$ %. Thus, once the ansatz is well conditioned, a short SR phase suffices to close the last few 10^{-4} Ha of variational gap without changing the qualitative structure learned during residual training.

The comparison between PINN and PINN+BF in Table 6.1 also confirms that backflow is *not necessary* to reach DMC accuracy for two electrons. Energies with and without backflow agree within error bars at all traps, and the only visible effect of backflow in this case is a mild variance reduction and smoother optimisation in shallow traps. For N=2 the nodal structure is adequately captured by the Slater–Jastrow correlator alone.

Scaling to N=6 and N=12. The more demanding tests are the many-electron dots in Table 6.2. For $N \in \{6, 12\}$ and the traps with DMC references ($\omega \in \{0.1, 0.5, 1.0\}$), the final PINN+BF energies are consistently close to DMC and always slightly higher, as required by variationality. Relative deviations fall in the band 0.026–0.048%, corresponding to absolute errors of order 10^{-3}–10^{-2} Ha for the interacting cases, with no sign of catastrophic growth in error with either particle number or trap strength. The worst relative deviations occur at intermediate ω and remain safely below the percent level.

One subtlety is worth stating precisely, because the message-passing (CTNN) ansatz occasionally sits *below* the quoted DMC reference (for example N=12, ω=1: 65.6956 vs. 65.7001 Ha). This does not violate the variational principle, which bounds the energy only from below by the *exact* ground state, $E_{\mathrm{VMC}} \geq E_0$. The DMC benchmarks are *fixed-node* calculations and therefore carry their own nodal (systematic) bias; a neural ansatz with a better nodal surface can legitimately lie below a fixed-node DMC value obtained with poorer nodes. We therefore do not treat sitting above the published DMC number as a requirement, and where the CTNN falls below it we read this as a plausibly improved nodal structure rather than an error — though establishing that rigorously would require the DMC trial nodes and a matched uncertainty analysis, which we have not undertaken.

This is non-trivial for two reasons. First, the grid spans qualitatively different physical regimes: compact, kinetic-dominated states at $\omega = 1.0$ and swollen, interaction-dominated states at $\omega = 0.1$. Second, the same architecture and training protocol are used across all (N, ω), aside from trivial scaling in trap units. The fact that a single Slater–Jastrow+backflow ansatz maintains sub-0.1% accuracy across this grid suggests that the chosen combination of (i) trap-unit normalisation, (ii) soft-core pair features, and (iii) an explicit analytic cusp is robust with respect to both particle number and confinement.

Ultra-weak traps and extrapolation to $\omega = 10^{-3}$. At $\omega = 10^{-3}$ there are no DMC references in Table 6.2, so the PINN+BF values ($E \approx 0.0138\,\mathrm{Ha}$ for N=2, $0.1409\,\mathrm{Ha}$ for N=6, $0.5158\,\mathrm{Ha}$ for N=12) should be viewed as variational predictions rather than benchmarked numbers. Several internal checks, however, support their reliability.

First, the energies connect smoothly to the DMC-validated points at $\omega = 0.01$ under the log–log scaling analysis of T, V_{int} and V_{trap} in Sec. 6.3, with exponents and virial ratios consistent with classical expectations in the Wigner regime. Second, for N=2 at $\omega = 10^{-3}$, independent MCMC diagnostics (stable r_{12} and $P(r)$ histograms, smooth local-energy distributions, consistent block estimates) indicate that sampling noise is well below the quoted uncertainties in Table 6.2. Third, for N=6 and N=12, the structural and energy-based diagnostics in Sec. 6.3 confirm that the states lie on the Wigner side of the crossover (correct shell topologies, near-classical virial ratios), with no indication of a mis-set energy scale.

Within these caveats, the $\omega = 10^{-3}$ energies can be regarded as trusted anchors for the Wigner-molecule analysis in Sec. 6.3 and for the representation analysis in Sec. 6.2.

Efficiency and conditioning. From an algorithmic point of view, the benchmarks highlight the importance of conditioning. The SR tail uses only $\mathcal{O}(10^3)$ iterations with $\sim 3{\times}10^3$ samples each, i.e. a few million local-energy evaluations per (N, ω) point. This is modest compared to some earlier neural approaches to quantum dots that require orders of magnitude more samples to reach similar accuracy.

The key design choices are: (i) working in trap units, which keeps typical coordinates $\mathcal{O}(1)$ from tight to shallow traps; (ii) soft-core pair features with $ds/dr \to 0$ at coalescence, which control gradients and Laplacian spikes near $r_{ij} = 0$; and (iii) an explicit analytic cusp that imposes the correct short-range behaviour analytically rather than learning it numerically. Together, these keep local energies numerically tame even in ultra-weak traps where electrons are far apart and the wavefunction varies on long length scales.

Energetic role of backflow. Finally, the tables clarify the energetic role of backflow across regimes. For N=2, backflow is not needed to attain DMC-level accuracy. For $N \in \{6, 12\}$ at intermediate traps ($\omega \in \{0.1, 0.5\}$), backflow provides a modest but systematic improvement over residual-only baselines (not shown in Table 6.2), bringing final errors into the $\mathcal{O}(10^{-2}\text{–}10^{-1})$% band. In the deepest Wigner cases ($\omega = 10^{-3}$), however, the backflow correction becomes energetically tiny (Sec. 6.2): the Slater–Jastrow correlator already captures the dominant structure and the optimiser effectively *turns off* backflow. This regime dependence is mirrored in the representation analysis, where backflow appears as a high-dimensional, predominantly local correction channel that is only variationally active away from the ultra-classical limit.

8.2 Discussion of the Wigner–molecule regime

The structural diagnostics in Sec. 6.3 show that the weakest traps considered place the dots firmly on the Wigner side of the crossover, in a sense that goes beyond the loose label "strongly correlated". Three strands of evidence support this: (i) localization and energy partitioning, (ii) shell geometry and bond order, and (iii) topology statistics and reconstruction of $g(r)$.

A caveat on total spin. Our reference states are the closed-shell singlets of the non-interacting problem, with $N_\uparrow = N_\downarrow$ held fixed throughout. This is safe at weak coupling, but in the deep-Wigner regime the ground-state *spin* can change: for parabolic dots the interaction can drive spin transitions as it strengthens (for instance an $S = 0 \to S = 1$ change for $N = 6$ at strong coupling) [1]. Fixing $N_\uparrow = N_\downarrow$ does not by itself pin the total spin, since an $S = 1$ state also has an $S_z = 0$ component. We do not scan total spin here, so the weak-trap energies and structures below should be read as the lowest states we find *within the fixed spin-projection sector*, not as proven global-spin ground states. Determining the true spin ordering across the crossover, by comparing sectors of different S, is left to future work.

Localization and energy partitioning. On the one-body level, the weak-trap radial laws $P(r)$ are sharply peaked at large radii with relatively small fluctuations. For the two-electron dot at $\omega = 10^{-3}$ we find $r_{\text{mode}} \approx 64.3$, $\langle r \rangle \approx 64.0$ and $\gamma = \sigma_r / r_{\text{mode}} \approx 0.29$, with only $\sim 4.6 \times 10^{-3}$ of the single-particle mass inside $r \leq 0.25\, r_{\text{mode}}$ (Table 6.6). The pair distribution $p(r_{12})$ is peaked at $r_{12}^{\text{mode}} \approx 122$ with Lindemann ratio $\gamma_{r_{12}} \approx 0.20$, and the relative angle is sharply anti-aligned, $\sigma(\pi - \Delta\phi) \approx 0.35$, with $p(r_{12} \leq 0.25\, r_{12}^{\text{mode}}) \approx 2.6 \times 10^{-3}$. These numbers are characteristic of a dilute, essentially two-site Wigner dimer rather than a delocalized Fermi liquid.

For $N \in \{6, 12\}$ the same pattern appears in the aggregate. Both r_{mode} and r_{rms} grow roughly as $\omega^{-0.6}$, the localization ratio γ decreases only mildly with N and ω, and the extracted $\langle r^2 \rangle \propto \omega^{-\beta}$ exponents $\beta \simeq 1.2$–1.3 exceed the non-interacting value $\beta = 1$. At the same time, the energy partitioning approaches the classical virial limit: at $\omega = 10^{-3}$,

$$\frac{2\langle V_{\text{trap}} \rangle}{\langle V_{\text{int}} \rangle} \simeq 1.25,\ 1.12,\ 1.02 \quad \text{for} \quad N = 2, 6, 12,$$

with the N=12 dot satisfying $V_{\text{int}} \approx 2V_{\text{trap}}$ to within 2% (Fig. 6.4). Combined with the log–log scaling exponents $\alpha_T \approx 1$ and $\alpha_{V_{\text{int}}} \approx 0.66$, this indicates that the kinetic energy has become a small correction on top of a nearly classical potential-energy landscape, as expected in the Wigner-molecule limit.

Shell geometry and bond order. Beyond global localization, the weak-trap states exhibit robust shell structure with non-trivial bond order. For N=6 the dot retains a single pronounced shell across all $\omega \in [1, 10^{-3}]$; what evolves is the stiffness of angular correlations. The ring Lindemann index decreases from ~ 0.70 at $\omega = 1$ to ~ 0.64 at $\omega = 10^{-3}$, while the pair-distance Lindemann ratio drops from ~ 0.35 to ~ 0.27 (Table 6.7). After angular registration, the electrons form a rotating Wigner ring with clear sixfold order.

At $\omega = 10^{-3}$, the N=6 ensemble is dominated by two symmetry classes, $(1, 5)$ and $(0, 6)$, with fractions ≈ 0.801 and ≈ 0.199. Both show strong ring order, with $|\Phi_5| \approx 0.60$, $|\Phi_6| \approx 0.66$ and ring Lindemann indices ≈ 0.21. The shell itself never breaks; quantum fluctuations manifest primarily as rotations and occasional transitions between $(1, 5)$ and $(0, 6)$. This matches classical analyses where $(1, 5)$ is the ground-state geometry and $(0, 6)$ a proximate competitor.

For N=12 the structure is richer but equally ordered. At $\omega = 10^{-3}$ the two-shell fraction is very high for a broad range of radial-gap thresholds, confirming that shelling is robust rather than a threshold artifact. Within the two-shell subset the inner-ring occupancy histogram is sharply peaked at $(3, 9)$, with $(2, 10)$ and $(1, 11)$ as the leading subdominant topologies. On $(3, 9)$ frames we find $|\Phi_3| \approx 0.69$, $|\Phi_9| \approx 0.53$ and Lindemann indices ≈ 0.22 on both rings, i.e. clear threefold and ninefold bond order with small angular fluctuations. This is precisely the commensurate, multi-shell crystalline order expected for a twelve-electron Wigner molecule.

Topology statistics and reconstruction A more stringent test of the Wigner picture comes from shell-resolved reconstruction of the pair distribution. For N=12 at $\omega = 10^{-3}$ we partition two-shell configurations into II/IO/OO pairs, form their individual histograms and weight them by the observed topology mixture (0.045:0.409:0.545). The resulting synthetic $g(r)$ has cosine similarity 0.9982 to the global $g(r)$ (Fig. 6.7), i.e. is indistinguishable at our statistical resolution. The same holds for N=6, where the global $g(r)$ is reproduced essentially exactly by an OO/IO mixture of $(1, 5)$ and $(0, 6)$ sectors.

This has two important implications. First, the structure seen in $g(r)$ is unambiguously geometric (shells and topologies) rather than a subtle finite-size or sampling artifact: once the shell occupancies and topology statistics are fixed, the global $g(r)$ is determined. Second, it shows that the neural ansatz is not merely "getting the energy right" but also reproducing the detailed two-body correlation structure expected from classical and quantum studies of parabolic Coulomb clusters.

From dimer to multi-shell Wigner molecules. Viewed across $N = 2, 6, 12$ and decreasing ω, a coherent Wigner-molecule picture emerges. The two-electron dot evolves into a dilute, strongly anti-aligned dimer with negligible overlap near the origin. The six-electron dot forms a single, stiff Wigner ring whose lab-frame symmetry is restored by rotation but whose co-rotating frame exhibits clear sixfold order and a quantum mixture of classically known $(1, 5)$ and $(0, 6)$ topologies. The twelve-electron dot, finally, develops a robust two-shell structure with a commensurate $(3, 9)$ split as the modal configuration and topology fractions that track classical ground-state expectations.

All of this structure is obtained from samples drawn from the same variational neural ansatz whose energies match or closely track DMC benchmarks. This gives strong evidence that the states analysed in Sec. 6.2 are bona fide Wigner molecules, and not artefacts of an over-flexible wavefunction. The next section discusses how this rich real-space structure is compressed into a remarkably low-dimensional latent geometry by the learned correlator, and how backflow behaves across the same regimes.

8.3 Discussion of learned representations

The representation analysis in Sec. 6.2 reveals a surprisingly simple internal organization of the ansatz: the correlator W_θ lives on a low-dimensional manifold whose leading axes track global size and fluctuations of the electron cloud, while the BackflowNet implements a high-dimensional but predominantly local correction whose net energy becomes small in the deepest Wigner regimes. This mirrors the physical picture developed in Sec. 6.3. One caution, developed at length in Chapter 7, applies throughout what follows: because the Jastrow and the backflow are partial substitutes, quantities that live on a single component—effective ranks, displacement magnitudes, per-branch loadings—are gauge-like, fixed by the training route rather than by the state. They are faithful descriptions of *this* trained model, and we read them as such; the conclusions that must hold regardless of training route rest instead on the invariants (energy, overlap, and ablation).

Low-dimensional correlator manifold. Across all (N, ω), the entropy effective rank $r_{\text{eff}}(Z)$ of the correlator features remains close to 1 and rarely exceeds 2 (Table 6.3). In the most extreme case, N=2 at $\omega = 10^{-3}$, we find $r_{\text{eff}}(Z) \approx 1.00$ with head–PC1 correlation ≈ 0.998 and PC1 block power almost entirely in the pair branch ψ. PC ablations show that projecting onto the top-k PCs rapidly recovers the head, while random k-dimensional subspaces leave most of the error intact: for $(N, \omega) = (2, 10^{-3})$ the relative MAE drops from $\sim 4.5 \times 10^{-2}$ with k=1 to $\sim 2.0 \times 10^{-5}$ with k=12, whereas random projections of the same dimension do not achieve comparable reductions.

For $N \in \{6, 12\}$ and for both quantum and Wigner regimes, the picture is similar. Effective ranks cluster between ~ 1.3 and ~ 2.5, head–PC1 correlations remain $\gtrsim 0.95$, and a small number of principal directions captures the bulk of sensitivity. From a representation-learning perspective, the correlator behaves more like a low-dimensional order-parameter model than like a generic high-dimensional feature extractor.

What the leading axes encode. Linear probes from the correlator PCs to coarse physical summaries clarify what these axes represent. For N=2 (both at $\omega = 0.01$ and $\omega = 10^{-3}$), the first few PCs predict the mean radius and its variance essentially perfectly ($R^2(r_{\text{mean}}) \approx 1$, $R^2(r_{\text{var}}) \approx 1$), and even near-origin masses such as $\Pr(r \leq 0.25\, r_{\text{mode}})$ are accurately captured. For $N \in \{6, 12\}$, the leading PCs still track global size with $R^2(r_{\text{mean}})$ close to unity and radial fluctuations with $R^2(r_{\text{var}})$ in the range 0.4–0.85 across regimes, while shell contrast and angular order remain only weakly linearly encoded. This is consistent with the structural analysis: the correlator primarily encodes how big the dot is and how widely its shells fluctuate, whereas detailed bond order and topology require explicit angular registration and conditioning on shell occupancy.

Backflow as a local, high-dimensional correction. Backflow behaves very differently. For N=6 and N=12 the displacement field Δx has $r_{\text{eff}}(\Delta x)$ in the range $\sim$ 10–22 (Table 6.4), with relatively flat PCA spectra and gradual reconstruction under PC ablation. Linear probes from Δx PCs to global observables such as r_{mean}, r_{var} or shell contrast yield negligible R^2, indicating that the backflow field does not act as a simple global rescaling or shape parameter.

Near-field diagnostics of ΔE and $\|\Delta x\|^2$ support a predominantly local interpretation. At intermediate traps (e.g. N=6 or 12 at $\omega = 10^{-2}$), the lowest q% of configurations by $r_{\min}$ carry a disproportionately large fraction of the backflow energy correction: for N=12 at $\omega = 10^{-2}$ the lowest 5% bin accounts for about six times its fair share of ΔE while the corresponding share of $\|\Delta x\|^2$ remains close to unity. Thus, many spatial modes contribute to the displacement field, but the *energetic leverage* of backflow is concentrated in short-distance configurations where small nodal adjustments have a large impact on the local energy.

Deep Wigner limit: the backflow's net contribution shrinks. The extreme Wigner points provide a clean limit. For N=2 at $\omega = 10^{-3}$, the backflow correction to the energy is $\Delta E \sim 10^{-7}$ Ha with comparable uncertainty, head–PC1 alignment between no-BF and BF correlator features is essentially perfect (cosine = 1), and the ranks of Z and ΔZ are both ≈ 1. PC ablations of Δx show that even though $r_{\text{eff}}(\Delta x)$ can be nominally high, the field is effectively constant over the sampled manifold (PC ablation error $\sim 10^{-6}$), and near-field shares of both ΔE and $\|\Delta x\|^2$ are ≈ 1. In this regime, the optimiser has effectively *learned away* backflow: the Slater–Jastrow correlator alone is sufficient to represent the Wigner dimer.

For N=6 and N=12 at $\omega = 10^{-3}$ the pattern is similar, albeit less extreme. The correlator manifold remains low-rank with PC1 almost unchanged between no-BF and BF models, $r_{\text{eff}}(\Delta Z)$ stays close to $r_{\text{eff}}(Z)$, and the energetic effect of backflow is very small. Near-field shares of ΔE revert to values close to one. Once the system is deep in the Wigner-molecule regime and the static correlator has adjusted to the interaction-dominated landscape, there is simply little left for backflow to correct.

Two qualifications keep this from being over-read, both from the tangent-kernel analysis of Chapter 7. First, "idle" describes the small *net energy* of the message-passing backflow at these points, not the structure a backflow must supply: a *conventional* backflow does not fall idle so gracefully. It collapses to a single collective mode (rank $\to$ 1) and *loses* accuracy at the Wigner crystal, because the residual correlation there is relative and full-rank, and its single round of pairwise messages does not sustain it, whereas the CTNN's copresheaf transport does. Second, the ranks and displacement magnitudes quoted here are gauge-like: they record how this particular trained model divided the labour between Jastrow and backflow, and the statements that survive a change of training route are the energetic ones—the ablation price, and the gap between a conventional and a copresheaf message-passing backflow—not the component ranks themselves.

Regime dependence and architectural implications. Taken together, the representation results show a clear regime dependence:

- In *weakly to moderately correlated* traps (larger ω, smaller r_s), the correlator remains low-rank but backflow carries a non-trivial fraction of the energy correction, focused on near-field configurations. Here it is beneficial to allocate capacity to a reasonably wide BackflowNet with rich pairwise inputs, while keeping the correlator compact.
- In the *deep Wigner* regime (ultra-weak traps such as $\omega = 10^{-3}$), the correlator alone captures most of the global size/shelling and the relevant near-field structure, and the backflow's *net* energy contribution becomes small. Adding *width* to an already message-passing backflow buys little here; the architectural *kind* of backflow, however, still matters—a conventional single-round-message network collapses and loses accuracy exactly where the copresheaf CTNN holds full rank and does not (Chapter 7).

From a design perspective, this suggests that Slater–Jastrow PINNs with a small, smooth correlator and moderately wide backflow layers are a natural default for parabolic dots. The correlator behaves like a learned set of collective coordinates summarizing dot size and shell fluctuations, while backflow functions as a flexible local correction channel that is activated only where near-field physics is delicate. In the Wigner limit, the model naturally reduces to a static Wigner-molecule ansatz with low-rank latent geometry.

Connection to the physical picture. Finally, the learned representations dovetail with the structural analysis in Sec. 6.3. The same correlator that reproduces shell topologies, bond order and virial partitioning in the Wigner regime is found to live on a one- or two-dimensional manifold whose leading axis correlates almost perfectly with global size and radial spread. Backflow, in turn, is high-dimensional and local exactly in the regimes where shell structure is less rigid and close encounters are more frequent. This supports a physically meaningful decomposition: a small set of global "collective coordinates" captured by W_θ and a many-body, predominantly local correction channel implemented by BackflowNet. The success of this decomposition in capturing both energies and detailed Wigner-molecule structure suggests that similar architectures may be effective for other strongly correlated finite fermion systems.

8.4 Discussion of collocation-assisted training

The collocation experiments reported in Sec. 6.4 demonstrate that a purely importance-sampled, MCMC-free training pipeline can produce wavefunctions of near-DMC accuracy for moderate particle numbers and confinement strengths, while revealing systematic limitations at large N and ultra-weak confinement. This section interprets these findings, identifies the mechanisms behind successes and failures, and discusses their broader implications.

8.4.1 The central result: MCMC-free training to DMC accuracy

The most significant finding is that collocation training—using only analytic Gaussian proposals and importance resampling, with no Markov chain at any stage of optimisation—achieves +0.009% accuracy relative to DMC for N=6 at ω=1.0, and +0.018% for N=12 at the same confinement. These errors are comparable to the statistical uncertainty of the DMC references themselves and are achieved with a training budget that is modest by neural-network VMC standards.

This result is non-trivial because the prevailing assumption in the neural quantum state community is that accurate wavefunction optimisation requires equilibrated MCMC sampling from $|\Psi|^2$ throughout training. The collocation approach sidesteps this by treating the local energy as a *reward signal* in a REINFORCE policy gradient, drawing points from a fixed analytic distribution and reweighting them. The price is higher gradient variance (because the proposal only partially overlaps $|\Psi|^2$), but this is compensated by the absence of autocorrelation, burn-in, and mixing overhead.

For N=6 across all five confinement strengths tested ($\omega \in \{1.0, 0.5, 0.1, 0.01, 0.001\}$), the worst collocation error is +0.244% at ω=0.001, where electrons are separated by $\sim 100\, a_0^*$ (about four oscillator lengths) in a 12-dimensional configuration space. That a simple Gaussian mixture with three width components can maintain usable overlap with $|\Psi|^2$ at such extreme aspect ratios speaks to the effectiveness of the $1/\omega$ scaling in the proposal width and to the robustness of the importance-resampling pipeline. The Gaussian mixture is only the first proposal one might try, however. Where it does eventually fail—the origin-centred form aims where a deep Wigner molecule no longer has any mass—Chapter 7 replaces it with a proposal built from the Wigner shell itself and carries the same MCMC-free objective down to $\omega = 0.0035$, a regime the Gaussian mixture cannot reach.

8.4.2 Why REINFORCE succeeds where residual losses fail

The catch-22 analysis (Appendix C) identified a fundamental obstacle for direct PDE-residual training of the many-body Schrödinger equation: the backflow learning signal is concentrated near nodes and coalescence points where the local energy E_L diverges, but these are precisely the configurations where second-derivative pathways are ill-conditioned. Backpropagating through the Laplacian in such regions produces pathological gradients that destabilise training.

The REINFORCE formulation resolves this by construction. The Laplacian enters only through the *reward* (the local energy in Eq. 6.2), and gradients act exclusively on the score function $\nabla_\theta \log |\Psi|$. The stability is subtler than the score simply being bounded. For a *parameter-dependent* (backflow) node the score itself can diverge— $\nabla_\theta \log |\Psi| \supset (\nabla_\theta D)/D$ grows like $1/d_\theta$ as the learned determinant $D \to 0$ near its node. What matters is that this divergence is *integrable* under the $|\Psi|^2$ sampling measure, which weights it by $|\Psi|^2 \sim d_\theta^2$ (and the near-node points are further suppressed by the importance screening). REINFORCE avoids the *stronger*, non-integrable divergence of the strong-form residual, in which the Laplacian term $\nabla^2 D/D$ enters the gradient directly (Appendix C). Keeping the Laplacian in the *reward* but out of the *gradient flow* is what lets the collocation pipeline train backflow and Jastrow parameters stably in regimes where direct residual minimisation fails.

The superiority of pure REINFORCE (β=0) over the hybrid objective (β>0) has a related explanation. The direct weak-form term $\tilde{E} = \frac{1}{2}\|\nabla \log \Psi\|^2 + V$ is cheaper to evaluate (no Laplacian needed) but minimises a different functional: it drives the wavefunction toward low kinetic energy rather than low total energy. At nonzero β, the resulting gradient conflict between the two objectives manifests as oscillatory behaviour in the energy, with the REINFORCE term pulling toward the variational minimum and the direct term pulling toward a kinetic-energy minimum that is generally distinct.

8.4.3 The sampling bottleneck and its regime dependence

The central limitation of collocation training is that importance-sampling quality degrades systematically with particle number and with decreasing confinement. This degradation is governed by two competing effects:

1. **Dimensionality.** The configuration space is Nd-dimensional. A Gaussian proposal in Nd dimensions has an overlap with $|\Psi|^2$ that shrinks exponentially with Nd unless the proposal is well

matched to the target distribution. For N=6 (Nd=12), a three-component mixture is sufficient; for N=20 (Nd=40), it is marginal.

2. **Correlation structure.** At low ω, the many-body wavefunction develops strong inter-particle correlations (Wigner-molecule structure) that make $|\Psi|^2$ highly non-Gaussian: it concentrates on shell-like geometries with specific inter-particle spacings. An isotropic Gaussian proposal has no mechanism to capture this structure.

The ESS quantifies this overlap. Empirically, the ESS per epoch scales roughly as:

- N=6: ESS $\sim$ 50–500 at ω=1.0, $\sim$ 5–50 at ω=0.01, $\sim$ 2–10 at ω=0.001.
- N=12: ESS $\sim$ 30–200 at ω=1.0, $\sim$ 5–15 at ω=0.1, $\sim$ 1 at ω=0.01 (training fails).
- N=20: ESS $\sim$ 2–15 at ω=1.0, $\sim$ 1–5 at ω=0.1.

When the ESS approaches 1, the importance-weighted batch degenerates to a single effective sample, and the REINFORCE gradient becomes essentially a random vector. The ESS-adaptive oversampling mechanism delays this collapse but cannot prevent it: eventually, the number of proposal samples needed to maintain a given ESS grows exponentially, making the approach intractable.

This explains the systematic pattern in Table 6.8: accuracy degrades smoothly from excellent (+0.009%) at (N=6, ω=1.0) to poor (+5.68%) at (N=20, ω=0.1), following the trajectory of ESS collapse.

8.4.4 The optimiser regime boundary

The sharp failure of natural-gradient preconditioning (CG-SR) below $\omega \lesssim 0.01$ reveals an important structural feature of the optimisation landscape.

The Fisher information matrix $F_{ij} = \mathbb{E}[\partial_i \log|\Psi|^2 \cdot \partial_j \log|\Psi|^2]$ characterises the curvature of parameter space as seen by the wavefunction. In strong confinement ($\omega \geq 0.1$), the wavefunction is compact and all parameters contribute to its structure in a comparable manner: F is well-conditioned and its inverse provides an effective preconditioning of the gradient.

At ultra-weak confinement ($\omega \leq 0.01$), two effects conspire to make F ill-conditioned:

1. **Scale separation.** The wavefunction extends over $\sim 1/\sqrt{\omega}$ oscillator lengths, creating a large separation between the scale of the Jastrow correlator and the scale of the orbital envelope. Parameters controlling short-range correlations (cusp, pair features) contribute very differently to F than parameters controlling long-range structure (orbital widths, backflow offsets).

2. **Sampling starvation.** The Fisher matrix is estimated from importance-resampled points. When the ESS is low, the estimate of F is dominated by a few high-weight samples, making it noisy and rank-deficient. Tikhonov damping can stabilise the inversion, but at the cost of washing out the very curvature information that motivates natural gradients.

Adam, by contrast, maintains per-parameter adaptive learning rates using only first and second moment estimates of the gradient. It makes no attempt to model parameter-space curvature globally, and therefore does not fail when the curvature is pathological. The price is slower convergence in well-conditioned regimes (where CG-SR can take larger, more informed steps), but the benefit is robustness in ill-conditioned regimes where CG-SR diverges.

This regime boundary has practical implications for future work: any natural-gradient or second-order method applied to weakly confined many-body systems must explicitly address the conditioning of the Fisher or Hessian matrix in the low-overlap regime.

8.4.5 Why Langevin refinement fails

The Langevin proposal-refinement strategy was motivated by a natural intuition: if the Gaussian proposal has poor overlap with $|\Psi|^2$, then evolving samples toward high-$|\Psi|^2$ regions via the score function $\nabla_{\mathbf{R}} \log|\Psi|^2$ before resampling should improve overlap.

The failure of this approach reveals a subtlety that is well known in the MCMC literature but easy to overlook in the importance-sampling context: K steps of overdamped Langevin dynamics from an out-of-equilibrium starting distribution produce a distribution that is neither the proposal q nor the target $|\Psi|^2$, but an intermediate non-equilibrium distribution that depends sensitively on K and on the step size ε.

In the importance-resampling framework, the weights are computed as $w_i = |\Psi(\mathbf{R}_i)|^2/q(\mathbf{R}_i)$, which is only correct if the samples $\mathbf{R}_i$ are actually drawn from q. After K Langevin steps, the effective sampling distribution $\tilde{q}$ differs from q, but the weights are still computed using q—introducing a systematic bias in the estimated gradient. This bias is not merely noise: it systematically pushes the wavefunction toward regions favoured by the Langevin dynamics, creating a positive-feedback loop in which the wavefunction concentrates further around the Langevin-biased distribution.

The result is dramatic: at N=20, ω=0.1, Langevin refinement produced a wavefunction with +153% error, compared to +5.7% from standard importance resampling. Even at N=6, ω=0.001, where the overlap problem is less severe, Langevin added +0.06% of additional error.

The correct resolution would require either (i) running the Langevin chain to equilibrium ($K \to \infty$, equivalent to MCMC and defeating the purpose), (ii) a Metropolis–Hastings correction at each step (also MCMC), or (iii) computing annealed importance-sampling weights that account for the non-equilibrium path (computationally expensive and technically involved). None of these is practical within the collocation paradigm.

8.4.6 Architectural trade-offs at scale

The reversal of the backflow advantage at N=20 highlights an important architectural trade-off. The copresheaf backflow network (`CTNNBackflowNet`) stores edge features of size $\mathcal{O}(N^2 \times h_{\mathrm{bf}})$, which at N=20 with h_{bf}=128 requires ∼10× the memory of the Jastrow-only network. This forces a choice: reduce h_{bf} (losing expressivity), reduce the collocation budget (losing gradient quality), or reduce the oversampling factor (losing ESS). All three options degrade the result.

The Jastrow-only network, with ∼25k parameters and $\mathcal{O}(N^2)$ edge cost (no hidden-dimension multiplier), permits full-size collocation budgets (n_{coll}=4096, oversample=8–16) even at N=20. The resulting gradient is less noisy, and the simpler loss landscape (fewer parameters) makes optimisation more reliable.

This does not mean that backflow is unnecessary at N=20; it means that the current training pipeline cannot supply enough gradient signal to train backflow effectively at this scale. The implication is that scaling collocation to larger systems requires either (i) memory-efficient backflow architectures (e.g. sparse attention, low-rank message passing), or (ii) improved sampling that provides higher ESS at constant computational cost.

8.4.7 Comparison with MCMC-based results

A direct comparison with the SR-refined results in Table 6.2 illuminates the relative strengths of the two approaches:

- At (N=6, ω=1.0), collocation (+0.009%) matches the SR-refined result (+0.026–0.048% range for N=6) and is competitive with or slightly better than it. This is the regime where collocation is most effective, because the configuration space is low-dimensional and the Gaussian proposal overlaps well with $|\Psi|^2$.
- At (N=12, ω=0.1), collocation (+0.122%) is roughly 2–4× worse than SR (+0.040% in Table 6.2). Here the sampling bottleneck is beginning to bite, and the MCMC sampler's ability to draw equilibrated samples from $|\Psi|^2$ gives it a clear advantage.
- At N=20, collocation results (+1.4–5.7%) are significantly worse than what an MCMC-based approach would be expected to achieve, reflecting the exponential ESS degradation in 40 dimensions.

The SR pipeline also has its own scaling challenges: MCMC mixing times grow with particle number and correlation strength, and the SR step itself requires solving a $P \times P$ linear system (P=number of parameters). But these challenges manifest at much larger N than the collocation ESS collapse, because the MCMC sampler adapts to $|\Psi|^2$ throughout training rather than relying on a fixed proposal.

The practical conclusion is that collocation is competitive with MCMC for $N \lesssim 12$ and $\omega \gtrsim 0.01$, and that the two approaches are complementary: collocation for fast, parallelisable pretraining and exploration, MCMC for final refinement at larger N or lower ω.

8.4.8 Implications and future directions

The collocation experiments have several implications for the broader effort to apply neural wavefunctions to many-body quantum systems.

MCMC is not necessary for training. The demonstration that REINFORCE with importance resampling can match DMC accuracy for $N{=}6, 12$ challenges the assumption that MCMC is essential for neural wavefunction optimisation. This opens the door to fully parallel training pipelines that scale efficiently on modern hardware, particularly GPUs with large batch sizes.

What "MCMC-free" does and does not mean. Three distinctions are worth stating precisely, since they are easily conflated. (i) The pipeline is *Markov-chain-free*: no configuration is ever drawn from a Markov chain during training—samples come from an analytic proposal $q(\mathbf{R})$ and are importance-reweighted. It is not free of Monte Carlo in the broader sense: importance sampling and multinomial resampling are themselves Monte-Carlo operations. (ii) It is not, in general, *reference-free*. Where a DMC energy exists ($\omega \geq 0.1$), the residual pretraining anneals its target toward E_{DMC} and the instability-rollback monitors the relative error against it; the DMC value therefore enters the training signal, so those runs should be read as "trained without a Markov chain," not as an unsupervised solution of the Schrödinger equation. (iii) The deep-Wigner cascade of Chapter 7 is the one regime that is both Markov-chain-free *and* reference-free: below $\omega = 0.1$ no DMC value exists, so the cascade is guided only by the variational objective and cross-checked against the physical energy-scaling law, never against an external benchmark.

The proposal distribution is the bottleneck. The systematic pattern of ESS collapse with increasing N and decreasing ω identifies the proposal distribution as the critical component to improve. Possible directions include normalising flows trained to approximate $|\Psi|^2$, diffusion-model proposals, or variational inference techniques that can adapt the proposal to the evolving wavefunction during training.

Natural gradients require careful conditioning. The CG-SR failure at low ω is not specific to collocation: any natural-gradient method applied to weakly confined systems will encounter the same ill-conditioning. This suggests that preconditioning strategies (block-diagonal Fisher approximations, Kronecker-factored curvature, or adaptive damping schedules) deserve attention in the neural quantum state context.

Backflow scaling needs architectural innovation. The $\mathcal{O}(N^2 \times h_{\mathrm{bf}})$ memory scaling of current backflow networks limits their applicability beyond $N{\approx}12$. Memory-efficient architectures—sparse message passing, low-rank approximations, or local backflow corrections that avoid all-to-all edge features—are needed to extend the collocation approach to larger systems while retaining the nodal-surface corrections that backflow provides.

Collocation as a pretraining stage. Perhaps the most practical near-term use of collocation is as a fast pretraining stage that conditions the wavefunction before MCMC-based refinement. The collocation-trained checkpoints already contain useful structural information (orbital shapes, Jastrow correlations, approximate nodal surfaces) that accelerates subsequent SR convergence. This two-stage protocol—collocation pretraining followed by SR polish—combines the embarrassing parallelism of collocation with the asymptotic accuracy of MCMC-based methods.

Part VI

Conclusion

8.5 Conclusions for quantum dots

The quantum-dot benchmarks in this chapter serve a dual purpose: they test the numerical stability and expressivity of the Slater–Jastrow PINN+backflow ansatz, and they provide a controlled setting in which to study Wigner-molecule formation and the internal representations learned by the network.

On the *energy* side, the results in Sec. 8.1 show that a single, trap-unit–normalised architecture with soft-core pair features and an explicit cusp achieves DMC-level accuracy for two-electron dots and maintains sub-0.1% relative errors for $N \in \{6, 12\}$ across $\omega \in \{0.1, 0.5, 1.0\}$. The same setup extrapolates smoothly to ultra-weak traps at $\omega = 10^{-3}$, where no DMC data are available, with internal consistency checks (scaling exponents, virial ratios and structural diagnostics) supporting the reliability of the resulting variational predictions. Crucially, these accuracies are obtained with a relatively modest SR tail on top of a residual-based pretraining stage, emphasizing that conditioning and inductive bias matter at least as much as raw sample count.

On the *structural* side, Sec. 6.3 establishes that the weak-trap states are genuine Wigner molecules rather than vaguely "strongly correlated" dots. As ω decreases, all three systems ($N = 2, 6, 12$) undergo a smooth crossover in which (i) radii and localization ratios evolve towards large, stiff configurations, (ii) the energy partitioning approaches the nearly classical virial form $V_{\text{int}} \approx 2V_{\text{trap}}$, and (iii) shell structures and bond orders emerge that match classical Coulomb-cluster topologies. For N=6, the dot forms a single rotating Wigner ring with a quantum mixture of $(1, 5)$ and $(0, 6)$ sectors; for N=12, a robust two-shell Wigner molecule with a commensurate $(3, 9)$ split becomes modal. Shell-resolved reconstructions of $g(r)$ confirm that the full pair structure is explained by these topologies and their combinatorics, with cosine similarities ≈ 1 between reconstructed and global $g(r)$.

On the *representation* side, Sec. 8.3 shows that this rich real-space physics is carried by a remarkably simple latent geometry. The correlator W_θ lives on a low-dimensional manifold ($r_{\text{eff}}(Z) \sim$ 1–2) whose leading axes behave as collective coordinates for global size and radial fluctuations, with finer angular and topological information held in conditional distributions on that manifold; the identity of the leading axis shifts across regimes in a $\phi \leftrightarrow \psi$ recomposition rather than by growing in dimension. The backflow, by contrast, realises a high-dimensional displacement field whose energetic leverage sits in near-field configurations at intermediate traps. One earlier reading of the weak-trap limit—that the optimiser simply "switches backflow off"—proves to be architecture-specific, and is corrected here: it is the *conventional* backflow that collapses to a single collective mode (rank $\to$ 1) at the Wigner crystal—its single round of pairwise messages failing to sustain the relative displacement the crystal demands—whereas the copresheaf message-passing backflow retains full rank and keeps its accuracy there (Chapter 7). How these internal numbers should be read is itself a result: quantities split between the Jastrow and the backflow are gauge-like, fixed by the training route rather than the state, so the load-bearing conclusions rest on the invariants—energy, overlap, and ablation—not on the component ranks themselves.

On the *method* side, reading the ansatz through its tangent kernel (Chapter 7) connects these observations to the design choices that produced them. Message passing earns its cost by supplying a relational channel: it compresses the correlator onto fewer effective tangent directions and sustains a full-rank backflow where a separable network cannot, and at the Wigner crystal the inter-particle messages account for essentially the entire backflow contribution. Natural-gradient (SR) preconditioning helps in proportion to how ill-conditioned the tangent metric is—little for $|\Psi|^2$-sampled VMC, decisively for the collocation objective—and the same conditioning story explains why an MCMC-free, importance-sampled collocation scheme equals VMC when its sampling is healthy and, with a proposal built from the Wigner shell itself, reaches a genuine Wigner molecule at $\omega = 0.0035$ without any Markov-chain Monte Carlo during training.

Taken together, these findings converge on a single theme. What makes the difference across architecture, optimiser, and training paradigm is not raw capacity but alignment with the low-dimensional, physically collective structure of the problem—the same structure that, in real space, *is* the Wigner molecule. A compact correlator supplies learned collective coordinates; a message-passing backflow supplies the relational corrections a single-round-message network cannot sustain; natural gradients and $|\Psi|^2$ sampling keep the optimisation pointed along those directions. The quantum-dot results thus provide both a stringent validation of the method and a template for the same ideas in more complex finite fermion systems, where the trade-offs between global structure, local correlation, and representation are expected to recur.

Part VII

Appendices

Chapter A

Laplacian conditioning

A.1 Residual training for Schrödinger

We consider the stationary many-body Schrödinger equation on configuration space $\mathbf{R} = (\mathbf{r}_1, \ldots, \mathbf{r}_N) \in \Omega \subset \mathbb{R}^D$ with $D = dN$ (for our quantum dots, $d = 2$):

$$\hat{H}\Psi(\mathbf{R}) = E\Psi(\mathbf{R}), \qquad \hat{H} = \hat{T} + \hat{V}, \qquad \hat{T} = -\frac{1}{2}\sum_{i=1}^{N} \Delta_i, \tag{A.1}$$

where $\hat{V}$ collects external and interaction potentials.

In *strong residual minimization* one introduces

$$L := \hat{H} - E \quad \Rightarrow \quad L\Psi = 0, \tag{A.2}$$

and minimizes a squared residual loss of the schematic form

$$\mathcal{L}_{\text{res}}(\theta) = \frac{1}{2}\|L\Psi_\theta\|^2_{L^2(\Omega;w)} = \frac{1}{2}\int_\Omega |(L\Psi_\theta)(\mathbf{R})|^2\, w(\mathbf{R})\, d\mathbf{R}, \tag{A.3}$$

for a nonnegative weight w induced by the sampling strategy. A key message of the numerical-analysis viewpoint is that strong residual losses can be *severely ill-conditioned*, and training speed is tightly linked to spectral properties of an operator/matrix induced by L [38].

A.2 Linearized geometry and the matrix A

To expose the mechanism, follow the standard linearization argument used in conditioning analyses [38]. Fix an expansion point θ_0 and define tangent (feature) functions

$$q_j(\mathbf{R}) := \partial_{\theta_j}\Psi_\theta(\mathbf{R})\big|_{\theta=\theta_0}, \qquad j = 1, \ldots, P. \tag{A.4}$$

For small parameter increments $\delta\theta$, we have the first-order model

$$\Psi_{\theta_0+\delta\theta}(\mathbf{R}) \approx \Psi_0(\mathbf{R}) + \sum_{j=1}^{P} q_j(\mathbf{R})\,\delta\theta_j, \qquad \Psi_0 := \Psi_{\theta_0}. \tag{A.5}$$

Applying the residual operator gives

$$(L\Psi_{\theta_0+\delta\theta})(\mathbf{R}) \approx (L\Psi_0)(\mathbf{R}) + \sum_{j=1}^{P} (Lq_j)(\mathbf{R})\,\delta\theta_j. \tag{A.6}$$

Introduce the weighted inner product (complex-valued wavefunctions allowed)

$$\langle f, g\rangle_w := \int_\Omega f(\mathbf{R})^*\, g(\mathbf{R})\, w(\mathbf{R})\, d\mathbf{R}. \tag{A.7}$$

Let the Jacobian J be the linear map whose jth column is Lq_j. Then the loss (A.3) becomes (to second order in $\delta\theta$)

$$\mathcal{L}_{\mathrm{res}}(\theta_0 + \delta\theta) \approx \frac{1}{2}\|r_0 + J\,\delta\theta\|_w^2, \qquad r_0 := L\Psi_0. \tag{A.8}$$

with normal-equation matrix

$$A := J^* J, \qquad A_{ij} = \langle Lq_i,\, Lq_j\rangle_w. \tag{A.9}$$

Here A is Hermitian positive semidefinite; in practice it can be close to singular, so one often effectively works with a damped variant $A + \lambda I$ (implicit or explicit).

Now consider gradient descent on the quadratic model (A.8) (with step size η). The error in parameter space evolves as

$$\delta\theta^{(t+1)} - \delta\theta^\star = (I - \eta A)\left(\delta\theta^{(t)} - \delta\theta^\star\right), \tag{A.10}$$

so each eigen-direction of A contracts by a factor $(1 - \eta\lambda)$. Hence convergence speed is governed by the eigenvalue spread of A, typically summarized by the (effective) condition number

$$\kappa(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} \tag{A.11}$$

(over the nonzero/regularized spectrum). Large $\kappa(A)$ implies stiffness: a step size small enough to be stable for $\lambda_{\max}$ yields slow progress along directions associated with $\lambda_{\min}$. This is precisely the conditioning bottleneck emphasized in physics-informed training analyses [38].

A.3 Bi-Laplacian scaling

The defining feature of Schrödinger is that L contains a second-order differential operator. To isolate the kinetic contribution, consider the regime where the residual is dominated by the Laplacian part,

$$L \approx -\frac{1}{2}\Delta_{\mathbf{R}}. \tag{A.12}$$

From L to L^*L. The matrix A in (A.9) is a Gram matrix in the feature space $\{Lq_j\}$. The corresponding operator-level object is the *Hermitian square* L^*L (with adjoint taken in the $L^2(\Omega; w)$ inner product) [38]. For general non-constant w, the adjoint L^* differs from the formal differential adjoint by weight-induced lower-order terms; the cleanest picture is obtained for constant w and standard boundary conditions (periodic/Dirichlet), where the Laplacian is self-adjoint.

In the Laplacian-dominant regime (A.12), one has (up to a constant factor)

$$L \approx -\tfrac{1}{2}\Delta \quad \Rightarrow \quad L^*L \approx \tfrac{1}{4}\,\Delta^2, \tag{A.13}$$

i.e. a bi-Laplacian scaling. The prefactor does not affect conditioning ratios.

Fourier/eigenmode derivation of k^4. On a periodic domain (or, more generally, in a Laplacian eigenbasis) $-\Delta\phi_{\mathbf{k}} = |\mathbf{k}|^2\phi_{\mathbf{k}}$, so $\Delta^2\phi_{\mathbf{k}} = |\mathbf{k}|^4\phi_{\mathbf{k}}$. If an error $\delta\Psi$ expands as $\delta\Psi = \sum_{\mathbf{k}} c_{\mathbf{k}}\phi_{\mathbf{k}}$ and $L \approx -\frac{1}{2}\Delta$, then

$$\|L\,\delta\Psi\|_{L^2}^2 \approx \frac{1}{4}\sum_{\mathbf{k}} |\mathbf{k}|^4 |c_{\mathbf{k}}|^2. \tag{A.14}$$

Thus the squared residual penalizes high-frequency components with a *quartic* weight $|\mathbf{k}|^4$. If the relevant spectrum spans $|\mathbf{k}| \in [k_{\min}, k_{\max}]$, the induced eigenvalue ratio scales like

$$\kappa \sim \left(\frac{k_{\max}}{k_{\min}}\right)^4, \tag{A.15}$$

which is a sharp mathematical expression of “Laplacian-driven stiffness”. This quartic growth is consistent with the Laplacian/Poisson toy examples and frequency-dependent conditioning effects emphasized in [38].

Length-scale interpretation. Let ℓ denote the smallest length scale that must be represented. Then $k_{\max} \sim 1/\ell$ and (A.15) implies

$$\kappa \text{ grows rapidly as } \ell \downarrow 0. \tag{A.16}$$

In many-body Schrödinger problems, small ℓ arises from nodal structure, strong localization, and short-range correlation features; in all such regimes, the Laplacian makes the strong residual extremely curvature-sensitive.

A.4 Barron/spectral perspective

The conditioning story above can be rephrased in approximation terms. In spectral Barron-type viewpoints, function classes are controlled by norms that weight the Fourier transform by powers of frequency (or related spectral weights). A key implication is that applying derivatives systematically increases the importance of high-frequency tails [38]. Since Δ multiplies a Fourier mode by $|\mathbf{k}|^2$, and the squared residual effectively involves L^*L (hence Δ^2 in Laplacian-dominant regimes), the residual can require substantially more "spectral budget" than the wavefunction itself. In practical terms: even if Ψ is approximable in a given network class, *approximating its strong residual* can be much harder, because the loss implicitly demands accurate high-frequency curvature (cf. (A.14)). This is one of the motivations, in numerical analysis of PINNs, for considering weak/variational formulations that reduce derivative order when the strong form is too stiff [38].

A.5 Implications for Stochastic Reconfiguration

Stochastic reconfiguration (SR) updates parameters using a natural-gradient-like preconditioner [28, 30, 41]. Writing $O_i = \partial_{\theta_i} \log \Psi_\theta$, one common SR update is

$$S(\theta)\,\delta\theta = -g(\theta), \qquad S_{ij} = \mathrm{Cov}_{p_\theta}[O_i, O_j], \qquad g_i = \mathrm{Cov}_{p_\theta}[E_L, O_i], \tag{A.17}$$

where $p_\theta \propto |\Psi_\theta|^2$ and $E_L = (\hat{H}\Psi_\theta)/\Psi_\theta$. SR improves conditioning in parameter space by using the (quantum) information geometry of the variational family [28, 41]. However, the Laplacian-induced stiffness described above is an *operator-level* phenomenon: strong residuals magnify curvature errors via Δ and Δ^2 (cf. (A.14)), which remains a fundamental challenge even when parameter updates are geometrically preconditioned.

Chapter B

Coulomb conditioning

B.1 Coulomb structure and singularity

For our two-dimensional electronic systems ($\mathbf{r}_i \in \mathbb{R}^2$), the interaction contains Coulomb terms of the form

$$\hat{V}_{\mathrm{C}}(\mathbf{R}) = \sum_{1 \le i < j \le N} \frac{1}{r_{ij}}, \qquad r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|. \tag{B.1}$$

This is a *singular multiplication operator*: it diverges at particle coalescence $r_{ij} \to 0$. The exact eigenfunction remains finite (or vanishes by antisymmetry) but develops non-smooth short-distance structure (a cusp) so that divergences in kinetic and potential contributions cancel in physically meaningful quantities such as the local energy [30].

The central point for learning is:

> Coulomb does not merely add "another term" to the residual; it introduces *singular short-range physics* that forces delicate cancellations with the Laplacian. Strong residual objectives square these effects and become extremely sensitive to tiny cusp errors.

B.2 2D cusp conditions: antiparallel vs parallel spins

We present the 2D cusp logic in a way that separates the two spin channels. The cleanest statement (especially for practical ansätze) is in terms of the short-distance behaviour of a *Slater–Jastrow* factorization

$$\Psi(\mathbf{R}) = D(\mathbf{R})\, J(\mathbf{R}), \qquad J(\mathbf{R}) = \exp\Big(\sum_{i<j} u(r_{ij})\Big), \tag{B.2}$$

because (i) the Slater part D enforces antisymmetry and determines whether Ψ vanishes at coalescence, and (ii) the Jastrow slope $u'(0)$ is what cancels the $1/r$ Coulomb singularity in the local energy [30].

Fix a pair (i,j) and introduce relative/center-of-mass coordinates in 2D:

$$\mathbf{R}_{ij} := \frac{\mathbf{r}_i + \mathbf{r}_j}{2}, \qquad \mathbf{r} := \mathbf{r}_i - \mathbf{r}_j, \qquad r := |\mathbf{r}|. \tag{B.3}$$

A direct computation gives, in any dimension,

$$\Delta_i + \Delta_j = \frac{1}{2}\Delta_{\mathbf{R}_{ij}} + 2\,\Delta_{\mathbf{r}}, \tag{B.4}$$

hence the singular short-distance kinetic behaviour is controlled by the relative term $-\Delta_{\mathbf{r}}$.

A key 2D identity is

$$\Delta_{\mathbf{r}} r = \frac{1}{r} \qquad (r > 0,\ \mathbf{r} \in \mathbb{R}^2). \tag{B.5}$$

Antiparallel spins (distinguishable in the Slater part). If the two electrons have opposite spin, they enter different determinants in D and generically D does *not* vanish at $\mathbf{r} \to 0$. Then $\Psi(0) \neq 0$ and a direct cusp expansion in 2D is legitimate:

$$\Psi(\mathbf{r}, \ldots) = \Psi(0, \ldots)\Big(1 + a\, r + \mathcal{O}(r^2)\Big), \qquad r \to 0. \tag{B.6}$$

Using (B.5), the singular part of the relative Laplacian is

$$\Delta_{\mathbf{r}}\Psi \approx \Psi(0)\, a\, \Delta_{\mathbf{r}} r = \Psi(0)\, \frac{a}{r}, \qquad \frac{\Delta_{\mathbf{r}}\Psi}{\Psi} \approx \frac{a}{r}. \tag{B.7}$$

Since the pair kinetic contribution contains $-\Delta_{\mathbf{r}}$, the singular part of the local energy is

$$E_L(\mathbf{R}) = \frac{\hat{H}\Psi}{\Psi} \supset \left(-\frac{a}{r} + \frac{1}{r}\right) = \frac{1-a}{r}. \tag{B.8}$$

Thus finiteness of E_L at coalescence requires

$$a = 1, \qquad \text{i.e.} \qquad \partial_r \log \Psi|_{r=0} = 1 \quad \text{(2D, antiparallel spins)}. \tag{B.9}$$

Equivalently, in a Slater–Jastrow form with $D(0) \neq 0$, this is the same as $\partial_r \log J|_0 = 1$ for the ij-pair.

Parallel spins (identical fermions, node at coalescence). For same-spin electrons, antisymmetry forces D (and hence Ψ) to vanish as $\mathbf{r} \to 0$. Generically one has a *linear* zero:

$$D(\mathbf{r}, \ldots) \;=\; \boldsymbol{\eta}(\ldots) \cdot \mathbf{r} \;+\; \mathcal{O}(r^2), \qquad r \to 0, \tag{B.10}$$

with some nonzero vector $\boldsymbol{\eta}$ depending on the other coordinates. In this case, the cancellation of the Coulomb singularity is controlled by the Jastrow factor through cross terms in $-\Delta_{\mathbf{r}}(DJ)/(DJ)$. A standard short-distance analysis (carried out explicitly for 2D quantum dots) shows that the divergent kinetic contribution behaves like

$$-\frac{\Delta_{\mathbf{r}}(DJ)}{DJ} \supset -\frac{(2k+d-1)}{r}\, \partial_r \log J \qquad \text{with} \qquad d = 2, \quad k = 1 \text{ (identical particles)}, \tag{B.11}$$

while the Coulomb potential contributes $+1/r$ to E_L as before. Hence finiteness of the local energy requires

$$\partial_r \log J|_{r=0} = \frac{1}{2k+d-1} = \frac{1}{3} \quad \text{(2D, parallel spins)}. \tag{B.12}$$

This 1 (antiparallel) versus 1/3 (parallel) 2D cusp statement is the standard quantum-dot/Jastrow–Slater cusp condition [42].

What to remember. In 2D:

$$\text{antiparallel spins:} \quad \partial_r \log \Psi|_0 = 1 \quad \Longleftrightarrow \quad \partial_r \log J|_0 = 1, \tag{B.13}$$

$$\text{parallel spins:} \quad \Psi(0) = 0 \text{ (linear node)}, \quad \partial_r \log J|_0 = \frac{1}{3}. \tag{B.14}$$

B.3 Why Coulomb is ill-conditioned (2D emphasis)

The cusp conditions above show *how* Coulomb creates stiffness: it forces kinetic and potential contributions to be individually large and singular, while their *sum* is finite only under a short-distance constraint. Strong residual minimization is therefore a *cancellation problem*.

Sensitivity amplification in the local energy. If the learned antiparallel-spin cusp slope is $a_\theta = 1 + \delta a$, then (B.8) implies

$$E_L(\mathbf{R}) \supset -\frac{\delta a}{r}, \tag{B.15}$$

so an arbitrarily small cusp mismatch produces an arbitrarily large local-energy spike as $r \to 0$.

For parallel spins, if the learned Jastrow slope is $\partial_r \log J|_0 = \frac{1}{3} + \delta b$, then the same short-distance analysis leading to (B.12) implies

$$E_L(\mathbf{R}) \supset -\frac{3\,\delta b}{r}. \tag{B.16}$$

Thus both channels generate $1/r$ spikes unless the cusp is satisfied.

Why squaring is especially dangerous in 2D. A $1/r$ defect in E_L (or in $(\hat{H}-E)\Psi$) is concentrated near coalescence. But in 2D the squared singularity is not integrable—it diverges logarithmically:

$$\int_{r<\varepsilon}\left(\frac{1}{r}\right)^2 d^2\mathbf{r} = \int_0^{\varepsilon}\frac{1}{r^2}\,(2\pi r\,dr) = 2\pi\int_0^{\varepsilon}\frac{dr}{r} = +\infty.$$

Therefore, without effectively resolving the cusp, *strong* squared-residual objectives can be dominated (or even made ill-defined in the continuum limit) by the coalescence region. In Monte Carlo practice, this manifests as heavy-tailed fluctuations and extreme sensitivity to rare close-pair samples.

Residual expansion and cancellation structure. Writing $L = \hat{T} + (\hat{V} - E)$ and expanding the squared residual norm:

$$\|L\Psi\|_w^2 = \|\hat{T}\Psi\|_w^2 + \|(\hat{V}-E)\Psi\|_w^2 + 2\,\mathrm{Re}\langle\hat{T}\Psi,\ (\hat{V}-E)\Psi\rangle_w, \tag{B.17}$$

we see that near coalescence both $\hat{T}\Psi$ and $\hat{V}_{\mathrm{C}}\Psi$ are large. The exact solution relies on cancellation between these contributions (cf. (B.8) and (B.12)), which means (B.17) demands high *relative* accuracy in the cusp region: if either term is off by a small fraction, the net residual can be large in absolute magnitude. This is a classic recipe for ill-conditioning: the loss landscape contains directions where small parameter changes cause huge residual changes in a small region.

Operator-level picture (L^*L). From the same viewpoint as Appendix A, strong residual training induces matrices/operators tied to L^*L [38]. For $L = \hat{T} + (\hat{V} - E)$, one has (formally)

$$L^*L = \hat{T}^2 + (\hat{V}-E)^2 + \hat{T}(\hat{V}-E) + (\hat{V}-E)\hat{T}. \tag{B.18}$$

Coulomb enters here in two destabilizing ways:

1. the positive term $(\hat{V}-E)^2$ involves a $1/r^2$-type singular weight;
2. the mixed terms contain derivatives of V through product rules, e.g. $\hat{T}(V\Psi)$ generates $\nabla V\cdot\nabla\Psi$ and ΔV contributions, which are even more singular/distributional at coalescence.

This formal expansion explains why the Coulomb singularity can severely inflate the effective spectral spread of the training operator, aligning with the operator-conditioning viewpoint [38].

B.4 Barron/spectral perspective

Coulomb forces the *true* 2D solution to have cusp-like, non-smooth short-range structure, so approximating the *strong residual* requires representing high-frequency content that is not well captured by globally smooth/low-complexity function classes.

In spectral/Barron-style approximation theories, approximation quality is governed by spectral decay and norms that weight the Fourier tail [38]. A cusp (or a linear node times a cusp in J) implies slower spectral decay than a globally smooth function. Strong residual training then *further* amplifies the high-frequency tail because it applies Δ (and effectively L^*L), which multiplies Fourier modes by powers of $|\mathbf{k}|$ (Appendix A). Hence, even if Ψ can be approximated reasonably in function value, the strong residual can remain large unless the cusp structure is represented accurately. This is exactly the type of regularity mismatch that motivates weak/variational formulations in numerical analysis of PINN-type methods, since weak forms reduce the derivative order demanded of the approximation [38].

Practically, QMC wavefunctions therefore incorporate explicit short-range correlation structure (e.g. cusp-enforcing Jastrow factors), precisely to remove these singular residual pathologies and reduce the variance of local quantities [30, 42].

B.5 Implications for Stochastic Reconfiguration

SR performs a natural-gradient step using a (quantum) Fisher/metric matrix [28, 30, 41]. In one common formulation,

$$S_{ij} = \mathrm{Cov}_{p_\theta}[O_i, O_j], \qquad g_i = \mathrm{Cov}_{p_\theta}[E_L, O_i], \qquad S\,\delta\theta = -g, \tag{B.19}$$

with $O_i = \partial_{\theta_i} \log \Psi_\theta$.

Coulomb-induced cusp mismatch produces E_L spikes of the form (B.15)–(B.16), which has two consequences:

1. **Heavy-tailed Monte Carlo forces.** The estimator of $g_i = \mathrm{Cov}[E_L, O_i]$ becomes noisy when E_L has rare but extreme spikes (short-range events). This increases variance and can destabilize the linear solve unless one uses damping/regularization, as standard in SR practice [30].

2. **Coupling to curvature sensitivity.** Even if SR corrects anisotropy in parameter space (natural gradient) [28, 41], it does not remove the underlying operator-driven stiffness: the spikes originate from imperfect kinetic–potential cancellation enforced by Coulomb cusp physics.

Bottom line. Laplacian dominance creates stiffness through k^4-type amplification of curvature errors; Coulomb creates stiffness by enforcing cusp constraints (in 2D: 1 vs 1/3 depending on spin channel) and cancellation against the Laplacian. Both mechanisms worsen conditioning of strong residual objectives and increase statistical difficulty of SR through heavy-tailed local-energy fluctuations [30, 38, 42].

Chapter C

The collocation–backflow catch-22

This appendix formalizes a structural difficulty encountered when training backflow transformations under the *strong-form* collocation objective—the one that minimises the variance of the pointwise local energy E_L—and presents extensive numerical evidence for the N=6, ω=0.5 quantum dot. The core observation is a *catch-22*: backflow's learning signal is concentrated near the nodal surface, but the local energy diverges there for any imperfect wavefunction, and any sampling strategy that includes these points produces gradients with unbounded variance. The result is an irreducible accuracy floor—roughly 0.3%, set by the Jastrow alone—that is absent under VMC + SR training of the same ansatz.

Two things about the scope of this result should be stated before the details, so the floor is not misread. First, it is a property of the *strong form* specifically: the weak-form (score-function / REINFORCE) collocation objective used for the results in the main text never differentiates through the Laplacian and does not inherit this pathology, which is why it trains the same ansatz to far below the 0.3% floor. The value here is therefore a diagnosis of one objective's limits, not a measure of what MCMC-free training can achieve. Second, the analysis is what motivated the shift to the weak form in the first place; it is presented as a worked negative result—the kind that redirects a research programme—rather than as a headline number.

C.1 Setup and notation

We consider a Slater–Jastrow–backflow ansätz of the form

$$\Psi_\theta(\mathbf{R}) = \det \tilde{S}_\uparrow(\mathbf{R};\theta_{\mathrm{BF}})\ \det \tilde{S}_\downarrow(\mathbf{R};\theta_{\mathrm{BF}})\ \exp\big(J(\mathbf{R};\theta_J)\big), \tag{C.1}$$

where $\tilde{S}_\sigma$ is the Slater matrix evaluated at backflow-shifted coordinates $\tilde{\mathbf{r}}_i = \mathbf{r}_i + \Delta\mathbf{x}_i(\mathbf{R};\theta_{\mathrm{BF}})$, J is a neural Jastrow factor (PINN), and $\theta = (\theta_J, \theta_{\mathrm{BF}})$ are trainable parameters. A CTNN message-passing network produces the backflow shifts $\Delta\mathbf{x}_i$.

We train by minimizing the *screened collocation* objective

$$\mathcal{L}(\theta) = \mathrm{Var}_{\mathcal{S}}\big[E_L(\mathbf{R};\theta)\big], \qquad E_L = \frac{\hat{H}\Psi_\theta}{\Psi_\theta}, \tag{C.2}$$

where the collocation set $\mathcal{S}$ is obtained by importance-sampling from a Gaussian proposal q and retaining the top-K points ranked by $|\Psi_\theta|^2/q$ ("screened collocation"). This top-K selection naturally suppresses near-node configurations where $|\Psi|^2 \approx 0$.

C.2 The divergence of E_L near nodes

The local energy decomposes as

$$E_L = -\frac{1}{2}\frac{\nabla^2\Psi}{\Psi} + V = -\frac{1}{2}\Big[\nabla^2 J + |\nabla J|^2 + 2\,\nabla J \cdot \frac{\nabla D}{D} + \frac{\nabla^2 D}{D}\Big] + V, \tag{C.3}$$

where $D := \det\tilde{S}_\uparrow \det\tilde{S}_\downarrow$ and the last term $\nabla^2 D/D$ contains second derivatives of the backflow shifts $\nabla^2\Delta\mathbf{x}_i$ amplified by the inverse determinant.

For the exact wavefunction, E_L is constant. If Ψ is an eigenstate, $E_L = E$ everywhere, and the apparent D^{-1} singularity is cancelled by the numerator vanishing at the same rate as the denominator. Concretely, for a smooth wavefunction with a simple node at surface $\mathcal{N}$, both D and $\nabla^2 D$ vanish linearly in the signed distance d from $\mathcal{N}$:

$$D(\mathbf{R}) = \mathcal{O}(d), \qquad \nabla^2 D(\mathbf{R}) = \mathcal{O}(d), \qquad \frac{\nabla^2 D}{D} \text{ is finite on } \mathcal{N}. \tag{C.4}$$

For an approximate wavefunction, the cancellation generically fails. During training, the backflow parameters θ_{BF} are imperfect: the *learned* nodal surface $\mathcal{N}_\theta$ (where $D(\cdot;\theta) = 0$) does not coincide with the *true* nodal surface. Let d_θ denote the signed distance from the learned node and d^* that from the true node. Near a point where $d_\theta \approx 0$ but $d^* \neq 0$, the determinant $D \sim d_\theta \to 0$ while the kinetic contributions from the Jastrow and potential terms remain finite:

$$E_L(\mathbf{R}) \supset \frac{\nabla^2 D(\mathbf{R};\theta)}{D(\mathbf{R};\theta)} \sim \frac{c}{d_\theta} \to \pm\infty \qquad \text{as } d_\theta \to 0,\ d^* \neq 0, \tag{C.5}$$

for some configuration-dependent constant $c \neq 0$. Conversely, near a point on the true node ($d^* = 0$) but off the learned node ($d_\theta \neq 0$), $\Psi \sim d^* \to 0$ and the wavefunction assigns vanishing weight. The pathology (C.5) is therefore not an artifact of the nodal structure itself—it is a property of *mismatched* nodes, and is worst precisely when backflow still has something to learn.

C.3 The geometric mismatch argument

We now show that the variance of the local energy (and hence of the training gradients) generically diverges for imperfect backflow parameters.

Consider the contribution to $\mathrm{Var}[E_L]$ from a neighborhood U_ε of the learned nodal surface:

$$\mathrm{Var}[E_L] \geq \int_{U_\varepsilon} |\Psi(\mathbf{R};\theta)|^2 \left(E_L(\mathbf{R};\theta) - \bar{E}\right)^2 d\mathbf{R}. \tag{C.6}$$

When the node is "simple" (D vanishes linearly), we have

$$|\Psi|^2 \sim d_\theta^2, \qquad\qquad E_L^2 \sim d_\theta^{-2\alpha}, \tag{C.7}$$

where α characterises the severity of the mismatch. For the generic case where the numerator $\nabla^2 D$ does *not* vanish on $\mathcal{N}_\theta$ (because $\mathcal{N}_\theta \neq \mathcal{N}^*$), one has $\alpha = 1$ so that $E_L^2 \sim d_\theta^{-2}$.

In D-dimensional configuration space, the volume element in a tubular neighborhood of a codimension-1 surface scales as $dV \sim d_\theta^0\, d\sigma\, dd_\theta$. Thus the integrand in (C.6) behaves as

$$|\Psi|^2 \cdot E_L^2 \cdot dV \sim d_\theta^2 \cdot d_\theta^{-2} \cdot dd_\theta = dd_\theta, \tag{C.8}$$

which integrates to $\mathcal{O}(\varepsilon)$ and is therefore *finite*: a simple node does not, by itself, make $\mathrm{Var}[E_L]$ diverge under the $|\Psi|^2$ measure. The instability enters one derivative later, in the *parameter gradient*. But the key observation is that this is the *best case*. In practice, the mismatch between $\mathcal{N}_\theta$ and $\mathcal{N}^*$ is not uniform: there exist regions where the surfaces cross at angles, creating pockets where $|d^*|/|d_\theta|$ is large (the true wavefunction is non-negligible while the learned determinant is near-zero), and there the E_L^2 spike is not compensated by $|\Psi|^2$ suppression.

More concretely, the gradient $\partial\mathcal{L}/\partial\theta_{\mathrm{BF}}$ involves terms of the form

$$\frac{\partial E_L}{\partial\theta_{\mathrm{BF}}} \supset \frac{1}{D}\frac{\partial(\nabla^2 D)}{\partial\theta_{\mathrm{BF}}} - \frac{\nabla^2 D}{D^2}\frac{\partial D}{\partial\theta_{\mathrm{BF}}}, \tag{C.9}$$

both of which diverge as $D \to 0$. Even in the marginal case where $\mathrm{Var}[E_L]$ is finite, the *gradient variance* (which involves these higher-order poles) generically diverges, making gradient-based optimisation impossible at points near the learned node.

C.4 The catch-22

The preceding analysis reveals a three-sided dilemma:

1. **Backflow learns at the node.** The purpose of backflow is to reshape the nodal surface of the Slater determinant. The gradient signal $\partial E_L/\partial\theta_{\mathrm{BF}}$ is largest where the backflow shifts actually change which side of the node a configuration point lies on, i.e., where $D \approx 0$.

2. **E_L diverges at the node for imperfect θ.** As shown in §C.2–C.3, the local energy and its parameter gradient contain poles of the form D^{-1} and D^{-2} whenever the learned node does not coincide with the exact node. The resulting gradient variance is unbounded.

3. **Good collocation sampling must avoid the node.** Under screened collocation, the top-K selection ranks points by $|\Psi|^2/q \propto D^2 e^{2J}/q$. Since $D^2 \to 0$ at the node, near-node points are systematically excluded from the training batch. This is *protective*—it prevents gradient explosion—but it also removes the points where backflow's learning signal lives.

A "catch-22" emerges:

> *Including near-node points gives divergent, noisy gradients and backflow shrinks defensively or diverges. Excluding them gives stable training but removes backflow's learning signal. In either regime, collocation cannot effectively train backflow beyond an accuracy floor set by the Jastrow (PINN) quality alone.*

C.5 Why VMC + SR escapes

Under VMC, the sampling distribution *is* $|\Psi_\theta|^2$, which assigns vanishing measure to the nodal surface. Near-node configurations are strongly *suppressed*—for a simple node the probability of lying within a distance ϵ scales as ϵ^3—rather than strictly absent, which is enough to keep the D^{-1} singularity out of the gradient estimator in practice. However, VMC *still optimises the node position* through a subtle mechanism:

Indirect nodal information through bulk correlations. The variational energy $\langle E_L\rangle_{|\Psi|^2}$ is sensitive to nodal position because shifting the node changes the wavefunction *everywhere* (through normalisation and through the functional form of D). The SR force vector $g_i = \mathrm{Cov}_{|\Psi|^2}[E_L, O_i]$ (where $O_i = \partial_{\theta_i} \log \Psi$) receives contributions from the *bulk*—regions where both $|\Psi|^2$ and E_L are well-behaved—that are *correlated with* nodal position but do not require evaluating E_L at the node.

SR amplifies the signal. The Fisher metric $S_{ij} = \mathrm{Cov}[O_i, O_j]$ captures how sensitive $\log\Psi$ is to each parameter direction. By solving $(S + \varepsilon I)\,\delta\theta = -g$, SR amplifies exactly those subtle bulk signals that encode nodal information, compensating for the fact that the node itself has zero weight under $|\Psi|^2$.

Core asymmetry. VMC obtains nodal information *indirectly* through a well-conditioned integral (the covariance g_i is finite because E_L is sampled away from the node); collocation tries to obtain it *directly* through a divergent pointwise quantity (E_L evaluated near the node). None of the sampling, loss, or preconditioning strategies we tested (§C.6) overcame this asymmetry within the strong-form collocation framework; we present it as a structural obstruction of the tested pipeline rather than a theorem that no collocation method can ever avoid it.

C.6 Experimental evidence

We present a systematic study for N=6 electrons in a 2D harmonic trap with ω=0.5 (DMC reference: $E_{\mathrm{DMC}} = 11.78484(6)$ [39]). The ansätz uses a PINN Jastrow (∼15,000 parameters) and a CTNN backflow network (∼8,600 parameters) with bf_scale = 0.7. All energies are evaluated via MCMC with 15 000 samples after burn-in.

C.6.1 Baseline: PINN alone vs. PINN + CTNN

Table C.1: Baseline collocation results for N=6, ω=0.5. "PINN only" trains without backflow; "PINN + CTNN joint" trains both networks from scratch for 300 epochs with bf_scale = 0.7. The backflow provides only a marginal improvement of ∼0.1 percentage points.

Configuration	E (Ha)	σ_E	err (%)
PINN only (300 ep)	11.834	0.003	0.42
PINN + CTNN joint (300 ep)	11.823	0.003	0.33
PINN + CTNN separate opt	11.822	0.003	0.32
DMC reference	11.78484	—	0.00

The backflow reduces the error from 0.42% to 0.32–0.33%, far short of the $< 0.01\%$ accuracy achievable with VMC + SR on the same ansatz (Table 6.2).

C.6.2 Sampling strategies

We tested multiple strategies to bring near-node information into the collocation batch, hoping to give the backflow a stronger learning signal.

Table C.2: Effect of sampling strategy on collocation accuracy. All experiments use 50-epoch fine-tuning from the best collocation checkpoint (0.35%), except "hard-smooth (scratch)" which trains 300 epochs from random initialisation.

Method	Description	err (%)	Outcome
Top-K baseline	standard screened collocation	0.35	stable
Hard mining + Huber	top-K of residuals	0.41	neutral
Hard mining + smoothness	+ $\|\nabla^2 \Delta x\|^2$ penalty	0.38	slight help
SIR (single proposal)	$\|\Psi\|^2$-reweighted resampling	0.47	worse
SIR (mixture)	Gaussian mixture proposal	0.44	worse
Gumbel top-K	stochastic top-K selection	5.00	catastrophic

Every strategy that includes more near-node points (SIR, Gumbel) produced *worse* results—consistent with the catch-22: the additional points carry divergent E_L values that corrupt the gradient. Gumbel top-K (which replaces deterministic selection with stochastic Gumbel-softmax) was catastrophic, with early stopping triggered by rapid energy divergence.

C.6.3 Loss modifications

Table C.3: Effect of loss function modifications. The hybrid loss adds a mean-energy (variational) term; the det filter discards points with small $|\det \tilde{S}|$.

Method	Description	err (%)	Outcome
Variance loss (baseline)	Huber on $E_L - \bar{E}$	0.33	best collocation
Hybrid anneal	$\beta \cdot \mathrm{Var} + (1-\beta) \cdot \mathrm{Mean}$, $\beta : 0.7 \to 0.15$	2.59	catastrophic
Gumbel + hybrid	stochastic top-K + hybrid	5.00	catastrophic
Hybrid + det filter	β=0.2 fixed + $\|\det\|$ filter	24.75	broken

The mean-energy gradient $\nabla_\theta \langle E_L \rangle_{\mathcal{S}}$ under biased collocation sampling does *not* equal the VMC gradient—the sampling distribution is not $|\Psi|^2$—so the variational principle does not hold, and the backflow drifts to large, unphysical shifts ($|\Delta\mathbf{x}| \approx 0.83$, versus ∼0.23 under variance-only training). The variance loss's implicit constraint on backflow magnitude is *protective*, not pathological.

Table C.4: Regularisation of the E_L singularity (50-epoch fine-tune from 0.35% checkpoint). Each modification smooths or clips one of the divergent contributions to E_L.

Method	Modification	err (%)	Outcome
Soft Coulomb	$1/r \to 1/\sqrt{r^2+\varepsilon^2}$, ε=0.2	1.41	much worse
Regularised det	$\log\lvert D\rvert \to \mathrm{logaddexp}(\log\lvert D\rvert, \log\delta)$	0.49	worse
T clipping	$\max(\lvert T\rvert, T_{\max})$, $T_{\max}$=30	0.43	neutral

C.6.4 Gradient regularisation

We attacked the divergence sources in E_L directly:

Soft Coulomb trains on wrong physics (V is no longer the true potential). Regularising the determinant distorts the $\log\Psi$ landscape. Kinetic energy clipping is neutral because the clipped points are exactly those that carried the (noisy) backflow signal. None of the regularisations improve upon the baseline.

C.6.5 Gradient preconditioning

Table C.5: Gradient preconditioning experiments (300-epoch scratch training unless noted). K-FAC uses block-diagonal Kronecker-factored Fisher on the backflow network only.

Method	err (%)	Outcome
Vanilla Adam	0.33	baseline
K-FAC on CTNN	0.34	no improvement
K-FAC + smoothness	0.51	worse
Variance-minimising pretrain	0.54	worse
Adam fine-tune (30 ep, lr=1e-4)	0.34	no improvement
Adam burst (15 ep, lr=3e-4)	0.56	worse

K-FAC provides a block-diagonal approximation to the Fisher matrix and should, in principle, help with the ill-conditioned parameter-space geometry. In practice, it does not: the conditioning problem is *operator-level* (Appendices A–B), not merely parameter-space anisotropy, and K-FAC cannot address the fundamental divergence in E_L.

C.6.6 Summary of all interventions

The pattern is striking:

- Every intervention that *includes* more near-node points (SIR, Gumbel) *increases* the error—backflow shrinks defensively under gradient noise or diverges entirely.
- Every intervention that *changes* the loss to tolerate near-node contributions (hybrid mean-energy, det filter) produces catastrophic results because the collocation sampling is not $|\Psi|^2$ and the variational principle does not apply.
- The smoothness penalty $\lambda\|\nabla^2\Delta\mathbf{x}\|^2$ helps modestly (from 0.59% to 0.38% without it vs. with it) by damping the $\nabla^2 D/D$ poles, but cannot eliminate the fundamental divergence.
- Ironically, the only method that trains backflow effectively is MCMC-based VMC + SR—which sidesteps the entire collocation framework.

C.7 Connection to Appendices A and B

The catch-22 is a specific manifestation of the general conditioning difficulties analysed in Appendices A–B:

Table C.6: Complete summary of all collocation-based backflow experiments for N=6, ω=0.5. No intervention breaks the ∼0.32% accuracy floor. For comparison, the same PINN + CTNN ansatz achieves < 0.01% under VMC + SR (Table 6.2).

Experiment	err (%)	Category
PINN + CTNN sep opt (best)	0.32	baseline
PINN + CTNN joint 300 ep	0.33	baseline
K-FAC on CTNN	0.34	preconditioner
Adam fine-tune 30 ep	0.34	optimiser
Top-K baseline (retrained)	0.35	sampling
Hard mining + smoothness	0.38	sampling
Hard mining + Huber	0.41	sampling
PINN only	0.42	no backflow
T clipping	0.43	regularisation
SIR (mixture)	0.44	sampling
SIR (single)	0.47	sampling
Regularised det	0.49	regularisation
K-FAC + smoothness	0.51	preconditioner
Var-min pretrain	0.54	pretrain
Adam burst lr=3e-4	0.56	optimiser
Soft Coulomb	1.41	regularisation
Hybrid loss (annealed)	2.59	loss
Gumbel top-K + hybrid	5.00	sampling + loss
Hybrid + det filter	24.75	loss + filter
Same ansatz, VMC + SR	**<0.01**	**variational**

Laplacian stiffness (Appendix A). The k^4-type amplification of curvature errors (§A.3) is compounded by backflow, because $\nabla^2(\det \tilde{S})$ introduces second derivatives of $\Delta\mathbf{x}$ that are not present in a fixed-node ansatz. These additional high-frequency modes inflate the spectral spread of the training operator L^*L.

Coulomb singularity (Appendix B). The $1/r$ Coulomb potential (§B.1) adds a singular multiplication operator to $\hat{H}$. Near both the coalescence point ($r_{ij} \to 0$) and the node ($D \to 0$), E_L contains products of these two singularities, and the cusp cancellation (§B.2) must be maintained simultaneously with the nodal cancellation (C.4). In the strong residual squared $\|LE_L\|^2$, these compound singularities create an extremely stiff landscape.

The backflow-specific pathology. What distinguishes the backflow catch-22 from the general conditioning problem is the *self-referential* nature of the singularity: the backflow parameters θ_{BF} determine *where* the node is, so updating θ_{BF} moves the singularity in configuration space. Under the collocation loss, the gradient pushes θ_{BF} in a direction determined by E_L values at the current collocation points, but the resulting node shift invalidates those very points. VMC+SR avoids this because it samples from the *current* $|\Psi_\theta|^2$ at every step, so the sampling distribution co-adapts with the node position.

Chapter D

Post-catch-22 experimental history and philosophy

This appendix documents the full experimental program executed after the backflow catch-22 analysis in Appendix C. The goal here is not to present a single best number, but to make explicit the strategy, the failed branches, the partial recoveries, and the current status for difficult regimes ($\omega \in \{0.01, 0.001\}$ and larger N).

D.1 Program-level philosophy

After establishing that naive collocation backflow training can stall on nodal pathologies, the project moved to a pragmatic philosophy:

1. Preserve physically strong ansätze (Slater–Jastrow–backflow), but treat optimisation and sampling design as first-class objects.
2. Replace one-shot "hero runs" with staged continuation and transfer: first solve easier regimes, then warm-start harder ones.
3. Separate claims by evidence tier: online probe, restored VMC-best, and final heavy exact VMC evaluation.
4. Record negative results as signal: many variants that look plausible theoretically fail in practice because of gradient-noise geometry.

Operationally, this produced a sequence of campaign scripts rather than a single training script. The scripts define matrices of controlled variants over sampling, loss, architecture, and preconditioning.

D.2 What was tried (complete family map)

This section groups all post-catch-22 explorations by mechanism.

D.2.1 Sampling schemes

The following sampling and batch-formation ideas were tested:

- Gaussian and Gaussian-mixture proposal sampling with σ_f-mixtures scaled by $1/\sqrt{\omega}$.
- Oversample-then-resample pipelines (importance-resampled collocation) with effective-sample-size diagnostics.
- ESS-adaptive oversampling, including floor-triggered retries and oversample growth.
- Weight regularisation during resampling: temperature scaling and log-weight quantile clipping.

- Geometry-aware replay/hard-sample buffers (in regime-policy and robustness-style campaigns).
- Coalescence-focused diagnostics and regularisation campaigns for $N = 6$, $\omega = 0.01$.

Most important empirical conclusion: for low ω, regularising the resampling distribution (ESS control and top-mass reduction) was beneficial, while hard cusp gating alone had negligible effect at the tested scale.

D.2.2 Architecture families

The architecture search did not only vary width/depth; it also compared model classes:

- Jastrow-only families (CTNN variants, V-cycle forms, and alternatives) for robust baselines.
- Standard BF+Jastrow CTNN coordinate backflow (the most consistently competitive family in this phase).
- Wider/deeper BF variants to test whether capacity or optimisation was the bottleneck.
- Neural-Pfaffian branches were retained as exploratory alternatives, but were not the leading branch in the post-catch-22 campaign sequence.

The main pattern was that optimisation quality dominated architectural novelty: moderately sized BF+Jastrow models with stable training often outperformed more ambitious variants with weaker optimisation geometry.

D.2.3 Loss functions and objective styles

Three objective styles were actively explored:

1. FD-collocation residual variants (finite-difference Laplacian paths, with optional Huber/proximal terms).
2. Hybrid weak-form objective combining REINFORCE-style score-function gradient with an optional direct weak-form term.
3. Pure REINFORCE weak-form setting (hybrid with direct weight set to zero), which became the dominant reliable mode for key runs.

Crucially, the Laplacian term used to compute local-energy reward is kept in the forward reward path, while optimisation uses score-function gradients with robust clipping/selection controls. This was one of the practical stability levers after the catch-22 findings.

D.2.4 Optimiser and preconditioning styles

The project tested progressively richer preconditioning:

- Adam baselines and continuation schedules.
- Diagonal Fisher natural-gradient variants (multiple sweep rounds).
- Full SR-style preconditioners in collocation runs (Woodbury and CG modes) with damping annealing, trust-region controls, and parameter-change caps.
- CG-SR campaigns with staged continuation across (N, ω) targets.

Empirically, SR-style preconditioning in CG mode became central for transferring performance from solved regimes to harder ones.

D.2.5 Training styles

The workflow evolved from isolated runs to structured orchestration:

- Single-run sweeps (natgrad and architecture sweeps).
- Wave-based cascade campaigns with checkpoint inheritance.
- Long-running asynchronous campaigns where follow-up jobs launch from best available checkpoints.
- Regime-aware policy campaigns and seed-robustness protocols for reproducibility diagnostics.
- Stabilisation campaigns targeting known failure regimes (high N, low ω).

D.3 Campaign chronology and key outcomes

D.3.1 Natural-gradient sweeps

The natural-gradient program (baseline/high-LR/low-damping controls and continuations) established that diagonal Fisher alone was not sufficient for all hard regimes, but identified stable hyperparameter corridors and motivated the move to SR-style CG preconditioning.

D.3.2 SR sweep and refinement

Woodbury and CG SR variants were benchmarked, then refined with lower LR, larger SR subsamples, damping anneals, and tighter trust controls. These runs substantially improved reliability near the best $N = 6, \omega = 1.0$ basin and provided the best warm-start sources for cross-regime transfer.

D.3.3 Generalisation runs across ω

The SR generalisation campaign showed the strongest and weakest transfer cases simultaneously:

- $N = 6, \omega = 1.0$ and $N = 6, \omega = 0.5$ reached near-DMC quality.
- $N = 6, \omega = 0.1$ from scratch remained far from DMC, confirming that low ω requires transfer/cascade rather than cold-start SR.

This directly motivated multi-wave cascade campaigns.

D.3.4 Cascade campaigns

Cascade campaigns implemented structured inheritance:

$$\omega = 1.0 \to 0.5 \to 0.1 \to 0.01 \to 0.001, \tag{D.1}$$

and similarly along particle count for selected branches.

The strongest concrete successes to date in this phase include:

- $N = 6, \omega = 1.0$: final near-DMC performance ($E = 20.16167 \pm 0.00181$, error $+0.012\%$).
- $N = 6, \omega = 0.5$: essentially DMC-level agreement ($E = 11.78466 \pm 0.00199$, error -0.002%).
- $N = 6, \omega = 0.1$: major improvement from failed scratch regimes to transfer-level performance (best completed transfer around $E = 3.56525 \pm 0.00046$, error $+0.321\%$).
- $N = 12, \omega = 1.0$: near-DMC transfer success ($E = 65.70699 \pm 0.00595$, error $+0.010\%$).

D.3.5 Coalescence-focused matrix

The $2 \times 2 \times 3$ coalescence matrix (hard cusp gate on/off, regularised-resampling on/off, three seeds) provided a useful controlled test:

- Hard cusp gating alone produced no measurable improvement over baseline.
- Resampling regularisation was consistently beneficial: ESS and top-mass diagnostics improved sharply, and mean final error improved relative to unregularised variants.

This result narrowed the practical hypothesis space: in the tested low-ω setting, sampling regularisation mattered more than structural hard gating.

D.4 Current frontier: low-ω and high-N stabilisation

The current unresolved front remains the combination of extreme correlation regimes and larger particle count:

- $\omega \in \{0.01, 0.001\}$: transfer helps, but reliability and reproducibility are not yet at the same confidence level as $\omega \in \{1.0, 0.5, 0.1\}$.
- $N \geq 20$: memory pressure, architecture-capacity constraints, and higher gradient noise make training significantly less stable than $N \in \{6, 12\}$.

Dedicated stabilisation runs introduced stricter SR trust constraints, heavier damping, ESS-adaptive sampling, rollback controls, and explicit process-level relaunch hygiene for GPU memory contention. These changes improved operational stability but have not yet closed the full accuracy gap for the hardest targets.

The N=20 collocation results. For completeness we record here the N=20 collocation energies referred to in §6.4.3; they are the first collocation-trained wavefunctions at this size, and they show the memory bottleneck in the clearest possible form. The backflow's edge-feature tensor scales as $\mathcal{O}(N^2 \times h_{\text{bf}})$, and at N=20 fitting it in memory forces the hidden dimension down to h_{bf}=48–64. The best backflow run at ω=1.0 then reached only $\sim$+18% error, while a Jastrow-only ansatz—freed of that tensor, and so able to afford a larger collocation budget and higher oversampling—reached +1.32% ($E = 157.934 \pm 0.018$). The architecture that should in principle be the more expressive is beaten by the simpler one, purely because the memory-forced capacity cut degrades its gradients more than its extra flexibility repays. Polishing the Jastrow checkpoints at reduced learning rate improved the ω=1.0 error from $\sim$+2.6% to +1.32% over $\sim$400 epochs and the ω=0.5 error from $\sim$+7.0% to +2.74% ($E = 96.447 \pm 0.015$); the best ω=0.1 result was +5.53% ($E = 31.637 \pm 0.008$). These are honest first numbers. The gap to the sub-percent accuracy reached at N=6, 12 follows from the memory-forced capacity reduction and the gradient noise it brings, not from the collocation objective; closing it is a question of a more memory-frugal backflow, not of a different method.

D.5 Interpretive summary

The post-catch-22 phase supports six robust conclusions.

1. The collocation program is not a single recipe but a coupled system: sampling, objective, and preconditioning must be tuned jointly.
2. Transfer/cascade is not optional for difficult regimes; it is the primary mechanism by which low-ω performance becomes attainable.
3. SR-style preconditioning is central for stable progress beyond early Adam-only basins.
4. Diagnostic metrics (ESS, top-mass concentration, rollback counts, probe-final gaps) are essential; scalar final energy alone is insufficient for method development.

5. Several intuitively attractive interventions (e.g., hard cusp gating in this tested range) did not deliver gains, and documenting these negatives was necessary to avoid circular search.
6. The method now demonstrates near-DMC quality in multiple nontrivial targets, but the strongest claims must still be bounded to regimes where repeated campaign evidence exists.

From a thesis perspective, the philosophical shift was from “find one better architecture” to “engineer a reliable optimisation-and-sampling pipeline that makes physically good ansätze trainable across regimes.”

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